

A MERGED RESEARCH SYNTHESIS AND CONTINUATION

Jordan Separation, Cauchy Theory, and Stokes Calculus

on the Surcomplex Plane

Topologies, admissible chains, polynomial and deformed Stokes,
multiscale contours, radius-free residues, and untransformed Leray cycles

September 21, 2026

Abstract

A counterpart of a complex-analytic theorem over $\mathbf{SC} = \mathbf{No}[z]$ must specify three things before it has a definite meaning: a topology, a regularity algebra for the integrand, and a category of chains with an integration pairing. The full fine topology admits no nonconstant continuous ordinary real-parameter paths; a fixed Hahn workspace has a different, nondiscrete but totally separated intrinsic topology. Neither supports the literal ordinary Jordan-curve picture.

We separate the question into four theories and prove each in its own category. First, two Jordan separation theorems: a deliberately coarse one in the standard-part topology, whose separator is the whole boundary monad rather than a curve, and the established semialgebraic Jordan theorem over real closed fields, which we import rather than prove. Second, three chain categories carrying an exact generalized Stokes theorem: polynomial simplices over any characteristic-zero field, Hahn differential forms on ordinary manifolds with the cohomology comparison $H^q(\Omega_H^\bullet(M)) \simeq H_{\text{dR}}^q(M; \mathbb{C})((t^\Gamma))$, and infinitesimally deformed chains with a support-controlled Taylor pullback. Third, Cauchy theory as a period theory: an exact endpoint-transport formula, integer winding at displaced centers, a Cauchy formula on deformed Jordan contours, and moving-pole residues. Fourth, a positive-monomial rescaling theorem that converts radius-free Hahn analytic germs into polynomial-coefficient families, and with it microscopic circle and torus realizations of the perturbation residue, including a determinant-transformed realization for an arbitrary isolated complete intersection. That realization left open the question of a cycle for the *untransformed* integrand, and the question is then settled: a ramified finite covering of the coordinate torus, built from the leading map and one positive valuation threshold alone, carries the original form $H dz/(F_1 \cdots F_n)$ to the perturbation residue for every perturbation above that threshold and every radius-free numerator, with an exact equation-adapted lift, a pole-free Hahn homotopy, a degree normalization, a finite-sheet trace, and an extension to formal coefficient germs.

A rational contour pairing built from semialgebraic winding numbers is characterized by a uniqueness property and compared with Hahn integration on explicitly verified overlaps. Root-of-unity sampling is analyzed on both sides: a coefficientwise recovery theorem under an explicit decay hypothesis, and an exact example showing that the hypothesis cannot be dropped. Berkovich, tropical, perfectoid, definable, tame, and resurgent integration theories are compared without identifying their distinct targets.

Status. Established results, statements reported in the two supplied manuscripts, and deductions proved here are distinguished throughout. This is not a claim of a unique accepted foundation for surcomplex analysis, an exhaustive priority determination, or a formally verified proof. Symbolic checks validate the displayed finite examples and are not machine verification of the general proofs. All sums and mathematical workspaces used in the constructions are set-sized. The literature search was conducted on September 21, 2026, and was targeted rather than exhaustive; no named published conjecture is claimed to be solved.

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1 The question, the sources, and the principal conclusions

1.1 Three theorems, three kinds of structure

The three classical theorems in the title concern different structures. The Jordan curve theorem describes separation by a topological embedding of a circle. Cauchy's integral theorem describes the vanishing of periods of holomorphic differential forms under a homological hypothesis. Generalized Stokes relates a differential on forms to a boundary on chains. They cooperate in ordinary complex analysis but are not logically interchangeable: even over $\mathbb{C}$, a Jordan curve need not be rectifiable, a contour need not be simple for Cauchy's theorem to apply, and being closed does not make a cycle null-homologous in a punctured domain. One should therefore not try to extend the three theorems to $\mathbf{SC} = \mathbf{No}[i]$ by a single replacement of $\mathbb{C}$ by a larger field.

Real closedness of $\mathbf{No}$ supplies algebraic closedness of $\mathbf{SC}$, but it does not select a topology, a function algebra, or a chain category. The useful answer is not one unrestricted transplantation but a collection of precisely compatible counterparts. The classical starting point is the usual contour, Cauchy, and differential-form theory [5, 33, 24].

The positive conclusions are these. A genuine Jordan analogue is available in *semialgebraic topology* over every real closed Hahn workspace. A second, intentionally coarse Jordan statement is available after passing to standard parts. An exact $\mathbf{SC}$ -valued Stokes calculus is available in three separate chain categories: polynomial simplices, Hahn-coherent chains over an ordinary base, and infinitesimal deformations of the latter. Cauchy theory becomes a closed-form period theory rather than a consequence of fine connectedness. Positive monomial rescaling makes radius-free analytic germs accessible to contour methods, and rational differentials admit a contour pairing based on algebraic winding numbers even when an ordinary leading contour meets the reduction of a pole. Finally, the untransformed integrand of an arbitrary isolated complete intersection has a genuine single cycle, once the chain category is allowed the ramified finite coverings of the coordinate torus that it always admitted; the cycle is not a coordinate torus, and could not have been. These answers coexist; they are not interchangeable.

1.2 What the supplied manuscripts provide

We refer to *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory* as the *analysis manuscript* [1]. It distinguishes fine-local power-series calculus from the algebra

$$\mathcal{H}(U) = \mathcal{O}(U)((t^\Gamma))$$

of well-ordered Hahn families of ordinary holomorphic functions on one common ordinary domain U . Its section on contour functionals defines integration coefficientwise on ordinary paths and proves Cauchy, Morera, primitive, and integral-estimate theorems; later sections prove moving-pole residue formulas, permit coherent affine changes of scale, and construct a global fine-analytic logarithm. These are existing results within the supplied framework.

The second source, *Infinitesimal Analytic Geometry over the Surcomplex Numbers: Radius-Free Hahn Germs, a Nullstellensatz, and Conservation of Singular Zeros*, is the *geometry manuscript* [4]. It replaces common-domain coefficient functions by

$$\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma)),$$

where each coefficient is a convergent complex germ but the coefficients need not share a radius. For isolated complete intersections it defines a perturbation residue by a strongly summable series

of ordinary local residues, and it explicitly does *not* define that functional by a surcomplex contour integral. The gap between a well-defined residue and a representing contour is one principal topic below.

Neither manuscript establishes a Jordan theorem for arbitrary fine-topological curves, nor generalized Stokes for arbitrary fine-smooth functions on arbitrary surcomplex domains. We do not silently attribute either assertion to them. We retain their distinction between ordinary coefficient regularity, strong summability, and fine differentiation. This article is not an independent audit of every theorem in either supplied manuscript.

1.3 Established external counterparts

There are relevant results beyond the supplied sources. Berarducci and Otero develop definable intersection numbers, winding numbers, and Jordan–Brouwer separation over o-minimal expansions of real closed fields [7, 8, 9]; Woerheide’s o-minimal homology is the other route to definable Jordan–Brouwer [38]. Eisermann gives an algebraic winding-number construction using Cauchy indices and Sturm chains over real closed fields [17], with corrections and boundary cases treated by Perrucci and Roy [32]. Ohmoto and Shiota give C^1 semialgebraic triangulations and a Stokes formula, and explicitly allow a larger saturated target field [29]; Maříková and Shiota measure definable sets with values in an ordered semiring [26]; Kaiser develops tame semialgebraic integration over real closed fields with Archimedean value group [23]. Chambert-Loir and Ducros construct a Stokes theory for real forms on Berkovich spaces [14], whose underlying topology is developed by Baker–Rumely and Temkin [10, 35]. Mihara constructs analytic singular homology and an integration theory over cosimplicial perfectoid spaces [25]. Cherry and Diamond develop Schnirelman-type and p -adic line integrals [13, 16]; a contemporary use of such averages appears in [18]. Ducros, Hrushovski, and Loeser construct a comparison between complex and non-Archimedean integrals [15]. Costin and Ehrlich study surreal extension and integration of distinguished classes of real functions [12].

These works answer substantial parts of the question in their own categories. A theorem about real-valued tropical forms is not already a theorem about K -valued holomorphic contour integrals. A definably connected interval is not an ordinarily connected interval. An integral valued in a saturated extension or an ordered semiring is not automatically a canonical element of a previously fixed Hahn field.

1.4 A map of the answers

Question	Valid counterpart	Essential qualification
Jordan separation	Definable circle in a real closed plane	Definably connected components, not ordinary ones
Coarse Jordan separation	Standard-part preimage of an ordinary Jordan curve	A thick wall in a non-Hausdorff topology
Generalized Stokes, algebraic	Polynomial simplices over any characteristic-zero field	A finite factorial identity, not a measure
Generalized Stokes, analytic	Hahn forms on ordinary or infinitesimally deformed chains	Support-controlled pullback and integration
Cauchy vanishing	Closed Hahn-holomorphic 1-form on an admissible null-homologous contour	Coherent coefficient data, or a specified algebraic pairing

Question	Valid counterpart	Essential qualification
Contour deformation	Closed-form homotopy invariance; exact endpoint corrections	The whole homotopy must be admissible and avoid singular coefficients
Radius-free residues	Microscopic circle, or torus after positive monomial rescaling	A determinant transform for a general complete intersection
Untransformed residues	One Hahn cycle on a ramified finite cover of the torus, built from f and one valuation threshold	A convergent isolated leading map and a positive support bound; the cycle is not a coordinate torus
Arbitrary rational loops	Residue functional weighted by algebraic winding	A definition with compatibility theorems, not a Riemann limit
Sampling a circle	Coefficientwise root-of-unity recovery, or a verified valuation-tail criterion	Coefficientwise, valuation, and fine convergence are not interchangeable

The affirmative rows are theorems in specified categories, not approximations to a nonexistent unrestricted fine-topological contour theory.

1.5 Conventions, and one name for one concept

Three of the source manuscripts merged here use three different names for one and the same topology, and several symbols are used inconsistently across them. We fix the following conventions and use them without exception.

1. **The standard-part topology.** The initial topology of the standard-part map on the finite elements — whose open sets are exactly the preimages of ordinary open subsets of $\mathbb{C}$, whose fibres are indiscrete, and whose Hausdorff quotient is $\mathbb{C}$ — is called the *standard-part topology* throughout. It appears in the literature merged here under the names *residue topology*, *reduction topology*, and *shadow topology*; these are synonyms for one and the same topology, and are not used again. The descriptive name is preferred here because “residue” is already committed to the residue functionals of [sections 10](#) and [11](#) and “shadow” is used in the analysis manuscript for the ordinary shadow of a zero.
2. **One symbol per concept.** We write st for the standard-part (reduction) map $K^\circ \rightarrow \mathbb{C}$ and never red ; $\mathfrak{m}_K$ for the infinitesimals of the workspace and $\mathfrak{m}$ only for the class of all infinitesimals of $\mathbf{SC}$; $U^\sharp = \text{st}^{-1}(U)$ for the thickening of an ordinary set; v for the valuation of a field element and v_H for the least exponent of a Hahn coefficient family; wind for ordinary complex winding and wind_{sa} for semialgebraic (definable, algebraic) winding, never Ind or w_F ; res for an ordinary local residue and Res_F for the perturbation residue of [\(2.5\)](#) and [\(2.4\)](#).
3. **Scales.** Rescaling exponents are always written $\rho_j \in \Gamma_{>0}$ and the corresponding radii $r_j = t^{\rho_j}$. The sources use ρ , β , and s for the exponent and r , ρ , and r for the radius; only the first convention appears below.
4. **Multiplicities.** Three senses are kept apart and never conflated: the order $m \geq 1$ of an ordinary one-variable germ at zero; the global colength $\mu = \dim_{\mathbb{C}} \mathbb{C}\{z\}/(f_1, \dots, f_n)$ of an isolated complete intersection; and the local multiplicity $e_a = \dim_K B_a$ at an actual displaced zero, with $\sum_a e_a = \mu$. The coordinate exponents in a coordinate-power system are written d_j .
5. **Every theorem names its ring.** Four coefficient rings occur below under nearly identical notation, and no theorem is transferred between them. They are tabulated in [section 2.3](#).

1.6 What is merged here, and from where

This article merges four manuscripts of September 21, 2026. Three of them share roughly two thirds of their content: *Stokes Calculus*, *Stokes Integration*, and *Stokes Formulae on the Surcomplex Plane*. The shared material — the fine-topology no-go results, the standard-part Jordan theorem, the imported semialgebraic Jordan theorem, the Hahn form complex and its cohomology, deformed Stokes, the endpoint-displacement formula, the rescaling lemma, and the one-variable microscopic circle — is stated once.

The differences are real and sectional. The polynomial-chain theory of [section 5](#), the support-preserving Poincaré lemma [proposition 6.2](#), the order-valued estimate [proposition 7.7](#), the sampling theory of [section 15](#), and the scope discussion of [section 11.4](#) come from the second manuscript alone. The uniqueness characterization of the algebraic pairing [theorem 14.2](#), the two compatibility results [propositions 9.1](#) and [14.4](#), the quantitative finite-partition obstruction [proposition 3.5](#), and above all the unconditional general contour representation [lemma 11.3](#) and [theorem 11.4](#) come from the first. The numbered admissible-deformation definition [definition 7.1](#), the independent separating-torus route of [theorem 11.5](#), and the finite-parameter appendix [section B](#) come from the third. The second and the third both number a winding-stability theorem; [theorem 8.4](#) is the second’s affine-scale form, which is the stronger of the two and is the one printed.

Two of these overlap in a way that is a genuine disagreement about status rather than about arithmetic, and one is a disagreement about emphasis. Both are resolved in the open in [section 11.4](#) and [section 15](#) respectively, rather than averaged away.

A fourth manuscript, *Untransformed Leray Cycles for Radius-Free Hahn Analytic Residues* [2], was written afterwards and against this article, and is integrated here as [sections 12](#) and [13](#). It does not overlap the other three: it takes from them the perturbation residue (2.4), the positive-monomial rescaling discipline, the coefficient rings of [section 2.3](#), the support lemma [lemma 2.2](#) with the sharpening recorded after [example 2.3](#), and the standing refusal to convert strong summability into fine convergence; and it adds the construction that [section 11.4](#) and [section 17.3](#) had asked for by name. Everything it contributes is new to this article: the ramified finite covering [theorem 12.7](#) and the target-monomial direction [lemma 12.6](#) that avoids the discriminant; substitution of a positive-support Hahn map into a radius-free or formal germ, [lemma 12.2](#), which is not [theorem 10.1](#); the rooted-tree implicit-function lemma [lemma 12.4](#); the universal protecting cycle and its period [theorem 12.10](#); the exact adapted lift [theorem 13.2](#) with the pole-free homotopy [theorem 13.5](#); the direct degree normalization [theorem 13.6](#), the finite-sheet trace [corollary 13.7](#) and the ideal annihilation [proposition 13.8](#); and the formal-coefficient extension [theorem 13.10](#). Accordingly [sections 11.4](#) and [17.3](#) are rewritten below as the record of a question that was open and of the exact hypotheses under which it is now closed — not deleted, because a reader must still be able to see what was asked and that [theorem 11.4](#) answered only the transformed form.

2 Workspaces, supports, and the four coefficient rings

2.1 A set-sized workspace

Write $\mathbf{SC} = \mathbf{No}[i]$ and use the surreal normal-form monomials $t^\gamma = \omega^{-\gamma}$. Let $\Gamma \subset \mathbf{No}$ be a nonzero divisible set-sized ordered additive subgroup and put

$$R_\Gamma = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)) = R_\Gamma[i], \quad t^\gamma = \omega^{-\gamma}. \quad (2.1)$$

An element is a series whose exponent support is well ordered. Its valuation $v(a)$ is its least exponent, with $v(0) = +\infty$. The Hahn-field theorem makes R_Γ real closed and K algebraically closed; the algebraic-closure statement is Corollary 4 of Poonen [31]. The normal-form interpretation places both fields inside **SC**; the surreal background and restricted analytic evaluation are discussed by van den Dries and Ehrlich [37]. Put

$$K^\circ = \{a : v(a) \geq 0\}, \quad \mathfrak{m}_K = \{a : v(a) > 0\}, \quad \text{st} : K^\circ \longrightarrow \mathbb{C},$$

where st extracts the coefficient of t^0 . Every finite element is uniquely $a = c + \varepsilon$ with $c \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_K$. For ordinary $U \subset \mathbb{C}$ write $U^\sharp = \text{st}^{-1}(U)$. The modulus $|x + iy| = \sqrt{x^2 + y^2}$ is valued in R_Γ , not in $\mathbb{R}$; it satisfies the algebraic triangle inequality and is not a real-valued non-Archimedean norm.

Enlargement rule. Every finite collection of surcomplex points, and every set-sized coefficient presentation used below, lies in a workspace of this kind, obtained by adjoining the groups generated by the relevant normal-form supports and taking the divisible hull. *Its limit.* This does not authorize a sum over a proper class, nor a proper-class polynomial ring, nor does it imply that every rescaling required later already lies in a previously chosen Γ . Differentiation in a variable z holds every element of K fixed; it is not a derivation on **No** regarded as a transseries field. Full-class neighborhood statements may be interpreted in a class theory such as NBG, while all rings, supports, and parameter manifolds used in the integration constructions are set-sized.

2.2 Strong summability is algebra, not a hidden limit

Definition 2.1 (Strong summability). A family of Hahn series is *strongly summable* when the union of its supports is well ordered *and* only finitely many members contribute to any one exponent. Its sum is defined by finite addition at each exponent. For a well ordered $E \subset \Gamma_{>0}$, let E^* denote the additive monoid it generates, including 0.

Both clauses are load-bearing, and neither implies the other.

Lemma 2.2 (Support mechanism). *If $S, T \subset \Gamma$ are well ordered, then $S + T$ is well ordered and each exponent has only finitely many decompositions $s + t$. If $E \subset \Gamma_{>0}$ is well ordered, then E^* is well ordered and a fixed exponent has only finitely many representations as a finite ordered word in E — including only finitely many possible word lengths. The same conclusions hold for $S + E^*$.*

These are the Hahn–Neumann facts used throughout [28, 31]. Three consequences are used constantly. First, a complex-linear map $L : V \rightarrow W$ extends coefficientwise by $L(\sum_\gamma v_\gamma t^\gamma) = \sum_\gamma L(v_\gamma) t^\gamma$; this cannot enlarge support and requires no norm continuity. It will be applied to ordinary integration, differentiation, residue extraction, and chosen linear splittings of differential complexes. Second, for positive-support A the Neumann inversion $(1 + A)^{-1} = \sum_{q \geq 0} (-A)^q$ is strongly summable with output support in E^* . Third, $S + E^*$ is the universal support certificate: every construction below whose contributions are indexed by one source exponent in S together with a finite word in E is legitimate, and its output support lies in $S + E^*$.

Well-ordering cannot be weakened to positivity: $\{1/r : r \geq 1\}$ is a positive subset of $\mathbb{Q}$ that is not well ordered, and E^* would then fail to be well ordered.

Example 2.3 (Strong sums without valuation limits). For $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$, every integer is less than ω . The strong sum $\sum_{n \geq 0} t^n = (1 - t)^{-1}$ exists, but its finite partial sums do not converge to it in the valuation topology: their nonzero tails never have valuation greater than ω . Consequently no

argument of the form “the valuation improves by a fixed positive amount at each iteration” justifies any of the constructions below.

A second negative companion is equally load-bearing, and is stated here once because several proofs depend on it. “*Only finitely many dependencies at a given exponent*” does not mean “*only finitely many exponents below it*.” The latter fails below ω in $\mathbb{Q} + \mathbb{Q}\omega$, where the interval of exponents below ω contains every positive integer. It is the former statement, supplied by [lemma 2.2](#), that legitimises the finite-parameter replacement arguments in the residue transformation proofs of [section 11](#) and [section B](#). It is therefore unsafe to argue that “truncating below γ is a finite calculation”; the correct statement is that a *fixed output exponent* has finite combinatorial dependency.

2.3 The four coefficient rings, named once and never interchanged

Four coefficient rings appear below under nearly identical notation. They carry nearly identical theorem statements, and confusing them is the most likely source of a false attribution in this subject. Every theorem in this article names the ring it is stated over, and no theorem is transferred from one to another.

Ring	Notation	Defining requirement
Common-domain coherent	$\mathcal{H}(U) = \mathcal{O}(U)((t^\Gamma))$, $U \subset \mathbb{C}$ ordinary open	Every coefficient is holomorphic on <i>one</i> common ordinary domain U , and all share <i>one</i> well-ordered support. This is the fixed-domain (in one variable, fixed-disk) ring.
Polynomial-coefficient	$\mathbb{C}[w_1, \dots, w_n]((t^\Gamma))$	Every Hahn coefficient is an ordinary polynomial; contained in, but strictly smaller than, the common-domain ring with $U = \mathbb{C}^n$, since e^w is entire and is not a polynomial. This is the uniform-support target of rescaling, theorem 10.1 .
Radius-free germs	$\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$	Every coefficient is an ordinary convergent germ at 0, but their convergence radii may tend to zero, so <i>no common polydisk is required</i> .
Formal	$K[[z]]$, and $\mathbb{C}[[z]]((t^\Gamma))$	No convergence of any coefficient is required. Used only for injectivity statements and for the formal residue class (14.4).

One variant of the first row is used later and is not a fifth ring but the same requirement on a larger base: for a fixed ordinary complex *manifold* X , possibly disconnected, $\mathcal{H}(X) = \mathcal{O}(X)((t^\Gamma))$ demands holomorphy of every coefficient on that one fixed X and one common well-ordered support. It appears in [sections 12](#) and [13](#), where X is a finite covering of a polyannulus; see ([12.3](#)). No theorem stated over $\mathcal{H}(U)$ for an ordinary open $U \subset \mathbb{C}$ is transferred to it.

The inclusions $\mathbb{C}[w]((t^\Gamma)) \subset \mathcal{H}(\mathbb{C}^n)$ and $\mathcal{A}_n \subset \mathbb{C}[[z]]((t^\Gamma))$ are strict, and the containment $\mathcal{H}(U) \subset \mathcal{A}_n$ (for U a polydisk about 0, after taking germs) is strict as well. The running witness for strictness of the last containment is

$$H(z) = \sum_{j \geq 1} \frac{t^j}{1 - jz} \in \mathcal{A}_1, \quad (2.2)$$

whose coefficient poles at $1/j$ accumulate at 0, so that no fixed positive ordinary radius about zero carries all of its coefficient germs. Integrating it on one fixed ordinary circle as though all coefficients were holomorphic inside that circle would be invalid. It nevertheless evaluates at every $z \in \mathfrak{m}_K$. The remedy is not a uniform ordinary bound but infinitesimal rescaling; see [section 10](#).

A related warning is recorded here once. Changing the order of formal completion and Hahn extension can change the domain of the resulting expression. For instance

$$\frac{1}{z-t} = \begin{cases} \sum_{n \geq 0} t^n z^{-n-1}, & v(t) > v(z), \\ -\sum_{n \geq 0} t^{-n-1} z^n, & v(z) > v(t), \end{cases} \quad (2.3)$$

and the second expression lies in $K[[z]]$ but *not* in the unscaled radius-free ring $\mathcal{A}_1$, since its combined coefficient support in t is not well ordered. The first is an outer expansion enclosing the displaced pole; the second is a Taylor expansion on a smaller region not enclosing it. Their respective contour residues about zero are 1 and 0, and there is no contradiction: the admissible regions differ.

2.4 The perturbation residue, and why it is not a contour integral

Fix an ordinary analytic germ $f = (f_1, \dots, f_n)$ at 0 with isolated common zero, and a perturbation $F = f + E$ with $E_j \in \mathcal{A}_n$ and $v_H(E_j) > 0$ for each nonzero E_j . Zero perturbations are permitted. For $H \in \mathcal{A}_n$ the geometry manuscript defines

$$\text{Res}_F(H) = \sum_{k \in \mathbb{N}^n} (-1)^{|k|} \text{res}_{(f_1^{k_1+1}, \dots, f_n^{k_n+1})} \left(H \prod_{j=1}^n E_j^{k_j} \right), \quad (2.4)$$

where res is the ordinary local Grothendieck residue, a complex-linear functional on analytic germs, extended coefficientwise, and the normalization $(2\pi i)^{-n}$ is included in res . In one variable this reads

$$\text{Res}_F(H) = \sum_{k \geq 0} (-1)^k \text{res}_{f^{k+1}}(H E^k), \quad \text{res}_{f^{k+1}}(q) = \text{Res}_{z=0} \frac{q(z) dz}{f(z)^{k+1}}. \quad (2.5)$$

The positive support of E and [lemma 2.2](#) prove strong summability, with output support contained in $\text{supp } H + (\text{supp } E)^*$.

We state emphatically, once, what these formulas are. *Equation (2.4) is a series definition, not a surcomplex contour integral.* The geometry manuscript defines it algebraically precisely because no contour was available on $\mathcal{A}_n$. Supplying a contour that represents it is a separate task, and is the subject of [sections 10](#) to [13](#).

One functional, defined in two companion reports by the same series. The report *Finite Hahn Deformations: Division, Conservation of Multiplicity, and Residue Duality* [3], in this repository at `docs/surcomplex/finite-deformations`, defines a Hahn local-cluster residue by exactly the expansion (2.4) — the same alternating sum of ordinary local residues of $H \prod_j E_j^{k_j}$ against $\prod_j f_j^{k_j+1}$, applied coefficientwise, with the same sign $(-1)^{|k|}$, the same factor $(2\pi i)^{-n}$, the same ordered tuple $F_1, \dots, F_n$ and the same orientation convention. That report says emphatically that *the series is the definition*, that no fine-topological Riemann integral over a surcomplex parameterized torus is intended, and that supplying a genuine representing contour is a separate question left to this report. Its functional is stated over the common-domain ring $\mathcal{H}(D)$ for a fixed ordinary polydisk D , whereas (2.4) is the radius-free version over $\mathcal{A}_n$; the two formulas agree wherever both are defined,

and neither is a residue invented to make a contour identity tautological. *This identification is the point of the whole exercise:* every contour theorem below — [theorems 10.4, 11.1, 11.4, 11.5](#) and [12.10](#) — earns its value by computing that independently defined functional and nothing else. The companion report’s residue duality, its perfect pairing, and its conservation of multiplicity are results about the functional and are cited, not reproved, here; the trace identity [\(11.9\)](#) below, which [section 11](#) takes from the supplied geometry manuscript, is among the results that report establishes as well. The sign $(-1)^{|k|}$ and the factor $(2\pi i)^{-n}$ on the contour side are the chosen normalization; they are fixed so that $\text{Res}_F(J_F) = \mu$ and $\text{Res}_F(H) = \sum_a H(a)/J_F(a)$ at simple zeros, where $J_F = \det(\partial F_i/\partial z_j)$.

Three normalizations involving $2\pi i$ are kept deliberately separate throughout, and are never silently identified.

1. **Hahn contour.** The microscopic circle functional carries the factor $1/(2\pi i)$ and the torus functional $1/(2\pi i)^n$ outside the integral, so that each equals the algebraically defined perturbation residue on the nose.
2. **Local one-variable.** The residue $\text{res}_a(R dz)$ is the bare Laurent or partial-fraction coefficient $c_{a,1}$, *after cancellations* in the local finite-meromorphic germ; it is not merely the residue of one ordinary coefficient at $\text{st}(a)$. The factor $1/(2\pi i)$ again appears outside.
3. **Formal or algebraic.** In [\(14.4\)](#) and in the algebraic pairing [\(14.1\)](#) the factor $2\pi i$ is a *chosen* normalization matching ordinary complex integration, using the distinguished copy of $\mathbb{C}$ inside K . It is not forced by ordered-field arithmetic; over an abstract real closed field with no distinguished copy of $\mathbb{R}$ one would normalize formally or work with $I/(2\pi i)$.

3 Three topologies, and why the naive constructions fail

3.1 The full fine topology and its restriction

The full fine neighborhoods in **SC** are the balls $B(a, r) = \{z : |z - a| < r\}$ for *every* positive surreal r . A statement about this topology is a class-topological statement; it is not a claim that all fine-open classes form a set. The following conclusions are proved in the analysis manuscript [\[1, §3\]](#); the short argument is repeated because its exact scope is crucial.

Proposition 3.1 (Full-class discreteness of sets). *Every set-sized subset of **SC** is closed and discrete in the full fine topology. Every convergent net with a set-sized index set is eventually equal to its limit. Every fine-continuous map from an ordinary connected real interval, or from an ordinary circle, into **SC** is constant. In particular, the topology induced from the full fine topology on a fixed set-sized field K is discrete.*

Proof. For a point p and a set A , the nonzero numbers $|p - a|$, $a \in A$, form a *set* of positive surreals. The cut $\{0 \mid |p - a| : a \in A, a \neq p\}$ supplies a positive surreal δ strictly smaller than all of them, so $B(p, \delta)$ misses $A \setminus \{p\}$. This proves closedness and discreteness. Applying the same argument to the set of nonzero distances of the values of a net from its proposed limit, eventual membership in the resulting ball forces eventual equality. Finally, the image of an ordinary connected parameter space is a set; a connected subset of a discrete space has at most one point. $\square$

This is *not* a theorem that a Hahn field with its own valuation topology is discrete. The radius supplied by the cut may lie outside the workspace. The usual real-parameter unit-circle map is not fine-continuous; calling it a contour below will therefore always mean an admissible parameterized integration object, never a fine-continuous loop.

3.2 The intrinsic topology of a fixed Hahn workspace

The internal ordered-field topology on K uses radii from R_Γ . It agrees with the natural valuation topology: monomial valuation balls refine ordered balls, and sufficiently small ordered monomial balls refine any prescribed valuation ball, since $v(x) > v(r)$ implies $|x| < r$ for $r > 0$, while $|x| < t^{\gamma+\eta}$ with $\eta > 0$ implies $v(x) > \gamma$.

Proposition 3.2 (Intrinsic total separation). *The intrinsic topology on K is nondiscrete and totally separated. The cosets $a + r\mathfrak{m}_K$, $r \neq 0$, are open and closed and separate any pair of distinct points. Every connected subset is a singleton, every continuous map from an ordinary real interval or ordinary circle to K is constant, and ordinary singular homology of K in positive degrees vanishes.*

Proof. Every neighborhood of zero contains a nonzero monomial of sufficiently large exponent, so the topology is nondiscrete. The subgroup $r\mathfrak{m}_K$ is the valuation ball $\{x : v(x) > v(r)\}$; it is open, and its complement is the union of the other open cosets, so it is clopen. For $a \neq b$ the coset $a + (b-a)\mathfrak{m}_K$ contains a and not b , so a connected subset has at most one point, and continuous images of connected ordinary spaces are singletons. Every ordinary singular simplex is therefore constant, and the singular chain complex is the direct sum over points of the singular chain complex of a point, whose positive-degree homology vanishes. $\square$

The same clopen separation works in the full fine topology, using $a + (b-a)\mathfrak{m}$ with the full class of infinitesimals. There is consequently no ordinary topological embedding $S^1(\mathbb{R}) \hookrightarrow \mathbf{SC}$ in either topology, and deleting a putative circle does not leave exactly two nontrivial ordinary connected components. The nonzero periods constructed below therefore do *not* come from that singular chain complex.

3.3 The standard-part topology

On K° — or on the affine arena $X_{a,r} = a + rK^\circ$ for $a \in K$ and $r \in R_\Gamma$ positive, via the surjection

$$q_{a,r} : X_{a,r} \longrightarrow \mathbb{C}, \quad q_{a,r}(z) = \text{st}((z-a)/r) \quad (3.1)$$

— declare the open sets to be exactly the sets $q_{a,r}^{-1}(V)$ with $V \subset \mathbb{C}$ ordinary open. This is the initial topology of the standard-part map, the *standard-part topology* of [section 1.5](#). Its fibres are indiscrete and it is not Hausdorff, but its quotient by indistinguishability is exactly the ordinary complex plane. It deliberately forgets all infinitesimal distinctions. Its role is limited but clean: it supports a coarse Jordan theorem, not microscopic separation.

Remark 3.3 (Three topologies must not be conflated). The intrinsic topology of K uses radii in R_Γ . The topology inherited by the set K from the full class $\mathbf{SC}$ also permits radii outside R_Γ and is discrete by [proposition 3.1](#). The standard-part topology is coarser still. A claim about convergence or connectedness in one of them is not automatically a claim in either of the others.

3.4 Unrestricted fundamental theorems are impossible

Proposition 3.4 (No endpoint integration for all fine-local primitives). *There is no family of linear endpoint integral functionals $I_{a,b}$, defined for all fine-locally analytic derivatives and arbitrary endpoints, satisfying $I_{a,b}(H') = H(b) - H(a)$ for every fine-locally analytic H on the full surcomplex plane.*

Proof. The class $\mathfrak{m}$ of all infinitesimals and each of its additive cosets are fine-open and fine-closed. Hence

$$q(z) = \begin{cases} 0, & z \in \mathfrak{m}, \\ 1, & z \notin \mathfrak{m} \end{cases}$$

is locally constant, so fine-locally analytic, with $q' = 0$. At $a = 0$, $b = 1$ the proposed identity gives $I_{0,1}(0) = q(1) - q(0) = 1$, contradicting linearity. $\square$

This is the obstruction proved in [1, §3]; it is *not* an obstruction to a smaller coherent calculus. A proposed Stokes theorem must restrict either the forms, the chains, or the differential. The Hahn theory of sections 6 and 7 restricts coefficient coherence; the semialgebraic theory of section 4 restricts definability; the polynomial theory of section 5 restricts to finite algebraic data.

The analysis manuscript gives a sharper diagnostic. It constructs a globally fine-locally analytic logarithm $\mathcal{L}$ on $\mathbf{SC}^\times$ with $\mathcal{L}' = 1/z$, using a selected argument cut [1, section on global logarithms]. Its restriction to ordinary points has that argument jump and is *not* a coherent primitive on $(\mathbb{C}^\times)^\sharp$. Accordingly the coherent period $\oint_{|z|=1} dz/z = 2\pi i$ of section 8 does not contradict the existence of $\mathcal{L}$. Applying the coherent endpoint theorem theorem 8.2 to this noncoherent global function would be an invalid change of category; a valid fundamental theorem must restrict the primitive and the integration category together.

3.5 Finite meshes cannot resolve infinitesimal tolerances

Proposition 3.5 (Finite-partition obstruction). *Let $0 = x_0 < x_1 < \cdots < x_N = 1$ be a partition in R_Γ with an ordinary finite number N of subintervals. Its mesh is at least $1/N$. For $f(x) = x$ the right and left endpoint sums differ by*

$$\sum_{j=1}^N (x_j - x_{j-1})^2 \geq \frac{1}{N}. \quad (3.2)$$

Consequently ordinary finite tagged partitions cannot make all tag choices agree to a fixed positive infinitesimal error on $[0, 1]_{R_\Gamma}$.

Proof. The increments $\Delta_j = x_j - x_{j-1}$ are positive and sum to one, so at least one is at least $1/N$. The difference of the two endpoint sums is $\sum_j \Delta_j^2$, and the algebraic identity

$$N \sum_j \Delta_j^2 - \left(\sum_j \Delta_j \right)^2 = \sum_{j < k} (\Delta_j - \Delta_k)^2 \geq 0$$

proves (3.2). For any positive infinitesimal ϵ one has $1/N > 2\epsilon$ for every ordinary N , so the two tag choices cannot both lie within ϵ of one proposed integral. $\square$

There is a related logical pitfall. A definition asserting that “all partitions of mesh less than δ have sums close to I ” becomes vacuous when δ is infinitesimal: there are no such ordinary finite partitions. One must specify an actual directed approximation system and prove it has enough elements. Note also what is *not* being claimed: a polynomial antiderivative still gives $\int_0^1 x \, dx = 1/2$, and the obstruction is to a particular unrestricted finite-mesh definition, not to all integration. Hyperfinite partitions in a nonstandard universe can be useful, but they require an additional universe, transfer structure, and a chosen relation between that universe and the surreal or Hahn workspace. *They are not supplied merely by the field operations in $\mathbf{No}$* , and taking only the ordinary standard part of such an integral would also discard the infinitesimal information sought here.

3.6 A contour avoiding every pole need not have an admissible pullback

Consider the ordinary unit-circle trace and $a = 1 + t$ with $t > 0$ infinitesimal. The function $1/(z - a)$ has no pole on that trace. Away from $z = 1$ its Hahn expansion begins with $1/(z - 1)$, but at $z = 1$ its value is $-t^{-1}$. It cannot have a single Hahn expansion with continuous coefficient functions on the entire ordinary circle: the negative-exponent coefficient would vanish at all other ordinary points but not at 1. For $a = 1 - t$ the same problem occurs.

Semialgebraic geometry distinguishes these two poles as outside and inside the full unit circle respectively. A coefficientwise contour integral based on the original macroscopic parameterization does not automatically do so. One needs a boundary-layer change of chart, a finer scale, or the algebraic pairing developed in [section 14](#). Avoidance of actual poles is necessary, but uniform admissibility of the pulled-back coefficients is an additional hypothesis. This example recurs in [section 14.3](#) as the explicit case where the two compatibility results must not be conflated.

4 Jordan separation: what survives, and in which sense

4.1 A thick-wall theorem in the standard-part topology

Theorem 4.1 (Jordan separation in the standard-part topology). *Let $J \subset \mathbb{C}$ be an ordinary Jordan curve with bounded interior D and exterior E . Fix $a \in K$ and $r \in R_\Gamma$ with $r > 0$, and use the map [\(3.1\)](#). Then*

$$X_{a,r} \setminus q_{a,r}^{-1}(J) = q_{a,r}^{-1}(D) \sqcup q_{a,r}^{-1}(E)$$

has exactly two connected components in the standard-part topology, and both have boundary $q_{a,r}^{-1}(J)$ relative to $X_{a,r}$. In the case $a = 0$, $r = 1$ this reads $K^\circ \setminus J^\# = D^\# \sqcup E^\#$ with common boundary $J^\#$.

Proof. For any subset $B \subset \mathbb{C}$ the subspace $q^{-1}(B)$ has precisely the saturated open sets pulled back from B , since every open set of the standard-part topology is saturated for the fibres of q and every fibre is nonempty. A separation of $q^{-1}(B)$ therefore descends to a separation of B , and conversely; so $q^{-1}(B)$ is connected exactly when B is. The ordinary Jordan theorem gives $\mathbb{C} \setminus J = D \sqcup E$ with both sets nonempty and connected, whence $q^{-1}(D)$ and $q^{-1}(E)$ are connected. They are disjoint relatively clopen subsets covering the complement, so they are exactly its components. Finally

$$\overline{q^{-1}(B)} = q^{-1}(\overline{B}), \quad \text{int}(q^{-1}(B)) = q^{-1}(\text{int } B),$$

directly from the neighborhood definition and surjectivity of q , and the ordinary boundary identities give the last assertion. $\square$

The removed object is the entire *boundary tube* $q^{-1}(J) = a + r(J + \mathfrak{m}_K)$, the monad of every point of J , and not a one-dimensional embedded curve or the thin trace $\{a + r\gamma(s) : s \in S^1(\mathbb{R})\}$ of one parameterization. A continuous coarse path whose endpoints have standard parts on different sides meets this thick wall. Several things are *not* asserted. Two infinitesimally separated points in the same boundary monad cannot be assigned different sides by this construction; different points in a fibre can lie on opposite sides of a finer-scale separator, and a finer scale is needed to distinguish two zeros in one standard-part fibre. Neither component is being called fine-connected. And “exterior” here means that the standard-part image is ordinary unbounded: all of K° is bounded by any positive infinite element of R_Γ , so we have *not* obtained an unbounded exterior in the internal ordered-field sense. This is an exact topological theorem, but it recovers only the geometry visible at the selected scale.

Example 4.2 (A boundary layer cannot be discarded silently). At $a = 0$, $r = 1$, both $1 - t$ and $1 + t$ reduce to the point 1 on the unit circle, so neither belongs to the thickened disk $\mathbb{D}^\sharp = q^{-1}(\mathbb{D})$, although $|1 - t| < 1 < |1 + t|$. The theorem’s interior is *not* the ordered-radius ball $\{|z| < 1\}$, and a Cauchy formula whose hypothesis is $z \in \mathbb{D}^\sharp$ does not automatically cover $z = 1 - t$.

This construction remains available at every infinitesimal or infinite affine scale, and it explains the topology underlying the ordinary-base coherence of [1] — without asserting that the coefficient functions of a coherent family are continuous for any of the other topologies under discussion.

4.2 The semialgebraic replacement

For this subsection let R be a real closed field containing $\mathbb{R}$ and $K = R[i]$; the case of interest is $R = R_\Gamma$. A set is *semialgebraic* if it is described by a finite Boolean combination of polynomial equations and inequalities with parameters in R . A map is semialgebraic when its graph is. A set is *semialgebraically connected* (equivalently, definably connected) when it has no separation into two nonempty relatively open semialgebraic subsets. Paths now have parameter domain the *field* interval

$$[0, 1]_R = \{s \in R : 0 \leq s \leq 1\},$$

not just the ordinary real points in it. Definable connectedness and definable path connectedness agree for semialgebraic sets; this is part of the standard finite-topology theory [6, 8].

Theorem 4.3 (Semialgebraic Jordan theorem; classical input). *Let $j : S^1(R) \rightarrow R^2$ be a semialgebraic homeomorphism onto its image J , where $S^1(R) = \{(x, y) : x^2 + y^2 = 1\}$. Then $R^2 \setminus J$ has exactly two semialgebraically connected components. They are semialgebraically path connected, one is bounded by an element of R and the other is unbounded, and J is the common boundary of both. Every semialgebraic continuous path between the two components meets J . More generally, smooth definable Jordan–Brouwer separation holds in the o-minimal manifold framework of Berarducci–Otero.*

We import this theorem, rather than claim to prove arbitrary topological Jordan separation by elementary first-order transfer. One route uses compatible finite triangulation and the combinatorial separation theorem for an embedded polygonal circle on a triangulated sphere: compactify the ambient plane semialgebraically, triangulate the pair consisting of the ambient sphere and the curve, observe that the curve is a finite subcomplex which is a circle and that the adjacent two-dimensional simplices split into two sides, and compute connectivity by chains of incident simplices. The component containing the point at infinity is the exterior; the other has bounded realization; local incidence at the curve gives the common-boundary conclusion. A path missing the curve cannot move from one relatively open definable side to the other, because its parameter interval is definably connected. The finite simplicial incidence data do not depend on which real closed field realizes them, nor on the Archimedean property of the coefficients. Alternatively, the definable Jordan–Brouwer theorem of Berarducci–Otero supplies the smooth-manifold case directly, and Woerheide’s o-minimal homology the general one; the semialgebraic triangulation framework handles the topological formulation [6, 7, 8, 38, 29]. The technical finite-topology input is being imported, not replaced by an informal appeal to transfer of arbitrary continuous maps.

Several qualifications are essential. The theorem does *not* say that those components are connected in the unrestricted intrinsic topology: by [proposition 3.2](#) all their ordinary connected components are singletons, and the nondefinable cuts that disconnect the ordered field are excluded from the definition of semialgebraic connectedness. This is why [theorem 4.3](#) does not contradict

proposition 3.1: $[0, 1]_R$ is definably connected but need not be connected as an ordinary topological space. “ R -bounded” does not mean bounded by an ordinary real number. The generalization in the last sentence refers to the differentiability and definability hypotheses of the o-minimal framework; it is not a statement about arbitrary class functions or arbitrary fine-topological subsets. And definable results concern a finite description and its real-closed extensions; they do not supply ordinary singular homology of a proper class. Finally, a semialgebraic problem involving full surreal parameters may always be placed in a real closed set-sized workspace containing those parameters, which makes this a genuine separation theorem for surcomplex data.

Example 4.4 (A microscopic circle with a genuine definable inside). For $a \in K$ and $r \in R_\Gamma$, $r > 0$, the circle $C_{a,r} = \{z \in K : |z - a| = r\}$ has semialgebraic inside $|z - a| < r$ and outside $|z - a| > r$. This remains true for $r = t^\rho$, for an infinite r , and for arbitrary a . The two points $a + r(1 - t^\eta)$ and $a + r(1 + t^\eta)$, $\eta > 0$, lie on opposite definable sides although their normalized standard parts are equal.

A distinction that will recur: the *full semialgebraic circle* $C_{a,r}$, the *coarse thick wall* $q^{-1}(J)$ of [theorem 4.1](#), and the *ordinary-parameter contour trace* $\{a + re^{i\theta} : \theta \in [0, 2\pi]_\mathbb{R}\}$ are three different objects, even when they are informally written with the same equation. The last is generally only a small subset of the first, and its parameterization is not fine-continuous.

4.3 An integer-valued index without a contour integral

For a semialgebraic loop γ avoiding b , define its index as the definable degree of

$$s \mapsto \frac{\gamma(s) - b}{|\gamma(s) - b|} \text{ into } S^1(R),$$

equivalently the corresponding definable homology invariant. With positive orientation it is 1 inside a semialgebraic Jordan curve and 0 outside, it takes values in $\mathbb{Z}$, and it is invariant under definable homotopies avoiding b . We write it $\text{wind}_{\text{sa}}(\gamma, b)$. *This is definable degree, not the ordinary topological degree of an ordinary real-parameter map into the intrinsic field topology.*

For piecewise polynomial or rational parameterizations there is a computationally explicit substitute. After a generic fixed rotation of the coordinate axes, write $\gamma - b = x + iy$ on each piece and form a Cauchy index; the rotation can be chosen so that no piece has identically zero y . With the convention that a jump from $-\infty$ to $+\infty$ contributes $+1$,

$$\text{wind}_{\text{sa}}(\gamma, b) = \frac{1}{2} \sum_{\text{pieces}} \text{Ind}\left(\frac{x}{y}\right), \quad (4.1)$$

with the usual half-endpoint conventions and subdivision at exceptional points; a constant piece contributes zero and may be removed first. Sturm chains compute these indices using field arithmetic and sign tests. Eisermann develops this over arbitrary real closed fields [17]; Perrucci and Roy supply corrected product formulae and a systematic treatment of boundary cases [32]. Below, every pole is disjoint from the loop, so boundary conventions are not needed, and fractional conventions for zeros on an edge or vertex are never interpreted as contour integrals through poles.

Three limitations are recorded. First, this degree is an ordinary integer, not a surreal number of turns. Second, *it does not by itself define the integral of a general differential form*: the primitive of a semialgebraic function need not be semialgebraic, and a chosen analytic expansion of a real closed field does not automatically contain all the radius-free Hahn germs of $\mathcal{A}_n$. Third, the availability of

a finite algebraic prescription does not mean arbitrary surreal inputs have computable normal forms or decidable sign tests; finite Sturm and sign algorithms use an ordered-field arithmetic oracle, and the construction becomes an algorithm only when the coefficient field and support arithmetic have effective presentations.

4.4 What Jordan separation does not supply

Separation by itself does not define an integral. Even in a definable setting, integrating a definable function can require new functions or a larger target object (see [section 16](#)). Conversely, a useful period theory can be defined on cycles with self-intersections without invoking Jordan separation at all. Below, the standard-part theorem interprets ordinary-base domains, the definable theorem interprets algebraic winding, and parameterized chains prove Stokes. *These are compatible uses, but not identifications of categories.*

5 A completely algebraic Stokes theorem for polynomial chains

There is a direct counterpart of generalized Stokes that needs neither ordinary fine paths nor infinite summation, and that lives over an arbitrary characteristic-zero coefficient field rather than over any of the four rings of [section 2.3](#). It also supplies a compatibility test for the Hahn construction of [section 6](#).

5.1 Polynomial integration on a simplex

For a characteristic-zero field L , *define* a linear functional on polynomials on the standard n -simplex by

$$\int_{\Delta^n}^{\text{pol}} u_1^{\nu_1} \cdots u_n^{\nu_n} du_1 \wedge \cdots \wedge du_n = \frac{\nu_1! \cdots \nu_n!}{(|\nu| + n)!}, \quad (5.1)$$

extended L -linearly. The geometric notation $\Delta_R^n = \{u_i \geq 0, \sum u_i \leq 1\}$ is available when L contains an ordered field R , but formula (5.1) is the definition, and the functional is therefore prior to any point set. The 0-simplex functional is evaluation. Give the simplex the ordered vertices $(0, e_1, \dots, e_n)$ and its boundary the alternating face orientation.

A *polynomial simplex* $\sigma : \Delta_R^n \rightarrow R^d$ is a polynomial map with field coefficients, and for a polynomial n -form α with coefficients in L we set $\int_{\sigma}^{\text{pol}} \alpha = \int_{\Delta^n}^{\text{pol}} \sigma^* \alpha$. Only finitely many algebraic operations are involved.

Theorem 5.1 (Polynomial-chain Stokes). *For every polynomial simplex σ and every polynomial $(n - 1)$ -form α ,*

$$\int_{\sigma}^{\text{pol}} d\alpha = \int_{\partial\sigma}^{\text{pol}} \alpha. \quad (5.2)$$

The identity extends to finite polynomial chains, and it holds over every characteristic-zero coefficient field L — in particular over every surcomplex workspace $K = \mathbb{C}((t^\Gamma))$ and over $\mathbf{SC}$ itself for finite data.

Proof. Pullback commutes with d by the polynomial chain rule, so it suffices to prove the identity on the standard simplex. Fix bounds on the degrees of the coefficients of α . Using (5.1) and the affine face maps, both sides are polynomials with rational coefficients in the finitely many coefficients

of α . Over $\mathbb{R}$, formula (5.1) is the ordinary monomial simplex integral, so ordinary Stokes proves their equality for every assignment of real coefficients. A polynomial over $\mathbb{Q}$ vanishing at every real tuple is zero, by induction on the number of variables. Hence the identity is a rational polynomial identity and holds over any characteristic-zero field. All boundary signs and face substitutions are finite, so linearity proves the chain assertion. $\square$

This proof uses classical Stokes to certify a finite algebraic identity; it does not transfer a statement quantifying over arbitrary integrable functions. Equivalent direct proofs expand the monomial factorials and cancel adjacent faces.

Example 5.2 (Infinite width, infinitesimal height). Let $L, H > 0$ be arbitrary elements of R_Γ and orient the triangle T with vertices $0, L, iH$ counterclockwise. Since $d(\bar{z} dz) = d\bar{z} \wedge dz = 2i dx \wedge dy$, polynomial Stokes gives

$$\int_{\partial T}^{\text{pol}} \bar{z} dz = iLH. \quad (5.3)$$

With $L = \omega$ and $H = \omega^{-1}$ the answer is the ordinary finite i , despite the extreme aspect ratio; the signed area functional is $LH/2$. No ordinary compactness of the entire field-valued triangle is asserted.

The construction also handles polynomial forms on triangulated polytopes whenever oriented internal faces cancel. It does *not* define an integral for every semialgebraic function, nor for a rational form whose denominator vanishes somewhere on the chain; rational contour periods will require residues or additional analytic data, which is exactly what [section 14](#) supplies.

6 A Hahn de Rham complex and coefficientwise Stokes

6.1 The differential graded algebra of globally supported Hahn forms

Let M be an ordinary smooth manifold, with corners allowed when integration is used, and let $\Omega^q(M; \mathbb{C})$ denote its ordinary smooth complex-valued differential forms. Define

$$\Omega_H^q(M) = \Omega^q(M; \mathbb{C})((t^\Gamma)). \quad (6.1)$$

An element is a family $\alpha = \sum_{\lambda \in S} \alpha_\lambda t^\lambda$ with *one global well-ordered support* S and ordinary smooth coefficient forms; smoothness up to a boundary means smooth extension in ordinary local corner charts. The word “globally supported” is load-bearing: a single well-ordered support serves the whole manifold. *This is a space of coefficient data over an ordinary manifold; it is not defined as the collection of all fine-smooth forms on a surcomplex point set. A derivative acts on α_λ , not on t^λ .* Put

$$d\alpha = \sum_{\lambda} (d\alpha_\lambda) t^\lambda, \quad \alpha \wedge \beta = \sum_{\nu} \left(\sum_{\lambda+\mu=\nu} \alpha_\lambda \wedge \beta_\mu \right) t^\nu.$$

The inner sum is finite by [lemma 2.2](#), so $\Omega_H^\bullet(M)$ is a differential graded algebra over K with $d^2 = 0$ and the graded Leibniz rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^q \alpha \wedge d\beta$ for α of degree q . Ordinary smooth pullback is coefficientwise, commutes with d , and does not enlarge support. This d differentiates ordinary variables while holding every element of K fixed.

For a compact oriented ordinary q -manifold C with corners and a smooth map $\phi : C \rightarrow M$, define

$$\int_{(C, \phi)}^H \alpha = \sum_{\lambda \in S} t^\lambda \int_C \phi^* \alpha_\lambda, \quad (6.2)$$

whose support is contained in S . Extend by finite linearity to finite piecewise smooth chains. Finite subdivision, orientation reversal, ordinary reparameterization, and cancellation of oppositely oriented faces have their usual meanings. *There is no infinite interchange justified by an unstated analytic bound*: the operation is the coefficientwise extension of an ordinary linear functional, and the output lies in K even when no real-valued bound on the entire coefficient family is available. Noncompact chains could be admitted with suitable support conditions, but compact chains suffice for everything proved here.

Theorem 6.1 (Hahn-valued generalized Stokes). *Let C be a compact oriented ordinary q -manifold with corners, with its boundary orientation given by the outward-normal-first convention, let $\phi : C \rightarrow M$ be smooth, and let $\alpha \in \Omega_H^{q-1}(M)$. Then*

$$\boxed{\int_{(\partial C, \phi|_{\partial C})}^H \alpha = \int_{(C, \phi)}^H d\alpha.} \quad (6.3)$$

The pairing is K -linear in the form and additive in the chain, the same identity holds for finite piecewise smooth chains with their chain boundary, and the integral is invariant under ordinary orientation-preserving smooth reparameterization.

Proof. For each exponent λ , ordinary Stokes [19, Thm. 4.6.1], [24] gives

$$\int_{\partial C} (\phi|_{\partial C})^* \alpha_\lambda = \int_C d(\phi^* \alpha_\lambda) = \int_C \phi^*(d\alpha_\lambda).$$

Both Hahn sums have support in one well-ordered containing set, so equality of every coefficient proves (6.3). For $A = \sum_\mu a_\mu t^\mu \in K$, the coefficient of $A\alpha$ at a fixed target exponent is a finite convolution, so ordinary complex linearity of integration applies to each finite sum, proving K -linearity. For a finite chain, internal faces have equal pulled-back forms and opposite signs and cancel coefficientwise. Ordinary change of variables at each coefficient proves reparameterization invariance. $\square$

The ordinary Stokes theorem, with its chain and orientation conventions, is the classical input [24, 19]. The contribution of theorem 6.1 is the precise Hahn coefficient setting and its compatibility with strong summability. Its content is *not* an assertion that the full surcomplex plane is an ordinary manifold: the chain is ordinary, while its coefficients and its integral may be genuinely nonstandard.

6.2 A support-preserving Poincaré lemma

On an ordinary star-shaped open $U \subset \mathbb{R}^d$ containing 0, define the radial homotopy operator, for $q \geq 1$, by

$$(h\alpha)_x(v_1, \dots, v_{q-1}) = \int_0^1 s^{q-1} \alpha_{sx}(x, v_1, \dots, v_{q-1}) ds, \quad (6.4)$$

and apply it to each Hahn coefficient. Ordinary smoothness at $s = 0$ makes the result an ordinary smooth form, and the Hahn support does not increase.

Proposition 6.2 (Support-preserving Poincaré lemma). *The coefficientwise extension of (6.4) to $\Omega_H^\bullet(U)$ satisfies $dh + hd = \text{id} - c_0^*$, where c_0 is the constant map to 0. Every closed Hahn form of positive degree on U is exact, with a primitive supported in the support of the original form. A Hahn function with zero differential on a connected ordinary manifold is a single constant in K .*

Proof. Differentiate the pullback under $x \mapsto sx$ with respect to s and use the ordinary chain rule; integrating from 0 to 1 gives the displayed homotopy identity for each coefficient, equivalently the ordinary fundamental theorem applied to the radial homotopy. Every operation is complex-linear, so its coefficientwise extension preserves support. In positive degree $c_0^* = 0$. In degree zero, a zero differential makes each smooth coefficient locally constant, hence constant on a connected ordinary base. $\square$

The explicitness of the operator matters: it produces a primitive with no support enlargement whatever, which is what later support certificates require.

6.3 Cohomology and periods

Theorem 6.3 (Cohomology of the globally supported Hahn complex). *For an ordinary smooth manifold M there is a canonical K -linear isomorphism*

$$H^q(\Omega_H^\bullet(M), d) \simeq H_{\text{dR}}^q(M; \mathbb{C})((t^\Gamma)). \quad (6.5)$$

In particular a closed Hahn form is exact if and only if each of its ordinary coefficients is exact. If the ordinary Betti number b_q is finite, the right side is isomorphic to K^{b_q} and naturally to $K \otimes_{\mathbb{C}} H_{\text{dR}}^q(M; \mathbb{C})$. Integration on ordinary cycles extends the usual de Rham period pairing, and when ordinary homology in degree q is finite dimensional that pairing is nondegenerate over K .

Proof. Write $Z^q = \ker d$ and $B^q = \text{im } d$ in the ordinary complex. A Hahn form is closed exactly when every coefficient lies in Z^q ; send its class to the Hahn series of ordinary cohomology classes, discarding zero coefficients. This has support contained in the original support, and exact Hahn forms map to zero. Choose a complex-linear section of $Z^q \rightarrow Z^q/B^q$; its coefficientwise extension lifts every Hahn series of classes without enlarging support, proving surjectivity. Choose also a complex-linear right inverse of d on B^q ; if all coefficient classes vanish, applying it coefficientwise produces a Hahn primitive with support in the same containing set, so the kernel is exactly the exact Hahn forms. The auxiliary choices establish existence; the induced isomorphism is the canonical coefficient-class map. Finite convolution proves K -linearity. For finite b_q , expansion in a fixed ordinary basis identifies Hahn families with scalar extension by K , a finite union of supports being well ordered; the ordinary de Rham theorem supplies a nonsingular finite period matrix against a basis of ordinary homology, and the same matrix is nonsingular over K . $\square$

The finite-dimensional qualification matters and is not a technicality. For an infinite-dimensional V , the space $V((t^\Gamma))$ generally contains families whose coefficients span infinitely many independent directions, whereas $K \otimes_{\mathbb{C}} V$ consists of finite sums of pure tensors. *One must not silently replace the first object by the second.* Nor have we claimed that the global uniform-support assignment (6.1) is a sheaf for arbitrary infinite covers: unlike holomorphic coefficient functions on a connected domain, smooth coefficients can vanish on entire open sets, and locally chosen supports need not have well-ordered union. For example, if ψ_j are smooth bumps supported near the positive integers, then $\sum_{j \geq 1} t^{-j} \psi_j(x)$ is locally a finite Hahn family on $\mathbb{R}$ but has global exponent set $\{-j : j \geq 1\}$, which is not well ordered, so it does not belong to (6.1). All chain integrals and homotopies below have explicitly common supports or involve finitely many charts.

On an arbitrary ordinary manifold the classical de Rham theorem translates (6.5) into a period criterion: a closed Hahn form of positive degree has a global Hahn primitive exactly when all its periods over ordinary smooth cycles vanish [24].

Example 6.4 (The surviving puncture class). On $\mathbb{C}^\times$ the class of dz/z spans $H_{\text{dR}}^1(\mathbb{C}^\times; \mathbb{C})$, normalized by its ordinary unit-circle period $2\pi i$; hence $H^1(\Omega_H^\bullet(\mathbb{C}^\times)) \simeq K[dz/z]$. This is ordinary-base cohomology with enlarged coefficients. It is *not* the positive-degree singular cohomology of the totally separated intrinsic $K^\times$, which vanishes by [proposition 3.2](#): the two complexes have different bases and different chains, and there is no contradiction.

6.4 Green, divergence, and area

For a Hahn one-form $P dx + Q dy$ on an admissible oriented planar 2-chain Σ , [theorem 6.1](#) reads

$$\int_{\partial\Sigma}^H (P dx + Q dy) = \int_{\Sigma}^H (Q_x - P_y) dx \wedge dy, \quad (6.6)$$

and in complex notation, for a C^1 Hahn coefficient family F ,

$$\int_{\partial\Sigma}^H F dz = 2i \int_{\Sigma}^H \partial_{\bar{z}} F dx \wedge dy. \quad (6.7)$$

These are exact identities for the specified pullbacks. They imply Cauchy's theorem for coherent holomorphic coefficients, but they also apply to nonholomorphic coefficients. In three dimensions the same construction applied to a Hahn family of vector fields X with $d(\iota_X \text{vol}) = (\text{div } X) \text{vol}$ gives the divergence identity; defining flux by this form avoids introducing an unproved independent theory of surreal surface area and unit normals.

Example 6.5 (An infinitesimal rectangle has nonzero area). Let $A, B \in R_\Gamma$ be positive and parameterize a rectangle with side lengths A, B by the ordinary unit square. For $\alpha = -y dx + x dy$ one has $d\alpha = 2 dx \wedge dy$; the right edge contributes AB , the top edge contributes AB , and the other edges contribute zero, so

$$\int_{\partial([0,A] \times [0,B])}^H \alpha = 2AB = \int_{[0,A] \times [0,B]}^H d\alpha. \quad (6.8)$$

The chain notation denotes the indicated affine parameterization; it does not presume a fine-continuous ordinary square. For $A = t$ and $B = t^\omega$ the value is $2t^{1+\omega} \neq 0$, whereas the standard part alone would be zero, losing the geometric content of the exact identity.

7 Infinitesimally deformed chains, homotopy, and estimates

7.1 Admissible deformations and the Taylor-jet pullback

Definition 7.1 (Admissible positive-support deformation). Let M be an ordinary *compact* smooth parameter manifold with corners and let $U \subset \mathbb{R}^d$ be ordinary open. An *admissible deformation* of a smooth map $\phi_0 : M \rightarrow U$ is

$$\Phi = \phi_0 + \eta, \quad \eta = \sum_{e \in E} \eta_e t^e, \quad (7.1)$$

where $E \subset \Gamma_{>0}$ is *one* well-ordered set, so the displacement has strictly positive support, and the η_e are smooth real vector-valued coefficient functions on M , extending to a neighborhood in each corner chart. For complex coordinates use real and imaginary parts; piecewise smooth parameterizations are handled by finite subdivision.

At each ordinary parameter value η is an infinitesimal vector. *The coefficient maps need not satisfy a uniform ordinary norm bound, and no assertion of fine continuity of the real-parameter map Φ is intended*; indeed a nonconstant ϕ_0 already prevents such continuity by [proposition 3.1](#). The notation Φ describes an admissible parameterized chain, not a fine-continuous map.

For a smooth coefficient function a on U , define the *Taylor-jet pullback*

$$T_\Phi a = \sum_{\alpha \in \mathbb{N}^d} \frac{(\partial^\alpha a) \circ \phi_0}{\alpha!} \eta^\alpha, \quad d\Phi_j = d\phi_{0,j} + d\eta_j, \quad (7.2)$$

and for $\omega = \sum_{\lambda, I} a_{\lambda, I} t^\lambda dx_{i_1} \wedge \cdots \wedge dx_{i_q}$ set

$$\Phi^* \omega = \sum_{\lambda, I} t^\lambda T_\Phi a_{\lambda, I} d\Phi_{i_1} \wedge \cdots \wedge d\Phi_{i_q}. \quad (7.3)$$

The result is a form on the *ordinary parameter manifold* with Hahn coefficients. For analytic a this agrees with the infinitesimal evaluation used in the supplied manuscripts. For merely C^∞ functions, (7.2) is the *specified jet prolongation*: we do not assert ordinary convergence of its Taylor series, do not identify it with every possible fine extension of a , and do not claim a canonical prolongation of every smooth function to every fine domain. Real analyticity, when assumed below, is imposed on the *target* coefficient functions; the parameter coefficients need only be smooth.

Lemma 7.2 (Pullback and differential with a support certificate). *For $\omega \in \Omega_H^q(U)$ with support S , formula (7.3) defines an element of $\Omega_H^q(M)$ with*

$$\text{supp}(\Phi^* \omega) \subset S + E^*,$$

with only finitely many contributions at each exponent. It respects products, wedge products, restrictions to boundary faces, ordinary smooth reparameterizations of M , and the differential:

$$d(\Phi^* \omega) = \Phi^*(d\omega). \quad (7.4)$$

It also respects composition whenever the intervening ordinary coefficient maps are analytic and the same positive-support substitution hypotheses hold, and it agrees with canonical real-analytic infinitesimal evaluation of the coefficients at every ordinary parameter point.

Proof. Every term is indexed by one source exponent in S together with a finite word of error exponents in E ; differentiating η_e changes its coefficient function but not its exponent, and exterior products introduce only finitely many coordinate indices. [Lemma 2.2](#) therefore gives a well-ordered total support and only finitely many contributing words at each exponent, so each output coefficient is a finite sum of products of ordinary smooth functions, forms, and derivatives of ordinary functions, hence smooth. This proves existence and the support bound.

For functions, ordinary Taylor product identities give $T_\Phi(ab) = (T_\Phi a)(T_\Phi b)$. Differentiating (7.2) produces two groups of terms — differentiating $(\partial^\alpha a) \circ \phi_0$ contributes $d\phi_{0,j}$ and differentiating η^α contributes $d\eta_j$ — and reindexing yields

$$d(T_\Phi a) = \sum_{j=1}^d T_\Phi(\partial_j a) (d\phi_{0,j} + d\eta_j),$$

the reindexing being finite at every Hahn exponent by the first paragraph. This is the chain rule needed for (7.4), and the exterior product rule completes it. Equivalently, one may fix a target

exponent, retain the finitely many coefficient functions and Taylor degrees that can contribute, and verify the identity as an ordinary finite Taylor-jet identity; no discarded higher degree can affect that exponent. Boundary restriction and ordinary reparameterization commute with ordinary derivatives and products, hence coefficientwise. Equality at every exponent proves the Hahn identities. $\square$

Definition 7.3 (Integral on a deformed chain). For Φ as in (7.1) and an admissible form ω of degree $\dim M$, put

$$\int_{\Phi}^H \omega := \int_M^H \Phi^* \omega.$$

Boundary parameterizations are obtained by restriction to the ordinary faces of M . A finite chain is a finite linear combination of such parameterizations, whose common faces are required to agree *as parameterized Hahn data* with the appropriate orientations.

This is a definition on a parameterized object, not an attempt to integrate over its image with a fine-topological measure.

Theorem 7.4 (Stokes for Hahn-deformed chains). *Let M be compact oriented of dimension n , let Φ be an admissible deformation, and let $\omega \in \Omega_H^{n-1}(U)$ on the ordinary coefficient neighborhood U — so its coefficient forms are ordinary smooth forms, with no analyticity assumed. Then*

$$\boxed{\int_{\partial\Phi}^H \omega = \int_{\Phi}^H d\omega.} \quad (7.5)$$

The same identity holds for finite sums of admissible chains with matching faces, with the support bound of lemma 7.2. The integral is invariant under ordinary orientation-preserving smooth changes of parameter and under finite admissible subdivision.

Proof. By lemma 7.2 the pulled-back form is a well-defined smooth Hahn form on M and commutes with d . Apply theorem 6.1 on M :

$$\int_{\partial M}^H \Phi^* \omega = \int_M^H d(\Phi^* \omega) = \int_M^H \Phi^*(d\omega).$$

Boundary compatibility identifies the first term with the integral on $\partial\Phi$, since restriction to a face is the same as pulling back by the restricted parameterization. Ordinary change of variables at every coefficient proves reparameterization invariance, and subdivision follows by finite boundary cancellation; internal faces cancel for finite chains. $\square$

Remark 7.5 (Why smoothness suffices). Two of the three merged manuscripts state this theorem for real-analytic coefficient forms. Smoothness is enough, and the reason is the one that runs through the whole subject: the Taylor-jet pullback (7.3) is not summed by any convergence argument. Because Φ moves the chain by an infinitesimal of strictly positive support, η^α has support bounded below by $|\alpha| \cdot \inf E$, so at each Hahn exponent only finitely many multi-indices α contribute, and only finitely many ordinary derivatives $\partial^\alpha a$ are ever read. Those exist for a smooth a . Requiring the ordinary Taylor series to converge would be importing exactly the kind of limit this calculus does without.

This is not merely a formal rewriting of $d^2 = 0$: it constructs a defined pairing with a chain complex and proves that the pairing intertwines d with the oriented boundary. Two cautions. First, one cannot replace “ordinary changes of parameter” by arbitrary fine-analytic reparameterizations

without proving a separate substitution theorem and checking that the new pulled-back family remains admissible. Second, meromorphic forms are allowed only when their denominators have nonvanishing leading coefficient on the base image, where geometric inversion supplies holomorphic Hahn data on a common ordinary neighborhood; *passing through a leading pole is not justified by calling its displacement infinitesimal.*

7.2 Affine scales and polynomial compatibility

At an arbitrary scale write $z = a + rw$ with $a \in K$ and $r \in R_\Gamma$ positive, and suppose the pulled-back form in the w chart has admissible Hahn coefficients. Define its integral by the chart pullback; for a one-form,

$$\int_{a+r\Phi}^H F(z) dz = \int_{\Phi}^H rF(a + rw) dw. \quad (7.6)$$

For a real area form the Jacobian is r^2 , and for an arbitrary constant invertible real linear map the usual determinant appears; in several real coordinates one inserts the ordinary algebraic Jacobian and wedge factors of the affine change. Such factors lie in K and do not disturb finite convolution. *The scale factor is not optional:* a k -dimensional uniform real scaling contributes r^k to a k -form, and forgetting it would incorrectly erase microscopic areas and volumes. This is the affine contour convention of [1], here extended to deformed chains and all form degrees. Cauchy and Stokes transfer through it with their correct differential factors; a pole at $a + r\xi$ contributes the same residue after substitution, since the factor r in dz cancels the scaling of a simple-pole denominator.

A polynomial map with arbitrary Hahn coefficients need not have the specific normalization $\phi_0 + \eta$ of (7.1). It still has a finite polynomial pullback and can be integrated coefficientwise; expanding its finite polynomial coefficients and applying ordinary simplex integration gives exactly (5.1). Thus theorem 5.1 and theorem 6.1 agree on their common domain, and polynomial parameterizations are a useful enlargement beyond a single positive-support normalization.

7.3 The homotopy formula

Let $\mathcal{H} : M \times [0, 1]_{\mathbb{R}} \rightarrow U^\sharp$ be admissible in the sense of definition 7.1, with one common positive support E on the product, and write $\mathcal{H}_u$ for its restrictions. Define the degree-minus-one operator

$$\mathcal{K}\omega = \int_0^{H^1} \iota_{\partial/\partial u}(\mathcal{H}^*\omega) du, \quad (7.7)$$

the integration being coefficientwise in u and leaving a form in the M variables.

Theorem 7.6 (Hahn homotopy formula). *For every admissible form ω with support S ,*

$$\mathcal{H}_1^*\omega - \mathcal{H}_0^*\omega = d\mathcal{K}\omega + \mathcal{K}d\omega, \quad (7.8)$$

and $\mathcal{K}\omega$ has support in $S + E^$. Consequently a closed form has equal integrals on two admissibly homotopic cycles, and integrals of closed forms on closed oriented parameter manifolds are invariant under such homotopies.*

Proof. Write a coefficient of $\mathcal{H}^*\omega$ as $\beta(u) + du \wedge \gamma(u)$, both forms using only the M variables. Exterior differentiation on the product gives the ordinary coefficient identity

$$\beta(1) - \beta(0) = d_M \int_0^1 \gamma(u) du + \int_0^1 \iota_{\partial_u} d(\mathcal{H}^*\omega) du,$$

and (7.4) converts the last term. The support bound of lemma 7.2 is uniform over the ordinary compact parameter cylinder, all coefficient integrations preserve support, and finite contributions at each exponent make the calculation valid in the Hahn complex. If $d\omega = 0$, the difference is exact and its integral on a closed parameter manifold vanishes by theorem 6.1; equivalently, apply theorem 7.4 to the cylinder with the usual opposite orientations on its two ends. $\square$

Ordinary deformations of the base map are allowed if their whole image lies in the same ordinary coefficient domain; an infinitesimal deformation with fixed base is automatically of this form. A homotopy passing through a pole is *not* admissible for a form singular there, and its period can change. For nonclosed forms there is a genuine nonzero swept-region contribution: a moving boundary applied to $\bar{z} dz$ changes the integral through $d(\bar{z} dz) = 2i dx \wedge dy$. *It would be incorrect to claim that all contour integrals are infinitesimally deformation invariant.* Closedness, not mere existence of a derivative, is the controlling condition; example 8.7 exhibits the effect exactly.

7.4 An order-valued parameter estimate

The coefficientwise integral does not discard the surreal order. If $A(s) = \sum_{\lambda} a_{\lambda}(s)t^{\lambda}$ has continuous real coefficients on an ordinary compact interval, one common support, and $A(s) \geq 0$ at every ordinary s , then either $A \equiv 0$ or its first nonzero coefficient function is nonnegative everywhere and positive somewhere, so its ordinary integral is positive; hence $\int^H A(s) ds \geq 0$.

Proposition 7.7 (An order-valued parameter estimate). *Write an admissible pullback as $B(s) ds$ on an ordinary interval. Suppose $m(s)$ is an ordinary nonnegative continuous function and $M \in R_{\Gamma}$ is nonnegative with $|B(s)| \leq M m(s)$ for all ordinary s . Then*

$$\left| \int^H B(s) ds \right| \leq M \int m(s) ds. \quad (7.9)$$

For an ordinary piecewise C^1 contour γ of ordinary length L and a uniformly supported continuous Hahn family F along it with $|F(\gamma(s))| \leq M$, this specializes to $|\int_{\gamma}^H F dz| \leq LM$.

Proof. If $I = \int^H B ds \neq 0$, put $u = \bar{I}/|I|$, so $|u| = 1$ and $\text{Re}(uI) = |I|$. The real Hahn function $M m(s) - \text{Re}(uB(s))$ is pointwise nonnegative and uniformly supported; positivity and K -linearity give (7.9). The case $I = 0$ is immediate, and piecewise regular data are treated on finitely many intervals. For the contour form take $m(s) = |\gamma'(s)|$, whose integral is L ; the integral of $M|\gamma'(s)| - \text{Re}(uF(\gamma(s))\gamma'(s))$ is then $LM - |I|$. $\square$

This is the estimate proved in the analysis manuscript, here given as a numbered order-valued statement, and it shows why a coefficientwise construction retains an exact surreal — rather than merely standard-part — inequality. The formulation deliberately avoids assuming that the speed $|\Phi'(s)|$ possesses a globally uniform smooth Hahn expansion through every possible zero of its leading speed. Ordered estimates do not imply convergence of set-indexed nets in the full fine topology.

8 Cauchy theory as a period theory

Throughout this section the coefficient ring is the *common-domain coherent ring* $\mathcal{H}(U) = \mathcal{O}(U)((t^{\Gamma}))$ of section 2.3, for an ordinary open $U \subset \mathbb{C}$. Every theorem below is stated over that ring and over no other; the radius-free ring $\mathcal{A}_n$ is treated separately in sections 10 and 11, and none of the statements here is transferred to it.

8.1 Coherent holomorphy and the vanishing theorem

An element $F = \sum_{\lambda \in S} f_{\lambda} t^{\lambda} \in \mathcal{H}(U)$ has one well-ordered support S and ordinary holomorphic coefficients $f_{\lambda} \in \mathcal{O}(U)$ on *one* common U . At $z = c + \varepsilon \in U^{\sharp}$, evaluation means

$$F^{\sharp}(c + \varepsilon) = \sum_{\lambda, n \geq 0} t^{\lambda} \frac{f_{\lambda}^{(n)}(c)}{n!} \varepsilon^n, \quad (8.1)$$

whose support is contained in $S + (\text{supp } \varepsilon)^*$, so it is strongly defined. Differentiation acts on the f_{λ} . The injectivity, composition, and gluing properties of $\sharp$ are supplied in [1, §5]. For an ordinary piecewise C^1 path γ in U ,

$$\int_{\gamma}^H F(z) dz = \sum_{\lambda} t^{\lambda} \int_{\gamma} f_{\lambda}(z) dz, \quad (8.2)$$

and the support cannot increase. In real coordinates

$$d(F dz) = 2i \partial_{\bar{z}} F dx \wedge dy = 0, \quad (8.3)$$

the last equality being coefficientwise ordinary holomorphy: the Cauchy–Riemann equation holds for each coefficient, not for an arbitrary fine-local function.

Theorem 8.1 (Coherent Cauchy theorem). *Let $F \in \mathcal{H}(U)$. If γ_0 is an ordinary piecewise smooth closed cycle null-homologous in U , then $\oint_{\gamma_0}^H F dz = 0$. If an admissible closed contour bounds a finite admissible 2-chain on which the coefficient functions of F are holomorphic, its period is likewise zero. In particular, for a Hahn-deformed closed contour $\gamma = \gamma_0 + \eta$ in the sense of [definition 7.1](#), the period equals the period of γ_0 , whether or not that period is zero; and every closed contour has zero period when U is simply connected.*

Proof. For an ordinary contour apply ordinary Cauchy to every coefficient, or use a smooth filling chain with (8.3) in [theorem 6.1](#). The bounding-chain assertion is [theorem 7.4](#) applied to the closed form $F dz$. For a deformed contour the homotopy $\gamma_u = \gamma_0 + u\eta$, $0 \leq u \leq 1$, is admissible with the same positive error support, so [theorem 7.6](#) applies to the closed form $F dz$; alternatively [theorem 8.2](#) identifies the two periods directly. The simply connected case follows from ordinary complex homology or from coefficientwise primitives. $\square$

Null-homology is a genuine hypothesis. On $U = \mathbb{C}^{\times}$ the coherent form dz/z has unit-circle period $2\pi i$, so “closed contour” alone does not imply a vanishing integral. Conversely, self-intersections do not obstruct Cauchy’s theorem when the cycle is null-homologous: neither the contour nor a filling chain need be embedded, and the Jordan theorem is not needed for this vanishing statement. The Jordan theorem is useful when identifying a particular simple boundary with its interior, or when turning winding into an inside/outside count.

8.2 Exact transport of an open contour

For $c \in U$ and $\varepsilon \in \mathfrak{m}_K$ define the infinitesimal primitive increment

$$T_F(c, \varepsilon) = \sum_{n \geq 0} \frac{F^{(n)}(c)}{(n+1)!} \varepsilon^{n+1}, \quad (8.4)$$

which involves no choice of global primitive.

Theorem 8.2 (Exact endpoint-transport formula). *Let $\gamma_0 : [a, b]_{\mathbb{R}} \rightarrow U$ be ordinary piecewise smooth, let $F \in \mathcal{H}(U)$, and let $\Gamma = \gamma_0 + \eta$ be an admissible complex deformation, piecewise on the same finite partition. Then*

$$\boxed{\int_{\Gamma}^H F dz - \int_{\gamma_0}^H F dz = T_F(\gamma_0(b), \eta(b)) - T_F(\gamma_0(a), \eta(a))}. \quad (8.5)$$

No simple-connectedness assumption on U is required. *In particular a closed deformed contour with matching endpoint displacement has exactly the period of its ordinary base contour, not merely the same standard part.*

Proof. Differentiate $T(s) = T_F(\gamma_0(s), \eta(s))$ coefficientwise. The derivative of the $F^{(n)}$ factor gives

$$\gamma'_0(s) \sum_{n \geq 0} \frac{F^{(n+1)}(\gamma_0(s))}{(n+1)!} \eta(s)^{n+1} = \gamma'_0(s) (F^{\sharp}(\Gamma(s)) - F(\gamma_0(s))),$$

and the derivative of the displacement gives $\eta'(s) F^{\sharp}(\Gamma(s))$, so

$$T'(s) = F^{\sharp}(\Gamma(s)) \Gamma'(s) - F(\gamma_0(s)) \gamma'_0(s).$$

All series have support in $\text{supp } F + E^*$, and [lemma 2.2](#) justifies every reindexing. Integrating the ordinary parameter coefficientwise and applying the ordinary fundamental theorem to each coefficient gives (8.5). Splitting into finitely many smooth pieces introduces endpoint terms which cancel at the joins; for a closed contour the two remaining terms coincide.

Equivalently, set $\gamma_u(s) = \gamma_0(s) + u\eta(s)$ and use the identity

$$\partial_u(F(\gamma_u) \partial_s \gamma_u) = \partial_s(F(\gamma_u) \partial_u \gamma_u), \quad (8.6)$$

all products and derivatives being admissible by [lemma 7.2](#); integrating coefficientwise over the ordinary rectangle $[0, 1] \times [a, b]$ and Taylor expanding identifies the two endpoint terms as (8.4). These derivatives differentiate the ordinary coefficient functions of the parameter, not a putative fine-continuous path. $\square$

Consequently the analysis manuscript's endpoint integral is the value of *every* admissible infinitesimal deformation of its base contour with those endpoints; its endpoint correction is not an arbitrary prescription. A primitive is not required, because only the homotopy strip and not an arbitrary filling disk is used. In contrast, choosing a different ordinary homology class can still change the answer by a period: a coherent global primitive eliminates those periods, merely fine-local primitives do not. The corresponding fundamental theorem for finite nonstandard endpoints $z_j = a_j + \varepsilon_j$ reads

$$-T_F(a_1, \varepsilon_1) + \int_{\gamma}^H F dz + T_F(a_2, \varepsilon_2) = \text{difference of a coherent primitive at the endpoints}, \quad (8.7)$$

which does not contradict [proposition 3.4](#) because the class of admissible primitives has been restricted.

Example 8.3 (The logarithm obstruction has not disappeared). The global fine-locally analytic logarithm $\mathcal{L}$ on $\mathbf{SC}^{\times}$ with $\mathcal{L}' = 1/z$ uses a selected argument cut and is *not* a coherent primitive on $(\mathbb{C}^{\times})^{\sharp}$. Accordingly $\oint_{|z|=1}^H dz/z = 2\pi i$ does not contradict its existence. Applying [theorem 8.2](#) to this noncoherent global function would be an invalid change of category.

8.3 Winding at a displaced center, and the Cauchy formula

Let $a = a_0 + \xi$ with $a_0 \in \mathbb{C}$, $\xi \in \mathfrak{m}_K$, and a_0 not on the ordinary contour γ_0 . On a common ordinary collar of the contour,

$$\frac{1}{z - a} = \sum_{n \geq 0} \frac{\xi^n}{(z - a_0)^{n+1}}$$

is an admissible Hahn family. Higher-pole terms are ordinary derivatives and have zero periods.

Theorem 8.4 (Winding stability at one scale). *Let γ_0 be an ordinary closed piecewise smooth contour, let $c \notin \gamma_0$, let η be an admissible closed positive-support deformation, and let $\xi \in \mathfrak{m}_K$, $a \in K$, $r \in R_\Gamma$ with $r > 0$. Set $\Gamma = a + r(\gamma_0 + \eta)$ and $b = a + r(c + \xi)$. Then the integrand is coherent on an ordinary neighborhood of the base contour and*

$$\frac{1}{2\pi i} \oint_{\Gamma}^H \frac{dz}{z - b} = \text{wind}(\gamma_0, c) \in \mathbb{Z}. \quad (8.8)$$

Proof. The affine factor r cancels by (7.6), reducing to $a = 0$, $r = 1$. Two proofs are available. By theorem 8.2 the closed period is unchanged under deformation from γ_0 to $\gamma_0 + \eta$, and on the ordinary base the $n = 0$ term of the expansion has integral $2\pi i \text{wind}(\gamma_0, c)$ while each $n > 0$ term is an ordinary derivative and has zero integral; strong summability justifies coefficientwise integration. Alternatively, on the parameter cylinder set $g_u(s) = \gamma_0(s) - c + u(\eta(s) - \xi)$. Its leading coefficient never vanishes, compactness of the ordinary contour supplying a positive ordinary separation from c , so $1/g_u$ is a uniformly supported Hahn family; direct differentiation gives $\partial_u(\partial_s g_u/g_u) = \partial_s(\partial_u g_u/g_u)$, whose integral along the closed parameter is zero coefficient by coefficient, so the period at $u = 1$ equals the ordinary winding integral at $u = 0$. $\square$

Thus an admissible contour has an exact *integer* period index, not an unspecified surreal winding number. The theorem treats only points whose standard part avoids the base curve; a point reducing to the curve lies in its boundary tube and requires a finer chart, an explicit detour, or the separate algebraic definition of section 14. This is precisely the hypothesis that fails for $a = 1 \pm t$ in section 3.6.

Theorem 8.5 (Cauchy formula at displaced points and on deformed contours). *Let D be a bounded ordinary Jordan domain with piecewise smooth positively oriented boundary and $\overline{D} \subset U$, let $F \in \mathcal{H}(U)$, and let Γ be either ∂D or an admissible closed infinitesimal deformation of it. For $z = c + \varepsilon$ with $c \in D$, $\varepsilon \in \mathfrak{m}_K$, and every $n \geq 0$,*

$$(F^{(n)})^\sharp(z) = \frac{n!}{2\pi i} \oint_{\Gamma}^H \frac{F(\zeta)}{(\zeta - z)^{n+1}} d\zeta. \quad (8.9)$$

The corresponding statement holds after any admissible affine rescaling.

Proof. Expand

$$(\zeta - c - \varepsilon)^{-n-1} = \sum_{k \geq 0} \binom{n+k}{k} \varepsilon^k (\zeta - c)^{-n-k-1}.$$

The ordinary point c has positive ordinary distance from the compact boundary, so all coefficient functions are holomorphic on one common collar, and lemma 2.2 permits multiplication by F and coefficientwise integration. Ordinary Cauchy gives the coefficient $f_\lambda^{(n+k)}(c)/k!$, yielding $\sum_{\lambda, k} t^\lambda f_\lambda^{(n+k)}(c) \varepsilon^k / k!$, which is (8.1) for $F^{(n)}$. The support is contained in $\text{supp } F + (\text{supp } \varepsilon)^*$. The kernel form is closed on the collar, so the deformed case follows from theorem 8.2; affine substitution gives the rescaled statement with the ordinary derivative and differential factors. $\square$

This proves a formula for genuinely displaced points and contours. It does *not* claim that the deformed trace is the fine-topological boundary of a fine-connected interior, and by [example 4.2](#) its hypothesis $c \in D$ is not the same as $|z| < 1$ for the unit disk.

On a simply connected ordinary domain, normalized ordinary primitives of all f_λ assemble into a coherent primitive with support in S ; on a general connected domain such a primitive exists exactly when all ordinary closed-cycle Hahn periods vanish, and all coherent primitives of one function differ by a single constant in K [[1](#), §6]. Morera’s criterion holds for common-support continuous coefficient families, since vanishing Hahn integrals around every ordinary triangle means vanishing of every ordinary coefficient integral.

8.4 Moving poles and actual zero clusters

Let $F, G \in \mathcal{H}(U)$ with $G \neq 0$, and suppose the ordinary leading coefficient function g_0 of G after monomial normalization has no zero on the ordinary boundary of a bounded Jordan domain D with $\overline{D} \subset U$ and finitely many zeros inside. The preparation and moving-pole theorem of [[1](#)] gives

$$\frac{1}{2\pi i} \oint_{\partial D}^H \frac{F}{G} dz = \sum_{b \in D^\#} \text{Res}_b \left(\frac{F}{G} dz \right), \quad (8.10)$$

the sum being over *actual surcomplex poles*, not merely their standard parts. The sum is finite; its summands may be infinite. As fixed in [section 2.4](#), the residue at b is the coefficient of $(z - b)^{-1}$ in the local finite-meromorphic germ *after cancellations*, and is not merely the residue of one ordinary coefficient at $\text{st}(b)$.

The decisive local reduction is worth recording, because it is the mechanism reused in [theorem 10.5](#). Preparation near a leading zero writes G as a coherent unit times a monic polynomial P whose lower coefficients are infinitesimal; division writes $F/G = J + R/P$ with J coherent holomorphic and $\deg R < \deg P$; the polynomial has finitely many infinitesimal roots, and rational partial fractions give $R/P = \sum_b \sum_{j=1}^{m_b} A_{b,j} (z - b)^{-j}$, where m_b is the order of b as a root of P . On a small ordinary protecting circle only the $j = 1$ terms have nonzero coefficientwise period, contributing $2\pi i A_{b,1}$, and the holomorphic part contributes zero. This argument is valid in higher rank because the expansions are strong sums, not sequential limits. It is stronger than calling a coefficientwise residue at the reduced center “the residue” and leaving its relation to the split points unspecified.

Corollary 8.6 (Residues on a deformed protecting contour). *Formula (8.10) remains true when ∂D is replaced by an admissible closed infinitesimal deformation lying in a coefficient collar on which g_0 is nonzero. The right side is still the divisor selected by the ordinary interior $D^\#$.*

Proof. The fraction is Hahn-holomorphic on that collar, so its closed period is unchanged by [theorem 8.2](#). $\square$

This describes the selected divisor without inventing a fine-connected interior of the deformed trace. The same theorem applied to F'/F gives the argument principle: its normalized period is the total zero multiplicity, or zeros minus poles for a fraction, with ordinary integer multiplicities. Positive-support perturbations preserving the leading function preserve the total multiplicity in each ordinary monad [[1](#), [4](#)]; individual roots can nevertheless split at finer scales, which is the subject of [sections 9](#) and [10](#).

Example 8.7 (A deformed ellipse and its exact area). Let $\varepsilon, \delta \in \mathfrak{m}_K \cap R_\Gamma$ and map the ordinary unit disk by $\Phi(x, y) = (1 + \varepsilon)x + i(1 + \delta)y$, with boundary $\Gamma(\theta) = (1 + \varepsilon)\cos\theta + i(1 + \delta)\sin\theta$. Since the pullback of $dx \wedge dy$ is $(1 + \varepsilon)(1 + \delta)dx \wedge dy$, [theorem 7.4](#) applied to $\bar{z}dz$ gives

$$\oint_{\Gamma}^H \bar{z}dz = 2\pi i(1 + \varepsilon)(1 + \delta) = 2\pi i(1 + \varepsilon + \delta + \varepsilon\delta). \quad (8.11)$$

Thus an infinitesimal deformation changes a nonclosed form's integral by a genuine infinitesimal, *including a second-order mixed term*. Setting $\delta = \varepsilon$ gives $2\pi i(1 + \varepsilon)^2$; further affine scaling by r multiplies by r^2 ; translation adds an exact one-form with zero closed period. In contrast $z^q dz$ is exact for every integer $q \geq 0$ and integrates to zero over both the circle and the ellipse. Writing $R = 1 + t$ and $S = 1 + t^\omega$ gives $2\pi i(1 + t + t^\omega + t^{1+\omega})$, a value visible only if all valuation ranks are retained. For the undeformed circle of radius $r \in R_{\Gamma, > 0}$ the same identity reads $\oint_{|z|=r}^H \bar{z}dz = 2\pi i r^2 = 2i \int_{r\mathbb{D}}^H dx dy$, giving a nonzero infinitesimal area for $r = t^\rho$ and an infinite one for infinite r . No measure on all fine-topological subsets is asserted.

Example 8.8 (Separating residues invisible at the coarse scale). Let $t = \omega^{-1}$ and $R(z) = 1/(z^2 - t^2)$. Its residues at $\pm t$ are $\pm 1/(2t)$, so

$$\frac{1}{2\pi i} \oint_{|z|=1}^H R(z) dz = 0 :$$

both poles reduce to zero and the coarse contour sees only their cancelling sum. Choose $r = t^\rho$ with $\rho > 1$ — *including a higher Archimedean class such as $\rho = \omega$* — and use the contour $z = t + rw$, $|w| = 1$ in the ordinary parameter. Then

$$R(z) dz = \frac{dw}{w(2t + rw)} = \frac{1}{2t} \sum_{k \geq 0} \left(-\frac{r}{2t}\right)^k w^{k-1} dw, \quad (8.12)$$

whose relative support is positive because $v(r/t) = \rho - 1 > 0$. Its normalized period is $1/(2t)$, so the actual integral is $\pi i/t$; the analogous circle about $-t$ gives $-\pi i/t$. The same support-controlled calculus therefore resolves the two infinite residues individually and recovers their cancellation on the larger contour. For $z/(z^2 - t^2)$ the two residues are $1/2$ and the coarse normalized period is 1. No all-scale coherence hypothesis on any transcendental function was used: a contour only needs a valid representation at its chosen scale, so the all-scale rigidity theorems of [\[1\]](#) pose no obstruction to this local multiscale calculus.

9 Affine scales, microscopic contours, and compatible charts

9.1 A precise compatibility criterion

Proposition 9.1 (Agreement on admissible overlaps). *Suppose two chart descriptions of the same parameterized oriented chain produce pointwise equal pulled-back differential forms at every ordinary parameter point, and both pulled-back forms are common-support Hahn families with smooth coefficients. Then their integrals agree. The same holds for a finite collection of chart pieces with matching oriented faces.*

Proof. Enlarge the workspace to contain both supports. Uniqueness of Hahn normal forms makes each coefficient of the difference vanish at every parameter point, so the coefficient forms are equal; their ordinary integrals agree, and so do their Hahn sums. For finitely many pieces use additivity and cancellation of matching faces. $\square$

This criterion is intentionally stronger than equality of the set-theoretic images of two curves: a chain includes multiplicity, orientation, and the pulled-back differential. It is also weaker than requiring one global chart for every scale, as finitely many verified overlap identities suffice. What is *not* justified is a patchwork of arbitrary local fine germs for which common-support pullbacks and their identities have not been established; equality of bare fine germs in unrelated, unverified summation descriptions is not a substitute.

9.2 Macroscopic and microscopic winding resolve different information

A macroscopic contour whose reduction avoids a cluster assigns the same winding to every point of that cluster. A contour centered on one actual pole and rescaled below the finite separation distances can resolve an individual contribution. This requires no surreal-valued winding number: ordinary integer winding numbers in different admissible charts suffice.

Example 9.2 (One coarse contour, two microscopic contours). The poles of $e^z/(z^2 - t)$ are $\pm t^{1/2}$, with residues $e^{t^{1/2}}/(2t^{1/2})$ and $-e^{-t^{1/2}}/(2t^{1/2})$. These are individually infinite, but their sum is

$$\frac{\sinh(t^{1/2})}{t^{1/2}} = \sum_{n \geq 0} \frac{t^n}{(2n+1)!}. \quad (9.1)$$

The ordinary unit circle and the protecting microscopic circle $|z| = t^{1/4}$ both have integral $2\pi i$ times (9.1). A circle centered at $t^{1/2}$ of radius t isolates the positive pole: in the coordinate $z = t^{1/2} + tw$ the second pole sits at $w = -2t^{-1/2}$, an infinite normalized distance. Each integral is computed in its own admissible chart, and the outer integral is the sum of the two inner residues even though the individual residues are infinite. *There is no assertion that these ordinary-parameter traces constitute ordinary topological circles in the fine plane.*

Example 9.3 (Two poles in different valuation ranks). Let $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ with its inherited surreal order and put $R(z) = 1/(z - t) + 1/(z - t^\omega)$, both residues being 1. Three positively oriented microscopic circles give

$$\frac{1}{2\pi i} \oint_{|z|=t^\rho}^H R(z) dz = \begin{cases} 2, & \rho = \frac{1}{2}, \\ 1, & \rho = 2, \\ 0, & \rho = 2\omega. \end{cases} \quad (9.2)$$

For the middle circle, t^ω lies inside but t lies outside, because $\omega > 2 > 1$ reverses the magnitude order of the monomials; the last circle is smaller than both pole scales. Each assertion follows by geometric expansion in the appropriate ratio, or by algebraic winding and [theorem 14.2](#). *A single real-valued valuation that discards the higher rank cannot retain all of these distinctions.*

10 Radius-free germs become coherent after microscopic rescaling

From here to the end of [section 11](#) the coefficient ring is the *radius-free germ ring*

$$\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$$

of [section 2.3](#), and the target of rescaling is the polynomial-coefficient ring $\mathbb{C}[w_1, \dots, w_n]((t^\Gamma))$. No theorem of [section 8](#), which is stated over $\mathcal{H}(U)$, is applied to $\mathcal{A}_n$ before the rescaling theorem has moved the data into a common-domain ring.

10.1 The missing common-radius problem

An element $H = \sum_{\lambda} h_{\lambda}(z)t^{\lambda} \in \mathcal{A}_n$ has ordinary convergent germs as coefficients, with no requirement that all h_{λ} extend to the same ordinary polydisk. The witness (2.2) is legitimate, but its coefficient poles at $1/j$ prevent a common ordinary disk of positive radius, so an ordinary fixed-circle coefficient integral at zero is unavailable; applying theorem 8.5 to it on a fixed ordinary circle would leave a gap. Evaluation at infinitesimal points is nevertheless strongly defined. The remedy is rescaling.

Theorem 10.1 (Positive rescaling gives polynomial coefficient families). *Let $\rho_1, \dots, \rho_n \in \Gamma_{>0}$, put $r_j = t^{\rho_j}$, and write $rw = (r_1w_1, \dots, r_nw_n)$. The Taylor substitution*

$$\sigma_r(H)(w) := H(rw) = \sum_{\lambda, \alpha} h_{\lambda, \alpha} t^{\lambda + \alpha \cdot \rho} w^{\alpha} \quad (10.1)$$

defines an injective K -algebra homomorphism

$$\sigma_r : \mathcal{A}_n \longrightarrow \mathbb{C}[w_1, \dots, w_n]((t^{\Gamma})).$$

Every fixed Hahn coefficient of (10.1) is an ordinary polynomial; the support is contained in $\text{supp } H + \{\rho_1, \dots, \rho_n\}^$; the image has coherent evaluation on the thickening of every ordinary w -domain; and $\partial_{w_j} \sigma_r(H) = r_j \sigma_r(\partial_{z_j} H)$. More generally, for $a \in \mathfrak{m}_K^n$ the translated substitution*

$$H(a_1 + r_1w_1, \dots, a_n + r_nw_n) \in \mathbb{C}[w_1, \dots, w_n]((t^{\Gamma})) \quad (10.2)$$

has support contained in $\text{supp } H + (E_a \cup \{\rho_1, \dots, \rho_n\})^$, where $E_a = \bigcup_i \text{supp}(a_i)$, and it evaluates correctly at every finite tuple $w \in (K^{\circ})^n$.*

Proof. Write the Taylor germ $h_{\lambda}(z) = \sum_{\alpha} h_{\lambda, \alpha} z^{\alpha}$. At a fixed Hahn exponent ν , the possible contributions satisfy $\lambda + \alpha_1 \rho_1 + \dots + \alpha_n \rho_n = \nu$. Lemma 2.2 gives finitely many source exponents λ and finitely many finite words in the positive alphabet $\{\rho_1, \dots, \rho_n\}$, hence finitely many multi-indices α contributing at ν ; the coefficient at ν is therefore an ordinary polynomial, and the total support is well ordered. Taylor multiplication and differentiation are valid at each exponent because the relevant regroupings are finite, which gives the algebra and derivative statements. The Taylor embedding $\mathcal{A}_n \hookrightarrow K[[z]]$ is injective, and diagonal substitution $z_j = r_j w_j$ is injective in that formal ring because $r_j \neq 0$; hence σ_r is injective. Polynomial coefficients are holomorphic on every ordinary domain, and evaluation at a finite normalized point agrees with evaluation of H at the infinitesimal point rw by the same finite-contribution Taylor identities. For (10.2), first translate by a using the strongly summable Taylor translation of [4, §3], whose support lies in $\text{supp } H + E_a^*$, then repeat the argument; evaluation at finite $w = c + \varepsilon$ is justified by adjoining the positive supports of the infinitesimal parts. $\square$

No coefficient radii enter this proof. The argument in fact works for arbitrary *formal* complex coefficient series at infinitesimal arguments, although the ring of interest here is the analytic-germ ring. Its analytically relevant conclusion is not ordinary convergence of the original germs on a uniform disk, but a *new coherent representation in an infinitesimal chart*; it neither establishes nor requires all-scale coherence of the original function. For (2.2) the substitution $z = t^{1/4}w$ gives $H(t^{1/4}w) = \sum_{j \geq 1} \sum_{q \geq 0} j^q t^{j+q/4} w^q$, at a fixed exponent a finite sum, so the singularities of the original ordinary coefficient representatives no longer obstruct coefficientwise contour integration in the w -plane. The resulting coefficients need not have any uniform degree or growth bound as the Hahn exponent varies.

Convention. For $r = t^\rho$ the symbol $\{|z| = r\}$ in a Hahn integral always means the ordinary parameterized circle $z = re^{i\theta}$, $0 \leq \theta \leq 2\pi$, with its pulled-back differential. *It does not mean the full class of points of that modulus equipped with a fine-topological integration theory*, and it is not the full semialgebraic circle of [example 4.4](#).

10.2 Microscopic Cauchy formulae over $\mathcal{A}_n$

Theorem 10.2 (Cauchy formula for a radius-free germ). *Let $H \in \mathcal{A}_1$ and $a \in \mathfrak{m}_K$. Choose $\rho > 0$ with $\rho < v(a)$ when $a \neq 0$, and put $r = t^\rho$; if $a = 0$, any $\rho \in \Gamma_{>0}$ is allowed. Then for every $k \geq 0$*

$$H^{(k)}(a) = \frac{k!}{2\pi i} \oint_{|z|=r}^H \frac{H(z)}{(z-a)^{k+1}} dz. \quad (10.3)$$

Proof. By [theorem 10.1](#), $G(w) = H(rw)$ lies in $\mathbb{C}[w]((t^\Gamma)) \subset \mathcal{H}(\mathbb{C})$ with ordinary entire polynomial coefficients, and the normalized point a/r is infinitesimal. Apply [theorem 8.5](#) to G on the ordinary unit circle. Since $G^{(k)}(a/r) = r^k H^{(k)}(a)$ and $dz = r dw$, the scaling factors yield (10.3). $\square$

The same proof gives a product-torus Cauchy formula for $H \in \mathcal{A}_n$, using one sufficiently large infinitesimal circle around each coordinate of the evaluation point. *A product of n circles is a real n -dimensional cycle, the distinguished boundary of a polydisk; it is not the real $(2n-1)$ -dimensional boundary of that polydisk.* There is also a local primitive theorem: ordinary germwise integration applied coefficientwise makes $D : \mathcal{A}_1 \rightarrow \mathcal{A}_1$ surjective with kernel K , and the ordinary analytic Poincaré homotopy applied coefficientwise makes the analytic differential-form complex of $\mathcal{A}_n$ exact in positive degrees. Thus microscopic periods of nonsingular forms vanish while poles supply residue obstructions. *These assertions use the prescribed Hahn-germ algebra, not all fine-analytic functions.*

10.3 Protecting radii and one-variable residues

Lemma 10.3 (A radius protecting an entire one-variable cluster). *Let $F = f + E \in \mathcal{A}_1$, where f has ordinary order $m \geq 1$ at zero, say $f(z) = z^m u(z)$ with $u(0) \neq 0$, and $\delta = v_H(E) > 0$ (with $\delta = +\infty$ if $E = 0$). Choose $\rho > 0$ with $m\rho < \delta$ and put $r = t^\rho$; if $E = 0$ every positive ρ works. Then $r^{-m}F(rw)$ has leading coefficient $u(0)w^m$ and no zeros on the ordinary unit-circle parameter, and every infinitesimal zero a of F satisfies $v(a) > \rho$.*

Proof. The higher powers in $u(rw) - u(0)$ have positive valuation, and every term of $r^{-m}E(rw)$ has valuation at least $\delta - m\rho > 0$; this proves the leading-coefficient and boundary assertions. For a nonzero infinitesimal zero with $v(a) \leq \rho$, the term $f(a)$ has valuation $mv(a) \leq m\rho < \delta$ while $E(a)$ has valuation at least δ , so they cannot cancel. The zero $a = 0$ satisfies the conclusion by convention. Divisibility of Γ supplies a strictly smaller positive ρ in the nonzero case, even for arbitrary rank. $\square$

The interpretation is a *protecting-radius condition*: the chosen circle must be infinitesimally large relative to the perturbation scales after accounting for the order m . Every zero in the originating infinitesimal cluster lies strictly inside the protecting circle.

Theorem 10.4 (Microscopic circle realization of the one-variable perturbation residue). *Let $F = f + E \in \mathcal{A}_1$ and $H \in \mathcal{A}_1$ be as in [lemma 10.3](#), with $r = t^\rho$ and $m\rho < v_H(E)$. Then the*

microscopic circle functional is well defined and

$$\boxed{\frac{1}{2\pi i} \oint_{|z|=r}^H \frac{H(z)}{F(z)} dz = \text{Res}_F(H),} \quad (10.4)$$

with Res_F the one-variable perturbation residue (2.5). The integral means substitution $z = rw$ followed by coefficientwise integration over the ordinary unit circle in w , and it is independent of every choice of ρ satisfying the displayed bound. Its support is contained in $\text{supp } H + (\text{supp } E)^*$.

Proof. Write $f(z) = cz^m + O(z^{m+1})$, $c \neq 0$. By theorem 10.1 both $H(rw)$ and $F(rw)$ have polynomial Hahn coefficients, and $r^{-m}F(rw) = cw^m + P(w)$ with $v_H(P) > 0$: higher terms of f acquire positive relative exponents $(j - m)\rho > 0$, and the perturbation terms have relative exponent at least $v_H(E) - m\rho > 0$. On an ordinary annulus around $|w| = 1$ the leading coefficient cw^m is nowhere zero, so geometric inversion gives a common-support holomorphic Hahn family — an element of $\mathcal{H}$ of that annulus — for $rH(rw)/F(rw)$. Expanding instead around $f(rw)$,

$$\frac{rH(rw)}{F(rw)} = \sum_{k \geq 0} (-1)^k \frac{rH(rw)E(rw)^k}{f(rw)^{k+1}},$$

which is strongly summable on the annulus since $E(rw)/f(rw)$ has strictly positive Hahn valuation there. Coefficientwise integration extracts the coefficient of w^{-1} . For an ordinary germ q the formal Laurent change of variable gives

$$[w^{-1}] \frac{rq(rw)}{f(rw)^{k+1}} = [z^{-1}] \frac{q(z)}{f(z)^{k+1}},$$

an identity that holds for a Hahn family of germs as well, by finite contributions after rescaling: in a term $g(z)dz$ of the ordinary meromorphic germ expansion, scaling sends $z^j dz$ to $r^{j+1}w^j dw$, and the circle residue retains only $j = -1$, with no surviving scale factor. Applying this to $q = HE^k$ and summing the strongly summable family gives exactly (2.5). Since the right side does not involve ρ , radius independence follows, and the support certificate is the one carried by (2.5). $\square$

Radius independence here is an *exact residue identity*. It does not require a convergent sequence of radii or a fine-continuous homotopy between scales whose ratio might be infinite.

Theorem 10.5 (Radius-free moving-pole residue theorem). *Let $N, G \in \mathcal{A}_1$ with $G \neq 0$. Normalize its leading exponent and choose a protecting microscopic circle as in lemma 10.3 when its leading germ vanishes at zero; if the leading germ is a unit, choose any sufficiently small infinitesimal circle. Then*

$$\frac{1}{2\pi i} \oint_{|z|=r}^H \frac{N(z)}{G(z)} dz = \sum_{a \in \mathfrak{m}_K} \text{res}_a \left(\frac{N}{G} dz \right), \quad (10.5)$$

a finite sum over actual poles, all of which lie strictly inside the chosen normalized circle. If $G = f + E$ with f of order m , there are exactly m denominator zeros counted with multiplicity, and the value agrees with (2.5).

Proof. If the leading germ is a unit then G is a unit of $\mathcal{A}_1$ and the assertion follows from theorem 10.2 or from coefficientwise primitives. Otherwise Hahn-germ preparation and division [4, §4] give $N/G = J + R/P$ with $J \in \mathcal{A}_1$, P monic of degree m with all roots infinitesimal, and $\deg R < m$, the normalization absorbing the leading monomial and unit factors into the numerator. These

roots are the zeros of G with multiplicities and satisfy the protecting bound of lemma 10.3. Factor P over the algebraically closed K and use finite partial fractions $R/P = \sum_a \sum_{j=1}^{m_a} c_{a,j}(z-a)^{-j}$. The J integral is zero. For each a the normalized position a/r is infinitesimal, so expansion on $z = rw$, $|w| = 1$, is strongly summable, with normalized integral 1 for $(z-a)^{-1} dz$ and 0 for every $(z-a)^{-j} dz$ with $j > 1$. Hence the integral is $\sum_a c_{a,1}$. That no other infinitesimal zero exists is lemma 10.3; agreement with (2.5) is theorem 10.4, and the same finite set of local residues proves independence of ρ . $\square$

This extends the common-domain moving-pole theorem (8.10) to the radius-free setting without requiring a common ordinary representative for the original infinitely many coefficient germs. It is *not* a claim that the same germ can be evaluated at arbitrary noninfinitesimal points, nor that its original coefficient functions have acquired a common ordinary analytic radius. An arbitrary Hahn family of meromorphic germs is a larger object than a quotient of two elements of $\mathcal{A}_1$, and *we do not assert that every such family admits the microscopic contour construction*.

Example 10.6 (An explicit radius-free contour value). Take H from (2.2) and $F(z) = z$. For any $\rho > 0$, theorem 10.4 gives

$$\frac{1}{2\pi i} \oint_{|z|=t^\rho}^H \frac{H(z)}{z} dz = H(0) = \sum_{j \geq 1} t^j = \frac{t}{1-t}. \quad (10.6)$$

After scaling the integrand is $\sum_{j \geq 1} \sum_{k \geq 0} j^k t^{j+k\rho} w^{k-1} dw$; at each exponent the coefficient is a finite Laurent polynomial and the residue selects exactly $k = 0$. This calculation is valid also when Γ has higher rank. The output is a strong sum, not a claim about convergence of its partial sums in any topology. With $\Gamma = \mathbb{Q}$, $a = t^2$ and $r = t^{1/2}$ the same mechanism gives

$$\frac{1}{2\pi i} \oint_{|z|=t^{1/2}}^H \frac{H(z)}{z-t^2} dz = \sum_{j \geq 1} \frac{t^j}{1-jt^2} = t + t^2 + 2t^3 + 3t^4 + 5t^5 + 9t^6 + 16t^7 + 31t^8 + \dots, \quad (10.7)$$

the coefficient at exponent ℓ being the finite sum of j^k over $j \geq 1$, $k \geq 0$ with $j + 2k = \ell$.

Example 10.7 (A radius-free numerator over a perturbed denominator). Take the numerator (2.2) and $F(z) = z^2 - t$, so $m = 2$ and $v_H(E) = 1$. For the protecting radius $r = t^{1/4}$, theorem 10.4 gives

$$\begin{aligned} \frac{1}{2\pi i} \oint_{|z|=r}^H \frac{H(z)}{z^2-t} dz &= \sum_{k \geq 0} t^k [z^{2k+1}] H(z) \\ &= \sum_{j \geq 1} \sum_{k \geq 0} j^{2k+1} t^{j+k} = \sum_{j \geq 1} \frac{j t^j}{1-j^2 t} \\ &= t + 3t^2 + 12t^3 + 64t^4 + 441t^5 + 3855t^6 + \dots \end{aligned} \quad (10.8)$$

Every rearrangement is strongly valid: the coefficient at t^ℓ has only the pairs $j + k = \ell$ with $j \geq 1$, $k \geq 0$. Equivalently the actual simple-pole formula $(H(t^{1/2}) - H(-t^{1/2})) / (2t^{1/2})$ gives $jt^j / (1-j^2t)$ for each summand of H , proving agreement. *No common ordinary disk for the original numerator coefficients exists*, so the period is genuinely supplied by the microscopic representation and not by an unavailable fixed-circle integral.

11 Microscopic tori and general complete-intersection residues

11.1 Coordinate-power leading systems

Let

$$F_j(z) = z_j^{d_j} + E_j(z), \quad d_j \geq 1, \quad E_j \in \mathcal{A}_n, \quad v_H(E_j) = \delta_j > 0 \quad (11.1)$$

for each nonzero E_j (with $\delta_j = +\infty$ if $E_j = 0$). Choose positive $\rho_j \in \Gamma$ with

$$d_j \rho_j < \delta_j \quad \text{whenever } E_j \neq 0, \quad r_j = t^{\rho_j}; \quad (11.2)$$

such choices exist by divisibility of Γ , and for a zero perturbation there is no upper restriction on ρ_j . Write T_r for the parameterized real n -torus $z_j = r_j e^{i\theta_j}$, $0 \leq \theta_j \leq 2\pi$, with ordinary parameter angles and orientation $d\theta_1 \wedge \cdots \wedge d\theta_n$; integration is defined by pullback to the ordinary product of unit circles. *It is a real n -dimensional cycle in an n -complex-variable calculation, not the $(2n-1)$ -dimensional sphere used in some other several-variable integral formulas.*

Theorem 11.1 (Coordinate-power torus realization). *For F as in (11.1), radii as in (11.2), and $H \in \mathcal{A}_n$, the integral*

$$\mathcal{R}_F(H) := \frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H(z) dz_1 \wedge \cdots \wedge dz_n}{F_1(z) \cdots F_n(z)} \quad (11.3)$$

is well defined and satisfies the exact coefficient formula

$$\mathcal{R}_F(H) = \sum_{k \in \mathbb{N}^n} (-1)^{|k|} [z_1^{d_1(k_1+1)-1} \cdots z_n^{d_n(k_n+1)-1}] \left(H \prod_{j=1}^n E_j^{k_j} \right) = \text{Res}_F(H), \quad (11.4)$$

the perturbation residue (2.4) for $f_j = z_j^{d_j}$. Its support is contained in $\text{supp } H + E^$, where $E = \bigcup_j \text{supp } E_j$, and it is independent of all protecting radii satisfying (11.2). Every infinitesimal common zero satisfies $v(z_j) > \rho_j$ for every j .*

Proof. After substituting $z_j = r_j w_j$ one has $r_j^{-d_j} F_j(rw) = w_j^{d_j} + P_j(w)$ with $v_H(P_j) \geq \delta_j - d_j \rho_j > 0$, and all coefficient functions are polynomials by [theorem 10.1](#). On a common ordinary product annulus about $|w_j| = 1$ all leading $w_j^{d_j}$ are nonzero, so each reciprocal has a strong geometric expansion; [lemma 2.2](#) proves strong summability of the entire pulled-back form, which may therefore be integrated coefficientwise. At multi-index k the denominator contributes $\prod_j z_j^{-d_j(k_j+1)}$ and the numerator is $H \prod_j E_j^{k_j}$; the ordinary iterated torus integral selects the Laurent monomial $w_1^{-1} \cdots w_n^{-1}$, so a source Taylor monomial z^α contributes only when $\alpha_j = d_j(k_j+1) - 1$ for every j , and for that monomial the scale factor in coordinate j is $r_j^{1+\alpha_j-d_j(k_j+1)} = 1$. The surviving coefficients are exactly (11.4), including the sign. Ordinary Taylor coefficient operators do not enlarge Hahn support and the perturbations have positive support, giving the support statement; the formula contains no r_j , giving radius independence. Finally, for an infinitesimal common zero, $d_j v(z_j) = v(E_j(z)) \geq \delta_j$ whenever $z_j \neq 0$, and a zero coordinate has valuation $+\infty$. $\square$

The theorem permits *different valuation ranks in different coordinates*: for example $\rho_1 \in \mathbb{Q}_{>0}$ and ρ_2 proportional to ω are both legitimate, and nothing in the proof replaces Γ by a real-valued norm. It does not assume that the perturbations are polynomial, nor that their coefficient germs share an ordinary neighborhood. It supplies a contour meaning to a residue that was deliberately defined algebraically in [\[4\]](#).

Example 11.2 (A two-variable torus and cancellation across valuation ranks). Take $F_1 = x^2 - Ay$, $F_2 = y^2 - Bx$ with $A, B \in \mathfrak{m}_K \setminus \{0\}$, and choose $2\rho_x < v(A)$, $2\rho_y < v(B)$. Then

$$\mathcal{R}_F(H) = \sum_{k, \ell \geq 0} A^k B^\ell [x^{2k+1-\ell} y^{2\ell+1-k}] H, \quad (11.5)$$

a coefficient with a negative exponent being zero. In particular $\mathcal{R}_F(1) = 0$, $\mathcal{R}_F(xy) = 1$, and $\mathcal{R}_F(4xy - AB) = 4$. The Jacobian is $J_F = 4xy - AB$, equal to $-AB$ at the origin and to $3AB$ at the three roots with $xy = AB$, so the four weights for numerator 1 are $-1/(AB)$ and $1/(3AB)$ three times, summing to zero. For $A = t$ and $B = t^\omega$ this cancellation involves weights of valuation $-(1 + \omega)$, whereas the torus result has no negative term at all.

11.2 Arbitrary isolated analytic complete intersections

Now let $f = (f_1, \dots, f_n)$ be an ordinary analytic germ at zero with an isolated common zero, i.e. $0 < \mu = \dim_{\mathbb{C}} \mathbb{C}\{z\}/(f_1, \dots, f_n) < \infty$, and let $F = f + E$ with all nonzero $E_j \in \mathcal{A}_n$ of strictly positive Hahn valuation. The perturbation residue is (2.4), where the ordinary local Grothendieck residue is a complex-linear functional on analytic germs whose coordinate-power case is Taylor coefficient extraction. We use the classical residue transformation law as presented in [36, 11].

Since (f) has finite colength, choose $d_1, \dots, d_n \geq 1$ and an *ordinary analytic* matrix $A(z)$ with

$$A(z)f(z) = (z_1^{d_1}, \dots, z_n^{d_n})^\top, \quad (11.6)$$

and set $G = AF$, so that $G_j = z_j^{d_j} + D_j$ with $D = AE$ and all nonzero D_j of positive Hahn valuation. *No invertibility of A is assumed.* The existence of such A and d_j uses finite colength alone; it is ordinary local analytic algebra and requires nothing from the geometry manuscript beyond the definition of Res_F .

Lemma 11.3 (Transformation of the Hahn perturbation residue). *With these hypotheses, for every $H \in \mathcal{A}_n$,*

$$\text{Res}_F(H) = \text{Res}_G(H \det A). \quad (11.7)$$

Proof. We first specify the classical input. For a finite analytic parameter deformation f_u of an isolated complete intersection, the sum of local residues over the nearby cluster is analytic in u ; if $g_u = Af_u$ also has an isolated central fibre, the classical transformation law gives

$$\text{res}_{f_u}^{\text{cluster}}(h) = \text{res}_{g_u}^{\text{cluster}}(h \det A).$$

The cluster for g_u can contain additional points, and *the determinant is part of the law.* Taylor expansion in u produces the geometric denominator expansions appearing in (2.4). This ordinary finite-family statement follows from the usual local residue transformation law and its parameter version [36, 11].

Now fix a Hahn exponent ν . On either side of (11.7), a coefficient is built from a source coefficient of H together with a finite word of positive perturbation exponents. Lemma 2.2 leaves only finitely many contributing words, hence finitely many ordinary coefficient germs — this is exactly the place where the distinction recorded after example 2.3 is load-bearing, since infinitely many *exponents* may lie below ν . Replace the finitely many retained perturbation terms by independent ordinary parameters. Together with the finitely many entries of A , their germs have a common ordinary representative neighborhood. Apply the finite-family identity there, compare the finitely many

required Taylor coefficients, and substitute the monomials t^e back for the parameters; exponent collisions cause only finite additions. Thus the two sides have equal coefficient at ν , and ν was arbitrary. A fully detailed truncation argument for a transformation matrix that is only *formal* in the parameters is given in [section B](#). $\square$

Theorem 11.4 (Microscopic contour representation of a general residue). *Choose A and d_j as in (11.6) and protecting radii for G as in (11.2). Then for every $H \in \mathcal{A}_n$*

$$\text{Res}_F(H) = \frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H(z) \det A(z) dz_1 \wedge \cdots \wedge dz_n}{G_1(z) \cdots G_n(z)}. \quad (11.8)$$

The value is independent of the admissible radii and of the choice of ordinary transformation data A, d_j .

Proof. The integrand is admissible by [theorem 10.1](#), and [theorem 11.1](#) identifies its normalized torus integral with $\text{Res}_G(H \det A)$. Apply [lemma 11.3](#). Every permitted choice gives the same independently defined left side, proving independence. $\square$

Two non-claims must survive this theorem, and are stated here as part of it. First, (11.8) is a contour representation *by a transformed form*. It does *not* assert that the untransformed form $H dz_1 \wedge \cdots \wedge dz_n / (F_1 \cdots F_n)$ has the same integral over that coordinate torus; in the classical theory the natural Grothendieck cycle is adapted to the equations, typically $\{|F_j| = \epsilon_j\}$, and replacing that cycle by a coordinate torus without the determinant transformation would in general be wrong. Second, the theorem does *not* claim that F and G have identical zero schemes: multiplication by a matrix with nonunit determinant can introduce additional zeros and multiplicities, and $\det A$ compensates for this in the residue identity — *it is not an optional Jacobian factor*. Formula (11.8) is useful precisely because it gives a controlled coordinate-torus substitute.

The logical cost of this theorem should be recorded exactly. It uses the strong residue expansion (2.4), ordinary finite-family residue transformation, and positive rescaling [theorem 10.1](#). *It does not require the Nullstellensatz or faithful-flatness theorems of [4], nor that manuscript's finite-algebra normal form.*

11.3 Relation to the finite-algebra conclusions of the supplied source

The geometry manuscript proves that the perturbed quotient $B_F = \mathcal{A}_n / (F_1, \dots, F_n)$ has K -dimension exactly μ , and that

$$\text{Res}_F(H J_F) = \text{Tr}(M_H) = \sum_a e_a H(a), \quad J_F = \det\left(\frac{\partial F_i}{\partial z_j}\right), \quad \sum_a e_a = \mu, \quad (11.9)$$

with Res_F annihilating (F) and e_a the local multiplicity at the *actual displaced* zero. For simple zeros this gives $\text{Res}_F(H) = \sum_a H(a) / J_F(a)$. These are reported companion results, not reproved here as part of the contour construction. Combining them with [theorem 11.4](#) identifies one functional simultaneously as a finite-algebra trace pairing and as a microscopic integral, and gives

$$\frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H J_F \det A}{G_1 \cdots G_n} dz_1 \wedge \cdots \wedge dz_n = \sum_a e_a H(a), \quad (11.10)$$

whose value at $H = 1$ is the ordinary integer μ . Thus this higher-dimensional contour has a finite local-degree interpretation even when the actual points split across several infinitesimal ranks. On a

common scaled neighborhood avoiding the divisor $G_1 \cdots G_n = 0$ the displayed meromorphic top form is holomorphic and closed, so its admissible deformations obey [theorem 7.4](#) and their periods agree when the intervening chain avoids that divisor — a genuine Stokes explanation of contour deformation in addition to the algebraic radius-independence proof. This is *not* a statement about arbitrary fine-continuous hypersurfaces in the full class plane.

In dimension one, Stokes, the residue of a Laurent series, and the integer winding of a circle already organize the calculation. In higher dimensions the determinant transformation and the real n -torus play the corresponding roles. *There is no need to postulate a new Jordan separation theorem for a $(2n - 1)$ -sphere in order to justify these residues.*

11.4 Scope: the transformed representation, the objection it drew, and the resolution of the untransformed question

This subsection records a disagreement about *status* among the three manuscripts originally merged here, and then records what became of the question that disagreement was about. *The history is kept deliberately: the objection below was raised, it was correct about the coordinate torus, and it identified by name a problem that a fourth manuscript later solved with a different cycle.* Nothing in the resolution reaches back and strengthens [theorem 11.4](#), whose own non-claim about the untransformed form stands exactly as printed above. The disagreement about status is the sharpest in the set, and it is not averaged. [Lemma 11.3](#) and [theorem 11.4](#) above are proved unconditionally and are printed with their proof. One of the three sources judged the corresponding statement *unproved* and declined to assert it, and a third reached a determinant torus formula only conditionally. Both positions are reproduced here, because dropping the objection would overstate how much is proved unconditionally, while dropping the theorem would understate it.

The objection, as raised. For arbitrary isolated f , a single diagonal rescaling need not make the leading equations coordinate powers, or even keep the leading homogeneous system isolated. Thus [theorem 11.1](#) cannot simply be applied after deleting its hypothesis. Moreover the usual cycle $\{|f_i| = \epsilon_i\}$ is not in general a coordinate product torus. What is available without further argument is an exact *coefficient-dependent cycle interpretation* of (2.4): at a prescribed Hahn exponent, [lemma 2.2](#) retains only finitely many numerator germs, perturbation germs, and multi-indices; those finitely many germs have a common ordinary representative neighborhood; their ordinary residues can be computed on suitable protecting cycles there; and the answer is independent of the representatives and radii by ordinary local residue theory. Doing this at each exponent gives precisely (2.4). *This is a compatible germwise procedure, not one fixed ordinary cycle for the entire original infinite family.* On that basis the source in question wrote: “The present article proves that realization for one variable and for the coordinate-power class above; it does not assume it for every complete intersection,” and listed a general realization theorem as an open problem: given arbitrary isolated f and radius-free positive-support E , construct *one* admissible n -cycle whose integral equals (2.4), possibly requiring a non-diagonal analytic change of variables, a finite resolution-type decomposition, or chains on an enlarged analytic space, and prove that no omitted summation or radius assumption is hidden in the parameterization. *That is the problem now solved in [sections 12 and 13](#)*; the last paragraph of this subsection states under exactly which hypotheses.

What the objection does and does not refute. It does not refute [theorem 11.4](#), and the two are consistent, because they answer different questions. The open problem asks for a cycle on which the *untransformed* integrand $H dz / (F_1 \cdots F_n)$ is integrated; [theorem 11.4](#) supplies instead a

coordinate torus for the *transformed* integrand $H \det A dz / (G_1 \cdots G_n)$, and, as stated immediately after that theorem, it explicitly does not claim the former. The objection is therefore best read as a statement about which of the two representations has been achieved. Nor does the objection's existence prove that no single suitably nonstandard cycle is possible. Conversely, [theorem 11.4](#) does settle the representation question in the transformed form, unconditionally, from finite colength alone. A reader should take away both: a general isolated complete intersection *does* have a controlled microscopic contour representation, and it is *not* the naive one. The objection's positive part is also confirmed rather than overturned by what follows: [example 13.12](#) exhibits an ordinary germ for which the original coordinate product torus returns 0 while the residue is $1/2$, so no theorem about the untransformed integrand on a coordinate torus was ever available. It is the *cycle* that had to change, not [theorem 11.4](#).

The conditional third route. A second and genuinely independent route to a determinant torus formula runs through univariate separating denominators. It is worth recording as an alternative proof, *provided its dependence is stated*: unlike [theorem 11.4](#), it assumes the finite-algebra and support-controlled normal-form results of the companion geometry manuscript [4], which is an unrefereed research manuscript. It is therefore conditional on that companion rather than independent of it.

Theorem 11.5 (Separating-torus realization; conditional on [4]). Assume the finite-algebra and support-controlled normal-form results of [4, §§7–9]: *that $B_F = \mathcal{A}_n / (F_1, \dots, F_n)$ has K -dimension μ , that any chosen ordinary monomial basis remains a basis, and that multiplication matrices and normal-form operations have explicit support in E^* . Let M_i be the multiplication matrix of z_i on B_F in that basis, and put*

$$Q_i(T) = \det(TI - M_i) = T^\mu + \sum_{j < \mu} q_{ij} T^j. \quad (11.11)$$

Its exponent-zero matrix is multiplication by z_i in the ordinary local algebra $\mathbb{C}\{z\}/(f)$, hence nilpotent, so every nonzero q_{ij} has positive valuation. By Cayley–Hamilton $Q_i(z_i) \in (F)$; applying the supplied normal form to it, with zero remainder, produces a matrix $C = (C_{ij})$ with

$$Q_i(z_i) = \sum_{j=1}^n C_{ij}(z) F_j(z), \quad C_{ij} \in \mathcal{A}_n, \quad (11.12)$$

all entries having nonnegative support contained in E^ . Choose $\rho_i \in \Gamma_{>0}$ with*

$$(\mu - j)\rho_i < v(q_{ij}) \text{ for every nonzero } q_{ij}, \quad r_i = t^{\rho_i}. \quad (11.13)$$

Then for every $H \in \mathcal{A}_n$

$$\boxed{\text{Res}_F(H) = \frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H(z) \det C(z)}{\prod_{i=1}^n Q_i(z_i)} dz_1 \wedge \cdots \wedge dz_n,} \quad (11.14)$$

independently of the allowed radii and of the chosen representing matrix C , with support contained in $\text{supp } H + E^$.*

Proof. Existence of the integral. By [theorem 10.1](#), $H(rw)$ and every entry of $C(rw)$ have polynomial complex coefficients at each Hahn exponent. The scaled denominator is $r_i^{-\mu} Q_i(r_i w_i) = w_i^\mu +$

$\sum_{j < \mu} q_{ij} r_i^{j-\mu} w_i^j$, whose non-leading coefficients have strictly positive valuation by (11.13); hence each reciprocal is a strong geometric expansion near the ordinary product unit torus, and products with the differential factors $r_i dw_i$ remain strongly supported. Moreover every root b of Q_i satisfies $v(b) > \rho_i$, since otherwise b^μ has strictly smaller valuation than each $q_{ij} b^j$ and cannot cancel; so every common zero of F lies strictly inside the separating coordinate circles.

Residue transformation. The classical rule $\text{Res}_F(H) = \text{Res}_Q(H \det C)$ for $Q = CF$ is standard residue theory [11, §0]. Its justification here is the finite-parameter argument of section B, which is needed because C arises from a linear splitting that need not be continuous in an analytic norm, so $C(u)$ cannot be asserted convergent as an analytic parameter series.

The separated residue is a torus integral. For the leading tuple $(z_1^\mu, \dots, z_n^\mu)$ the ordinary residues are coefficient extraction; expanding each Q_i^{-1} around $z_i^{-\mu}$ shows that Res_Q is the iterated coefficient of $z_1^{-1} \dots z_n^{-1}$, and after the admissible scaling, integration of an ordinary Laurent monomial on each unit circle extracts exactly those coefficients, the powers of r_i cancelling. All rearrangements are strongly summable by the verified support bounds. The left side is already independent of the radii and of C , so equality gives the remaining assertions; no uniqueness of C is needed. $\square$

Remark 11.6 (What was changed, and what was not). Formula (11.14) does *not* assert that the original quotient $H/(F_1 \dots F_n)$ is regular on the chosen product torus: a hypersurface $F_i = 0$ can meet that torus. The denominators were deliberately replaced by the univariate Q_i and $\det C$ was inserted. *Omitting the determinant is generally wrong.* Nor has a real n -torus been identified with the full boundary of a complex polydisk; the integration cycle and its real dimension are exactly those specified. Using univariate polynomials in a zero-dimensional ideal to compute multidimensional residues is a classical method [11, §0]; the additional conclusion here is that the method admits a legitimate support-controlled contour interpretation for $\mathcal{A}_n$. No claim is made that separating denominators or residue transformation are new ideas.

Which representation is sharper, and for what. Theorem 11.4 is sharper for *logical economy*: it is unconditional, needing only finite colength, ordinary residue transformation, and theorem 10.1, so it survives if the companion manuscript's finite-algebra package is withdrawn. Theorem 11.5 is sharper for *computation and for zero-scheme bookkeeping*: its denominators are explicit univariate characteristic polynomials whose roots and admissible radii are read off from (11.13), and the vanishing of $\det C$ at the spurious points of the separated system is visible, as example 11.7 shows. They are not competing proofs of one statement; they certify different things.

Example 11.7 (A four-point cluster and its separating torus). Take $F_1 = x^2 - Ay$, $F_2 = y^2 - Bx$ with $A, B \in \mathfrak{m}_K \setminus \{0\}$ as in example 11.2. The ordinary leading algebra is $\mathbb{C}\{x, y\}/(x^2, y^2)$ with basis $1, x, y, xy$ and $\mu = 4$. The coordinate characteristic polynomials are

$$Q_x = x^4 - A^2 Bx, \quad Q_y = y^4 - AB^2 y,$$

and an explicit transformation matrix with $(Q_x, Q_y)^\top = C(F_1, F_2)^\top$ is

$$C = \begin{pmatrix} x^2 + Ay & A^2 \\ B^2 & y^2 + Bx \end{pmatrix}, \quad D = \det C = x^2 y^2 + Bx^3 + Ay^3 + ABxy - A^2 B^2. \quad (11.15)$$

Hence

$$\text{Res}_F(H) = \frac{1}{(2\pi i)^2} \int_{T_r}^H \frac{H(x, y) D(x, y)}{(x^4 - A^2 Bx)(y^4 - AB^2 y)} dx \wedge dy, \quad (11.16)$$

admissible for any positive exponents with $3\rho_x < 2v(A) + v(B)$ and $3\rho_y < v(A) + 2v(B)$; with $A = t$ and $B = t^\omega$ both $\rho_x = \rho_y = 1/4$ work. The roots are $(0, 0)$ and $(\rho \zeta^j, \rho^2 \zeta^{2j}/A)$ for $j = 0, 1, 2$ with

$\varrho^3 = A^2B$ and ζ a primitive cube root of unity. The weights $-1/(AB)$, $1/(3AB)$, $1/(3AB)$, $1/(3AB)$ are individually infinite and sum to zero, while $\text{Res}_F(xy) = 1$ and $\text{Res}_F(J_F) = 4$. In the quotient algebra $D = ABJ_F$, the exact polynomial identity having a remainder in (F_1, F_2) .

The determinant's role is especially explicit here. *The separated system (Q_x, Q_y) has sixteen simple grid points, whereas the original system has only four.* At each of the other twelve points $CF = 0$ with $F \neq 0$, so $\det C = 0$ and the separated simple residue vanishes. Changing the denominators without the determinant would count the wrong zero scheme. For an independent check, expanding $Q_x^{-1} = x^{-4} \sum_{k \geq 0} (A^2B)^k x^{-3k}$ and $Q_y^{-1} = y^{-4} \sum_{\ell \geq 0} (AB^2)^\ell y^{-3\ell}$ on the chosen torus and extracting the coefficient of $x^{-1}y^{-1}$ reproduces

$$\text{Res}_F(x^p y^q) = \begin{cases} A^{(2p+q-3)/3} B^{(p+2q-3)/3}, & \frac{2p+q-3}{3}, \frac{p+2q-3}{3} \in \mathbb{N}, \\ 0, & \text{otherwise,} \end{cases} \quad (11.17)$$

in agreement with (11.5). The general equality is proved by [theorem 11.5](#), not inferred from the finite tests.

The resolution, and its exact hypotheses. The open problem quoted above is solved in [sections 12](#) and [13](#), and the solution should be read with its hypotheses attached rather than as a removal of the scope discussion. What is now proved, by [theorem 12.10](#), is that for

1. an ordinary *convergent* analytic map germ f with an isolated common zero of finite colength μ — the hypothesis of [section 11.2](#), unchanged;
2. one *specified* positive valuation lower bound $\delta \in \Gamma_{>0}$ on the perturbation support, $\text{supp } E \subset [\delta, \infty)$ — a hypothesis that (2.4) did not need, and the only genuinely new one, since a positive well-ordered support always has a least element and each individual E therefore qualifies with $\delta = \min \text{supp } E$; and
3. a radius-free numerator $H \in \mathcal{A}_n$, or a formal one by [theorem 13.10](#),

there is one compact oriented ordinary parameter n -manifold, a finite covering of the coordinate torus of total degree μ , and one Hahn parameterization of it, built from f and δ alone, on which the *untransformed* form $H dz_1 \wedge \cdots \wedge dz_n / (F_1 \cdots F_n)$ integrates to (2.4) for *every* E above the threshold. The problem's final demand — that no omitted summation or radius assumption be hidden in the parameterization — is met by the explicit support certificates (12.14) and (13.6), by the separation of the two finiteness principles in [section 12.5](#), and by [theorems 13.5](#) and [13.6](#), which together fix the degree of the cycle and identify the periods of the protecting and adapted cycles.

Four things remain exactly as they were. [Theorem 11.4](#) still does not assert that the untransformed form has the same *coordinate-torus* integral; that assertion is false, by [example 13.12](#). The new cycle is a ramified finite cover, in general disconnected, and its construction does not eliminate $\det A$ from [theorem 11.4](#) or $\det C$ from [theorem 11.5](#), both of which remain non-optional for their own formulas. The objection's germwise, coefficient-dependent cycle interpretation recorded above remains correct and remains the right description of what (2.4) does by itself. And the question is closed only under the three hypotheses just listed: an arbitrary Hahn tuple with no convergent isolated leading map, a non-isolated leading zero, and a perturbation support not bounded below by a positive element are all still outside the theorem, as [section 13.9](#) states in detail.

12 One cycle for the untransformed integrand

Section 11.4 distinguished two questions and answered one of them. Theorem 11.4 represents the perturbation residue of an arbitrary isolated complete intersection by a coordinate torus carrying the *transformed* integrand $H \det A dz / (G_1 \cdots G_n)$, and explicitly declines the untransformed form. The remaining question — one admissible n -cycle on which the original form $H dz_1 \wedge \cdots \wedge dz_n / (F_1 \cdots F_n)$ itself integrates to (2.4) — was recorded there and in section 17.3 as an open problem.

This section and section 13 construct such a cycle. The hypotheses are exactly those of section 11.2 together with *one* positive lower bound on the perturbation support: a convergent ordinary leading map with an isolated zero, a specified $\delta > 0$ with $\text{supp } E \subset [\delta, \infty)$, and a radius-free numerator. No determinant transformation is made, no equation is replaced, and no common convergence radius is assumed for the coefficient germs of E or of H . The cycle is built from f and δ alone and then serves *every* perturbation above that threshold and every numerator at once.

12.1 What “one cycle” is allowed to be

The answer depends on a decision about the chain category, and that decision is made explicitly here rather than left implicit in the word “cycle”.

Definition 12.1 (Admissible Hahn cycle on an ordinary cover). An *admissible Hahn n -cycle* consists of one compact oriented ordinary smooth n -manifold $\mathbb{T}$ without boundary — *possibly a finite disjoint union* — together with one Hahn parameterization $Z : \mathbb{T} \rightarrow K^n$ whose coordinates lie in $\mathcal{O}(\mathbb{X})((t^\Gamma))$ for a fixed ordinary complex manifold $\mathbb{X} \supset \mathbb{T}$ and have strictly positive support. Its integral is the coefficientwise integral (6.2) of the pulled-back top-degree form on $\mathbb{T}$, and the image is written $Z[\mathbb{T}]$.

Three features of this definition are the ones that make the construction possible, and each was already licensed by the chain formalism of sections 6 and 7: the parameter manifold is ordinary and compact, so theorem 6.1 applies to it; it need not be connected, so a finite covering of the ordinary torus is a legitimate single cycle; and the parameterization is a positive-support Hahn family, so it is an admissible deformation of the constant map to the origin in the sense of definition 7.1. What is *not* required is that $Z[\mathbb{T}]$ be a connected coordinate product torus in the original variables z ; the counterexample of example 13.12 shows that requiring that much would make the problem unsolvable. Nor is $Z : \mathbb{T} \rightarrow K^n$ asserted to be continuous for any of the three topologies of remark 3.3; by proposition 3.1 a nonconstant such map cannot be.

One point of care is that lemma 7.2 cannot supply the pullback certificate here. That lemma pulls back a form whose coefficients are ordinary smooth functions on *one* common ordinary target neighborhood; the integrands below have radius-free coefficient germs and no such neighborhood exists, by (2.2). The certificate is supplied instead by lemma 12.2. Once the pullback has been formed on $\mathbb{T}$, the ordinary-base Stokes theorems apply without change.

The data, and one threshold. Throughout sections 12 and 13 fix

$$f \in \mathbb{C}\{z_1, \dots, z_n\}^n, \quad f(0) = 0, \quad \mu = \dim_{\mathbb{C}} \mathbb{C}\{z\} / (f_1, \dots, f_n) < \infty, \quad F = f + E, \quad (12.1)$$

with $E \in \mathcal{A}_n^n$, and fix a *specified* $\delta \in \Gamma_{>0}$ with

$$S_E := \text{supp } E = \bigcup_{j=1}^n \text{supp } E_j \subset [\delta, \infty). \quad (12.2)$$

The set S_E is well ordered, and S_E^* is the monoid written E^* in [definition 2.1](#) and in [theorem 11.1](#); in the per-coordinate notation $\delta_j = v_H(E_j)$ of [\(11.1\)](#), condition [\(12.2\)](#) says $\delta \leq \min_j \delta_j$, but it is stated as a single threshold on the whole support because a single threshold is what the cycle is built from. Since a nonempty well-ordered subset of $\Gamma_{>0}$ has a least element, every individual positive-support perturbation satisfies [\(12.2\)](#) with $\delta = \min S_E$; the force of stating δ first is that one cycle then serves the whole valuation ball of perturbations with support above δ . *No bound of any kind is imposed on the complex coefficients of the germs $E_{j,\lambda}$ or H_λ , and their convergence radii may tend to zero.* As in [section 1.5](#), v_H denotes the least exponent carrying a nonzero coefficient *function*; it is not a pointwise minimum of valuations over a parameter domain, and no such pointwise notion is used.

The fixed-manifold coherent ring. The construction needs the common-domain ring of [section 2.3](#) with the ordinary open $U \subset \mathbb{C}$ replaced by a fixed ordinary complex manifold. For such an X , possibly disconnected, write

$$\mathcal{H}(X) = \mathcal{O}(X)((t^\Gamma)), \quad (12.3)$$

and $\mathcal{H}(X)_+$ for the elements of strictly positive support. The ring discipline of [section 2.3](#) is unchanged: every coefficient is holomorphic on *one* fixed X and all share one well-ordered support. No theorem of [section 8](#) is transferred to $\mathcal{H}(X)$ for a higher-dimensional or disconnected X , and nothing below asserts a sheaf property for $\mathcal{H}$ on arbitrary covers.

Notation reconciled with the source. The source manuscript of these two sections writes m_j for the target radial weights, M for their sum, b and c for the two ordinary radial bounds, T and X for the parameter manifold and its ambient complex manifold, $\langle P \rangle_{\mathbb{N}}$ for the monoid generated by P , and $\mathcal{R}_F$ for the perturbation residue. All six are renamed here to preserve the conventions already fixed in this article. The weights are κ_j with $\kappa_* = \max_j \kappa_j$ and $\kappa_\Sigma = \sum_j \kappa_j$, because m is reserved for the order of a one-variable germ and d_j for coordinate powers by [section 1.5](#). The radial bounds are b_{pole} and c_{sep} . The manifolds are $\mathbb{T}$ and X , because T_r is the coordinate torus of [theorem 11.1](#) and T_F the endpoint correction [\(8.4\)](#). The monoid is P^* . The residue is Res_F throughout, *never* $\mathcal{R}_F$: in this article $\mathcal{R}_F$ already denotes the coordinate-torus functional [\(11.3\)](#), and the entire content of the theorems below is that a *different* cycle computes Res_F .

12.2 Substitution of a positive-support Hahn map, and a nonlinear support lemma

[Lemma 2.2](#) is the only summability input, and the sharpening recorded after [example 2.3](#) — finitely many dependencies at an *exact* exponent, not finitely many exponents below it — is used at every step. The source proves [lemma 2.2](#) by a second route, through Higman’s lemma on words over a well-quasi-ordered alphabet [\[22\]](#): a well-order is a well-quasi-order, an embedding of words with each matched letter no larger than its partner cannot increase the sum, equality of sums forces identical words because every letter is strictly positive, and both a descending sequence of word sums and an infinite family of distinct words with one fixed sum therefore contradict Higman’s lemma. That is an independent proof of the same classical statement [\[28, 22\]](#); the statement itself is not restated here.

Two genuinely new lemmas are needed. The first replaces [theorem 10.1](#) in a situation where the substituted map is not a diagonal monomial rescaling but an arbitrary positive-support Hahn family of holomorphic maps.

Lemma 12.2 (Radius-free substitution into a Hahn map). *Let X be a fixed ordinary complex manifold, let $W = (W_1, \dots, W_n) \in \mathcal{H}(X)_+^n$, and let P_W be the union of the supports of its coordinates, a well-ordered subset of $\Gamma_{>0}$. For every $H = \sum_{\lambda \in S_H} t^\lambda h_\lambda$ with formal coefficient germs $h_\lambda \in \mathbb{C}[[z]]$ — in particular for every $H \in \mathcal{A}_n$ — the fully expanded family*

$$H(W) = \sum_{\lambda \in S_H} \sum_{\alpha \in \mathbb{N}^n} t^\lambda [z^\alpha] h_\lambda W^\alpha \quad (12.4)$$

is strongly summable, lies in $\mathcal{H}(X)$, and has support contained in $S_H + P_W^$. The assignment $H \mapsto H(W)$ respects sums and products and commutes with differentiation in the ordinary parameters of X . No common convergence radius for the h_λ is used, and none is produced.*

Proof. A fully expanded monomial $t^\lambda [z^\alpha] h_\lambda W^\alpha$ contributes at exponents λ plus a finite word of $|\alpha|$ letters drawn from P_W , one letter for each of the $|\alpha|$ factors. Fix a target exponent ν . By [lemma 2.2](#) only finitely many pairs consisting of an element of S_H and a finite word in P_W sum to ν , and only finitely many word lengths occur; hence only finitely many λ and finitely many multi-indices α can contribute, and for each there are finitely many ways to distribute the letters of the word among the n coordinates and their repeated factors. The coefficient of t^ν is therefore a finite sum of finite products of ordinary holomorphic functions on X , so it is holomorphic on X , and the total support is contained in the well-ordered set $S_H + P_W^*$. The same finiteness at each exponent licenses every regrouping, which gives the ring identities, and ordinary differentiation of a coefficient function leaves its exponent unchanged, so termwise differentiation is a finite operation at each exponent. *Convergence of the formal series h_λ was never used*; what is used is that each individual output coefficient is a finite polynomial expression in finitely many Taylor coefficients of finitely many h_λ and finitely many coefficient functions of W . $\square$

Corollary 12.3 (Radial evaluation of an ordinary family). *Let $W(q, s) = \sum_{\ell \geq 1} s^\ell W_\ell(q)$ be holomorphic on $X \times \mathbb{D}_\epsilon$ with $W(q, 0) = 0$. For every $\rho \in \Gamma_{>0}$ the substitution $s = t^\rho$ produces an element of $\mathcal{H}(X)_+^n$ with support in $\mathbb{N}_{>0}\rho$, and [lemma 12.2](#) applies to it. Each coefficient of the result is formed from finitely many ordinary Taylor coefficients of the W_ℓ , even when no single ordinary ϵ serves all the germs h_λ simultaneously.*

Proof. The exponents of the substituted family are the positive integer multiples of ρ , a well-ordered positive set, and $W(q, 0) = 0$ removes the exponent 0. Apply [lemma 12.2](#) with $P_W = \{\rho\}$; at a fixed output exponent only finitely many radial degrees ℓ occur, by [lemma 2.2](#). $\square$

This is where the direction of the argument differs from [theorem 10.1](#). Rescaling moves a radius-free germ into the polynomial-coefficient ring by substituting a monomial for each coordinate. [Lemma 12.2](#) instead substitutes an entire Hahn family of holomorphic maps on a fixed ordinary manifold, and lands in the coherent ring of that manifold. Neither statement implies the other, and neither produces a common ordinary radius for the original coefficient germs.

The second lemma solves the implicit equation that will produce the equation-adapted lift of [section 13](#). Its coefficient ring is not required to be a field, and its proof is combinatorial rather than iterative.

Lemma 12.4 (Positive implicit equation). *Let R be a commutative complex algebra and let $B(V) = (B_1(V), \dots, B_n(V))$ be a formal power series in finitely many variables $V = (V_1, \dots, V_n)$ whose coefficients lie in $R((t^\Gamma))$, and suppose the union of the supports of all coefficients of B is a well-ordered subset $W \subset \Gamma_{>0}$. Then the equation*

$$V = B(V) \quad (12.5)$$

has exactly one solution with strictly positive coordinate supports. Its support is contained in $W^* \setminus \{0\}$, it is given by a strongly summable family of finite rooted trees, and each of its coefficients is computed by finitely many coefficient operations in R .

Proof. Expand repeated substitution into finite ordered rooted trees. A vertex selects one coefficient of one component of B ; the multi-index of the selected monomial prescribes how many children it has and which component each child computes. Nullary vertices select constant terms. Assign to every vertex the exponent of its selected coefficient, an element of W , and give a tree the sum of its vertex exponents as its weight.

Fix a weight ν . By lemma 2.2 only finitely many finite ordered words in W sum to ν , and only finitely many lengths occur, so the number of vertices of a contributing tree is bounded. There are finitely many ordered rooted trees with at most that many vertices, finitely many ways to colour their vertices by components, and finitely many compatible multi-indices, since a vertex cannot have more children than there are other vertices. Hence the tree family is strongly summable, and its sum has support in $W^* \setminus \{0\}$; grafting all trees below a common root is a permitted regrouping and proves (12.5).

For uniqueness, suppose $V \neq V'$ both solve (12.5) with positive coordinate supports, and let ν_0 be the least exponent at which some coordinate of V and V' differ. Every term of $B(V) - B(V')$ contains such a difference multiplied by a coefficient of B , whose exponent is strictly positive, and possibly by further coordinates of positive support. No such term can contribute at ν_0 , so V and V' agree there, a contradiction. $\square$

Remark 12.5 (Not a Picard limit). Lemma 12.4 does not identify its support-defined sum with a limit of iterates. That distinction is the one recorded in example 2.3: the statement remains valid when the integer multiples of the least coefficient valuation are *not* cofinal in $\Gamma_{>0}$, where no iteration argument based on a fixed improvement of valuation is available. The lemma also needs no norm on R and no completeness assumption; it needs only positivity and well-ordering of the coefficient supports.

12.3 A fixed ramified family of ordinary Leray cycles

Everything in this subsection is ordinary complex analysis, and its radial variable s is an ordinary complex number. The substitution $s = t^\rho$ is made only in section 12.4. This order of operations is the whole mechanism: it prevents the interchange of ordinary analytic convergence with Hahn summability that section 2.3 and example 2.3 warn against, and it is the reason the construction never needs a common radius.

The classical input is the finite mapping theorem in its standard form [20, Chapters II–III]. For f as in (12.1) one may choose bounded representatives U_f of the source and V_f of the target with $f : U_f \rightarrow V_f$ proper and finite, $f^{-1}(0) = \{0\}$, and degree μ , and a nonzero convergent germ $\Delta(w)$ vanishing on the branch locus, so that f is an unramified covering of degree μ over $V_f \setminus \{\Delta = 0\}$. Concretely, take a small closed source ball whose boundary avoids $f^{-1}(0)$ and then a target polydisk whose inverse image stays clear of that boundary; properness follows, and the fibres are finite because a positive-dimensional compact analytic subset of a domain in $\mathbb{C}^n$ cannot occur. If f is already unramified near 0 take $\Delta = 1$. These are ordinary finite-map facts, and none of them is a statement about a Hahn analytic space.

Lemma 12.6 (A discriminant-avoiding monomial direction in the target). *For every nonzero convergent germ $\Delta(w_1, \dots, w_n)$ there are positive integers $q_1, \dots, q_n$, an integer $d \geq 0$, a multi-index*

α , and $c \neq 0$ such that on a fixed polyannulus $A = \{\varrho_1 < |\xi_j| < \varrho_2\}$ around the ordinary torus

$$\Delta(\tau^{q_1}\xi_1, \dots, \tau^{q_n}\xi_n) = \tau^d(c\xi^\alpha + \tau R(\xi, \tau)), \quad (12.6)$$

and, after shrinking the τ disk and if necessary the polyannulus, the bracket is nowhere zero on a neighborhood of $\bar{A} \times \{0\}$. Hence the displayed target points avoid the branch locus for every sufficiently small $\tau \neq 0$.

Proof. Let d_0 be the least total degree of a nonzero term of Δ . Choose positive rational weights very close to 1 and generic on the finitely many monomials of total degree d_0 , so that exactly one of them has least weighted degree; taking the weights close enough to 1 also makes every monomial of total degree at least $d_0 + 1$ have strictly larger weighted degree than all degree- d_0 monomials. Clearing denominators gives positive integers q_j and a unique least-weight monomial cw^α of weighted degree d . Substituting in the convergent series gives (12.6), every other weighted degree exceeding d by a positive integer, and the remainder R is holomorphic for bounded ξ and small τ . On a closed polyannulus $c\xi^\alpha$ is bounded away from zero, so compactness gives uniform nonvanishing of the bracket for small τ . $\square$

The monomial direction is chosen in the *target*, not in the source. This is what allows an arbitrary isolated f : the lemma does *not* require the lowest weighted homogeneous parts of the f_j to form an isolated complete intersection, and it is exactly the step at which the objection recorded in [section 11.4](#) — that a diagonal source rescaling need not keep the leading system isolated or of coordinate-power shape — is bypassed rather than contradicted.

Theorem 12.7 (Ramified Leray family; ordinary analysis). *For f as in (12.1) there exist a polyannulus A around the ordinary torus $(S^1)^n$, a finite unramified holomorphic covering $p : X \rightarrow A$ of total degree μ — with X possibly disconnected — positive integers $\kappa_1, \dots, \kappa_n$, and a holomorphic map*

$$Z : X \times \mathbb{D}_\epsilon \longrightarrow \mathbb{C}^n$$

with

$$Z(q, 0) = 0, \quad f_j(Z(q, s)) = s^{\kappa_j} p_j(q) \quad (1 \leq j \leq n), \quad (12.7)$$

where $p = (p_1, \dots, p_n)$. For $s \neq 0$ the fibre of p over ξ is identified with the set of all μ inverse images in U_f of $(s^{\kappa_j}\xi_j)_j$. The restriction $T = p^{-1}((S^1)^n)$ is a compact oriented real n -manifold without boundary, a finite disjoint union of tori, and $Z(T, s)$ is the ordinary Leray cycle of f for target radii $|s|^{\kappa_j}$. One may take $\kappa_j = Nq_j$ with q_j as in [lemma 12.6](#) and N dividing $\text{lcm}(1, \dots, \mu)$.

Proof. Pull the finite map $f : U_f \rightarrow V_f$ back along

$$A \times \mathbb{D}^* \longrightarrow V_f, \quad (\xi, \tau) \longmapsto (\tau^{q_j}\xi_j)_j.$$

By [lemma 12.6](#) the image avoids the branch locus, so the pullback is an unramified holomorphic covering of degree μ , and its monodromy is a permutation representation of $\pi_1(A \times \mathbb{D}^*) \simeq \mathbb{Z}^n \times \mathbb{Z}$. Let σ be the permutation induced by the radial generator and N its order, so N divides $\text{lcm}(1, \dots, \mu)$ by the elementary order bound for a permutation of μ letters. The substitution $\tau = s^N$ replaces σ by $\sigma^N = \text{id}$, so the resulting covering has trivial monodromy along the radial factor and is therefore pulled back from a finite covering $p : X \rightarrow A$; the classification of coverings gives an isomorphism of the total space with $X \times \mathbb{D}^*$, holomorphic because locally it identifies two holomorphic local inverses of the same base map. *No global labelling of all μ sheets over A is required, and X may be disconnected.*

Composing with the source-coordinate projection gives a holomorphic $Z : X \times \mathbb{D}^* \rightarrow U_f$ satisfying (12.7) for $s \neq 0$, with $\kappa_j = Nq_j$. It is bounded, because U_f was chosen bounded, so the removable-singularity theorem with holomorphic parameters extends it across $s = 0$: in a local chart, the Cauchy integral formula in s kills every negative Laurent coefficient, the remaining coefficients are holomorphic in q , and extensions agree on overlaps. Continuity gives $f(Z(q, 0)) = 0$, whence $Z(q, 0) = 0$ because $f^{-1}(0) = \{0\}$.

For fixed $s \neq 0$ every inverse image of the scaled target torus occurs exactly once in this covering, and the source-coordinate map is injective on $\mathbb{T}$: equality of source points forces equality of target points and then equality of the sheet. Orient $\mathbb{T}$ so that $p^*(d \arg \xi_1 \wedge \cdots \wedge d \arg \xi_n)$ is positive; multiplying the j th target coordinate by s^{κ_j} only adds a constant phase, so the induced orientation is the Leray orientation $d \arg f_1 \wedge \cdots \wedge d \arg f_n$ fixed in section 2.4. A finite covering of a compact torus is compact with finitely many components, each a torus. $\square$

Lemma 12.8 (Coefficientwise constant-period identity). *For $h \in \mathbb{C}\{z\}$ and $k \in \mathbb{N}^n$ set*

$$\beta_{h,k}(s) = \frac{h(Z(q, s)) d_q Z_1(q, s) \wedge \cdots \wedge d_q Z_n(q, s)}{s^{\sum_j \kappa_j(k_j+1)} p_1(q)^{k_1+1} \cdots p_n(q)^{k_n+1}}, \quad (12.8)$$

where d_q differentiates the angular parameters only. For each fixed h and k this form has a convergent Laurent expansion in s with finitely many negative powers, on one ordinary neighborhood of $\mathbb{T}$ and for sufficiently small s , and

$$\frac{1}{(2\pi i)^n} \int_{\mathbb{T}} \beta_{h,k}(s) = \text{res}_{(f_1^{k_1+1}, \dots, f_n^{k_n+1})}(h). \quad (12.9)$$

Equivalently, every nonconstant radial Laurent coefficient integrates to zero and the constant one integrates to $(2\pi i)^n$ times the ordinary residue. The admissible s disk may depend on h and on k ; no simultaneous disk for an infinite Hahn family of germs is asserted.

Proof. Compactness of $\mathbb{T}$ and $Z(q, 0) = 0$ give an ordinary s disk on which $h(Z(q, s))$ is holomorphic on one neighborhood of $\mathbb{T}$. The remaining numerator factors are holomorphic there, and the denominator contributes only the displayed finite negative power of s , since the p_j are nowhere zero on $\mathbb{A}$. Ordinary compact-uniform convergence justifies integrating that single Laurent expansion term by term. For small positive real s the cycle $Z(\mathbb{T}, s)$ is an ordinary regular Leray cycle for f with target radii $|s|^{\kappa_j}$, so its normalized integral is the ordinary local Grothendieck residue of $h dz / \prod_j f_j^{k_j+1}$ [21, 34]. Holomorphy in the punctured s disk and uniqueness of the Laurent expansion upgrade that single value to the coefficient statement. $\square$

12.4 The universal protecting cycle

Now, and only now, the radial variable becomes a Hahn monomial. Set

$$\kappa_* = \max_j \kappa_j, \quad \kappa_\Sigma = \sum_j \kappa_j, \quad (12.10)$$

and choose $\rho \in \Gamma_{>0}$ with

$$\kappa_* \rho < \delta; \quad (12.11)$$

divisibility of Γ supplies one, for example $\rho = \delta/(\kappa_* + 1)$. The direction of (12.11) is the one already recorded in lemma 10.3: a *smaller* positive exponent is a *larger* infinitesimal, so the cycle

must be large in valuation in order to protect the perturbation. Expanding the ordinary family of [theorem 12.7](#) as $Z(q, s) = \sum_{\ell \geq 1} s^\ell Z_\ell(q)$, put

$$Z_0 = \sum_{\ell \geq 1} t^{\ell \rho} Z_\ell \in \mathcal{H}(\mathbf{X})_+^n, \quad \text{so} \quad f_j(Z_0) = t^{\kappa_j \rho} p_j. \quad (12.12)$$

This is legitimate by [corollary 12.3](#), all coordinate supports lie in $\mathbb{N}_{>0}\rho$, and $(\mathbf{T}, Z_0)$ is an admissible Hahn cycle in the sense of [definition 12.1](#). *It depends only on f and δ .*

For the particular E under consideration put

$$P_0 = \{\rho\} \cup \bigcup_{j=1}^n (S_E - \kappa_j \rho), \quad \mathcal{M}_0 = P_0^*. \quad (12.13)$$

Finite unions and translates of well-ordered sets are well ordered, and [\(12.11\)](#) makes every element of P_0 strictly positive.

Proposition 12.9 (Admissibility of the untransformed pullback). *For every $H \in \mathcal{A}_n$ the pullback by Z_0 of the untransformed form $H dz_1 \wedge \cdots \wedge dz_n / (F_1 \cdots F_n)$ is a Hahn family of holomorphic n -forms on $\mathbf{X}$ with support contained in*

$$S_H - \kappa_\Sigma \rho + \mathcal{M}_0, \quad (12.14)$$

each coefficient being a finite sum of ordinary holomorphic n -forms on $\mathbf{X}$. The assertion holds for the fully expanded family, before any Taylor, perturbation, or radial index has been regrouped.

Proof. By [lemma 12.2](#), $H(Z_0) \in \mathcal{H}(\mathbf{X})$ with support in $S_H + \mathbb{N}\rho$, and likewise $\text{supp } E_j(Z_0) \subset S_E + \mathbb{N}\rho$. Put

$$Q_j = \frac{E_j(Z_0)}{t^{\kappa_j \rho} p_j},$$

whose support lies in $S_E - \kappa_j \rho + \mathbb{N}\rho$, a positive subset of $\mathcal{M}_0$; the functions p_j^{-1} are ordinary holomorphic units on the same $\mathbf{X}$. Hence

$$\frac{1}{F_j(Z_0)} = t^{-\kappa_j \rho} p_j^{-1} \sum_{r \geq 0} (-Q_j)^r \quad (12.15)$$

is a strongly summable identity in $\mathcal{H}(\mathbf{X})$, by the Neumann inversion recorded after [lemma 2.2](#). Each angular derivative of Z_0 still has support in $\mathbb{N}_{>0}\rho$, so multiplying the numerator and the differential factors by [\(12.15\)](#) gives [\(12.14\)](#).

For the stronger indexed assertion, expand a contribution carrying k_j factors from E_j . Its exponent has the form

$$\lambda - \kappa_\Sigma \rho + \sum_{j=1}^n \sum_{a=1}^{k_j} (e_{j,a} - \kappa_j \rho) + r\rho, \quad \lambda \in S_H, \ e_{j,a} \in S_E, \ r \geq 0, \quad (12.16)$$

and every letter after the fixed shift $-\kappa_\Sigma \rho$ belongs to P_0 . At an exact total exponent, [lemma 2.2](#) permits only finitely many words, hence finitely many multi-indices k , finitely many perturbation coefficient germs and finitely many radial degrees; and for a fixed radial degree, substitution into $Z(q, s)$ reads only finitely many Taylor monomials, because $Z(q, 0) = 0$. This is local finiteness before regrouping, which is what the integration step requires. $\square$

Theorem 12.10 (Universal untransformed contour). *Let f and δ be as in (12.1)–(12.2), and let $\mathbb{T}$, p , κ_j and Z_0 be as in [theorem 12.7](#) and (12.12). Then for every $E \in \mathcal{A}_n^n$ with $\text{supp } E \subset [\delta, \infty)$ and every $H \in \mathcal{A}_n$, the untransformed form has a well-defined coefficientwise pullback and*

$$\text{Res}_F(H) = \frac{1}{(2\pi i)^n} \int_{Z_0[\mathbb{T}]}^H \frac{H(z) dz_1 \wedge \cdots \wedge dz_n}{F_1(z) \cdots F_n(z)}, \quad (12.17)$$

with Res_F the perturbation residue (2.4). The cycle $(\mathbb{T}, Z_0)$ is constructed from f and δ alone, independently of E and of H ; no determinant transformation of F is made and no equation is replaced; and all resulting coefficient functions are holomorphic on one common ordinary neighborhood of $\mathbb{T}$, with the explicit support certificate (12.14).

Proof. Expand the reciprocal denominators by (12.15). The proposed normalized integral becomes

$$\frac{1}{(2\pi i)^n} \sum_{k \in \mathbb{N}^n} (-1)^{|k|} \int_{\mathbb{T}}^H \frac{(H E_1^{k_1} \cdots E_n^{k_n})(Z_0) dZ_{0,1} \wedge \cdots \wedge dZ_{0,n}}{\prod_{j=1}^n f_j(Z_0)^{k_j+1}}, \quad (12.18)$$

and the whole expanded family is strongly summable by [proposition 12.9](#), so integration and every regrouping are legitimate — the coefficientwise integral (6.2) being a complex-linear operation applied at each exponent, with no interchange of an infinite limit.

Fix one selection of Hahn coefficients from H and from the finitely many perturbation factors occurring in one term of index k . Their ordinary product is a single convergent germ h , and *before* the substitution $s = t^\rho$ the angular pullback of the corresponding summand is exactly the ordinary form (12.8). By [lemma 12.8](#) its integral kills every nonconstant radial Laurent coefficient and returns $\text{res}_{(f_1^{k_1+1}, \dots, f_n^{k_n+1})}(h)$. Multiplying by the selected Hahn monomial and summing the strongly summable family reproduces (2.4) term for term, including the sign $(-1)^{|k|}$ and the normalization $(2\pi i)^{-n}$ fixed in [section 2.4](#).

Only f and δ entered the construction of $\mathbb{T}$, p and Z_0 . The support certificate (12.13) depends on S_E , but the cycle does not, and H was arbitrary throughout. $\square$

Corollary 12.11 (Cancellation of the auxiliary scales). *Although the pulled-back form carries exponents in the larger set (12.14), the integral (12.17) has support contained in $\text{supp } H + (\text{supp } E)^*$ and is independent of the chosen discriminant Δ , the target weights q_j , the radial ramification order N , the identification of the covering, and the protecting exponent ρ .*

Proof. The value equals the independently defined series (2.4), whose support bound was established in [section 2.4](#) and whose definition mentions none of those choices. $\square$

This is independence of the period only. It does not assert that two differently scaled or differently ramified cycles are joined by a pole-free homotopy in some larger chain category; the only homotopy proved below is the one of [theorem 13.5](#), between the protecting cycle and its own adapted lift at a fixed scale. A chain-level comparison across weights, covers and scales remains open; see [section 13.9](#).

12.5 Why the missing common radius causes no gap

Two different finiteness principles are used in the proof of [theorem 12.10](#), and the construction works precisely because they are kept apart.

1. **For a fixed ordinary coefficient product**, ordinary compactness of $\mathbb{T}$ supplies a small ordinary s disk and proves the Laurent period identity (12.9). This is an ordinary analytic statement about one germ; the disk may shrink as the germ changes.
2. **For a fixed Hahn exponent**, lemma 2.2 supplies a finite list of contributing coefficient products, multi-indices and radial degrees. This is an algebraic statement about supports; it says nothing about any disk.

No step asks for one ordinary disk that serves all coefficient germs simultaneously, which is what (2.2) shows to be unavailable. The common parameter domain of the cycle comes from the finite map f and its covering — objects built before any perturbation is named — and not from an intersection of infinitely many convergence neighborhoods. This is also why the conclusion is stronger than a representation assembled separately for each numerator or each perturbation: one cycle serves an entire valuation ball of perturbations whose coefficient germs may have radii tending to zero. It is the same distinction as the one recorded after example 2.3 and used in lemma 11.3, applied here to a contour rather than to a transformation law.

13 The equation-adapted lift, its normalizations, and its limits

The protecting cycle of theorem 12.10 is adapted to the *leading* equations: it satisfies $f_j(Z_0) = t^{\kappa_j} p_j$ exactly, while $F_j(Z_0)$ differs from that by a positive-support error. This section corrects it, on the same finite cover, to a cycle on which the *perturbed* equations hold exactly, connects the two by a pole-free Hahn homotopy, and derives the normalizations that section 17.3 named as the acceptance tests for any proposed cycle. It closes with the example, the pathologies, the construction guide, and the exact boundaries of the result.

13.1 Two uniform ordinary radial bounds

The following bounds are ordinary, depend only on the finite map and its ramified family of theorem 12.7, and are fixed before any Hahn coefficient germ is named.

Lemma 13.1 (Finite radial bounds). *After shrinking the polyannulus and the ordinary s disk around the same compact $\mathbb{T}$, there are integers $b_{\text{pole}}, c_{\text{sep}} \geq 0$ such that*

$$(D_z f(Z(q, s)))^{-1} = s^{-b_{\text{pole}}} J(q, s) \quad (13.1)$$

with J holomorphic on the fixed product neighborhood, and such that for any distinct $q, q' \in \mathbb{T}$ with $p(q) = p(q')$ at least one coordinate of $Z(q, s) - Z(q', s)$ has a nonzero Taylor coefficient in s of degree at most c_{sep} . If $\mu = 1$ one may take $c_{\text{sep}} = 0$.

Proof. The determinant $a(q, s) = \det D_z f(Z(q, s))$ is holomorphic across $s = 0$ and nowhere zero for $s \neq 0$, because $Z(\mathbb{T}, s)$ then lies over regular target values. Near a point $(q_0, 0)$, Weierstrass preparation in s writes a as a unit times a monic polynomial in s ; every root of that polynomial must be 0, since no zero with $s \neq 0$ occurs in the product neighborhood, so locally $a = s^r u$ with u a holomorphic unit and r locally constant along $\mathbb{X} \times \{0\}$. A finite cover of the compact $\mathbb{T}$ by such neighborhoods, together with the adjugate formula for the inverse, gives a common finite pole order, and shrinking to a common collar of $\mathbb{T}$ and a common s disk gives (13.1).

For the separation bound, consider the off-diagonal fibre product $\mathbb{T} \times_{(S^1)^n} \mathbb{T} \setminus \text{diag}(\mathbb{T})$, which is compact, being a union of components of a finite cover. At each of its points the two holomorphic

source-coordinate vectors differ for $s \neq 0$, so some Taylor coefficient of their difference is nonzero, and nonvanishing of that coefficient persists on an ordinary neighborhood. A finite subcover gives a uniform upper bound c_{sep} on a degree at which a difference is already detected. $\square$

Fix now

$$L = \max\{b_{\text{pole}}, c_{\text{sep}}, \kappa_*\} + 1, \quad \rho \in \Gamma_{>0} \text{ with } (L + b_{\text{pole}})\rho < \delta, \quad (13.2)$$

for example $\rho = \delta/[2(L + b_{\text{pole}} + 1)]$, available by divisibility. Rebuild Z_0 by (12.12) with this possibly smaller exponent; since $L + b_{\text{pole}} \geq \kappa_*$, the protecting inequality (12.11) still holds and theorem 12.10 is unaffected. Put

$$P = \{\rho\} \cup (S_E - (L + b_{\text{pole}})\rho), \quad \mathcal{M} = P^*, \quad (13.3)$$

a well-ordered positive generating set with $P_0 \subset \mathcal{M}$, because $L + b_{\text{pole}} \geq \kappa_j$ for every j .

13.2 The scaled implicit equation and the exact lift

Write $s = t^\rho$ from here on as an algebraic abbreviation only — the ordinary analysis is finished — and set

$$J_0 = D_z f(Z_0), \quad \Phi = Z_0 + t^{L\rho} V.$$

The requirement $F(\Phi) = f(Z_0)$ is equivalent to the fixed-point equation

$$V = B(V), \quad (13.4)$$

$$B(V) = -t^{-L\rho} J_0^{-1} \left(f(Z_0 + t^{L\rho} V) - f(Z_0) - t^{L\rho} J_0 V + E(Z_0 + t^{L\rho} V) \right), \quad (13.5)$$

a formal power series in V with coefficients in $\mathcal{H}(\mathbf{X})$.

Theorem 13.2 (Equation-adapted lift on the same cover). *Equation (13.4) has exactly one solution with strictly positive coordinate supports. The correction $\Upsilon = t^{L\rho} V$ satisfies*

$$\text{supp } \Upsilon \subset L\rho + (\mathcal{M} \setminus \{0\}), \quad \text{supp } \Upsilon \subset S_E - b_{\text{pole}}\rho + \mathcal{M}, \quad (13.6)$$

so $v_H(\Upsilon) > L\rho$ and $v_H(\Upsilon) \geq \delta - b_{\text{pole}}\rho$ when $\Upsilon \neq 0$, and $\Upsilon = 0$ when $E = 0$. The parameterization $\Phi = Z_0 + \Upsilon$ lies in $\mathcal{H}(\mathbf{X})_+^n$ on the same fixed ordinary parameter domain, is an admissible Hahn cycle on the same finite cover $\mathbb{T}$ in the sense of definition 12.1, and satisfies the perturbed target equations exactly:

$$F_j(\Phi(q)) = t^{\kappa_j \rho} p_j(q) \quad (q \in \mathbb{T}, 1 \leq j \leq n). \quad (13.7)$$

It is the unique parameterization of the form $Z_0 + \Upsilon$ with $v_H(\Upsilon) > L\rho$ satisfying (13.7). No shrinking of the parameter domain depending on E occurs.

Proof. Consider the coefficients of (13.5). A coefficient of total V -degree $r \geq 2$ coming from the ordinary nonlinear part has exponents in $((r-1)L - b_{\text{pole}})\rho + \mathbb{N}\rho$, positive and contained in $\mathcal{M}$ because $L > b_{\text{pole}}$. A coefficient of total degree $r \geq 0$ coming from the perturbation has exponents in $S_E - (L + b_{\text{pole}})\rho + rL\rho + \mathbb{N}\rho$, again a positive subset of $\mathcal{M}$. Both assertions follow from Taylor expansion together with (13.1), and all coefficient functions are holomorphic on the same $\mathbf{X}$ by lemma 12.2. Hence the union of the supports of all coefficients of B is a well-ordered subset of $\mathcal{M} \setminus \{0\}$, and lemma 12.4 applies with $R = \mathcal{O}(\mathbf{X})$, giving existence, uniqueness in the positive ball, and $\text{supp } V \subset \mathcal{M} \setminus \{0\}$. The first bound in (13.6) is immediate.

For the second bound, every finite rooted tree in the expansion of lemma 12.4 has at least one nullary vertex, and the constant coefficient of B is exactly $-t^{-L\rho}J_0^{-1}E(Z_0)$, with support in $S_E - (L + b_{\text{pole}})\rho + \mathbb{N}\rho$. Removing one such vertex exponent from a tree's weight leaves exponents in $\mathcal{M}$, so $\text{supp } V \subset S_E - (L + b_{\text{pole}})\rho + \mathcal{M}$; multiplying by $t^{L\rho}$ gives the second bound. If $E = 0$ that constant coefficient vanishes, every tree has weight belonging to no admissible word, and $V = 0$.

All substitutions used to pass between $F(\Phi) = f(Z_0)$ and (13.4) are covered by the same support bounds and lemma 12.2, so reversing the algebra gives $F(\Phi) = f(Z_0) = (t^{\kappa_j\rho}p_j)_j$ *exactly*, not modulo a truncation. Uniqueness among corrections with $v_H(\Upsilon) > L\rho$ is uniqueness in lemma 12.4. $\square$

Corollary 13.3 (Regular protected sheets). *The matrix $D_z F(\Phi)$ is invertible in the Hahn matrix algebra over $\mathcal{H}(\mathbf{X})$. For each fixed ordinary $\xi \in (S^1)^n$ the μ values $\Phi(q)$ with $p(q) = \xi$ are pairwise distinct. In local holomorphic coordinates on $\mathbf{X}$ the angular derivative of Φ has full rank over the Hahn coefficient ring.*

Proof. Write $J_0^{-1}D_z F(\Phi) = I + J_0^{-1}(D_z F(\Phi) - J_0)$. The ordinary part of the difference contains at least one factor Υ , so after multiplication by J_0^{-1} its exponents exceed $(L - b_{\text{pole}})\rho > 0$; the perturbation derivative contributes exponents at least $\delta - b_{\text{pole}}\rho > 0$. A matrix Neumann series inverts the right-hand side, and J_0 is invertible by (13.1). For two distinct sheets, lemma 13.1 detects a nonzero coefficient of $Z_0(q) - Z_0(q')$ at an exponent at most $c_{\text{sep}}\rho$, while the difference of the corrections has all exponents strictly above $L\rho > c_{\text{sep}}\rho$ and cannot cancel it. Finally, differentiating (13.7) gives

$$D_z F(\Phi) d\Phi = \text{diag}(t^{\kappa_1\rho}, \dots, t^{\kappa_n\rho}) dp,$$

whose right-hand side has invertible determinant in local coordinates, since p is an unramified covering and the monomials are units in K . $\square$

These are statements about the protected sheets only. They do not count the zeros of F in any larger surreal domain, they do not assert that every zero of F is simple, and they do not assume that the displaced zero scheme is reduced; singular leading zeros are allowed throughout, and the cycle lies over regular nonzero target values. In particular (13.7) is an exact identity about *one* parameterized cycle. The image $\Phi[\mathbf{T}]$ is not identified with the entire class of surcomplex solutions of $|F_j| = t^{\kappa_j\rho}$, and no such identification is needed for any integral below. The distinction is the same one drawn in example 4.4 between a full semialgebraic circle and an ordinary-parameter contour trace.

13.3 A pole-free homotopy between the two cycles

Let $\Phi_u = Z_0 + u\Upsilon$ for $0 \leq u \leq 1$, with u an *ordinary* real homotopy parameter as in (7.7); it does not index a limit in K .

Lemma 13.4 (Admissible straight homotopy). *For each j and $0 \leq u \leq 1$,*

$$F_j(\Phi_u) = t^{\kappa_j\rho}p_j(1 + Q_{j,u}),$$

where $Q_{j,u}$ has positive support contained in $\mathcal{M}$ and coefficients smooth on $\mathbf{X} \times [0, 1]$. Every Taylor substitution, geometric inversion, differentiation and pullback for the untransformed form is strongly summable there, and the total pullback has support contained in $S_H - \kappa_{\Sigma}\rho + \mathcal{M}$.

Proof. The difference $f_j(Z_0 + u\Upsilon) - f_j(Z_0)$ carries at least one factor Υ , so its support lies in $L\rho + (\mathcal{M} \setminus \{0\})$, and division by $t^{\kappa_j\rho}p_j$ leaves positive support because $L > \kappa_j$. The perturbation part has support in $S_E + \mathcal{M}$, and division leaves positive support because $S_E - \kappa_j\rho \subset \mathcal{M} \setminus \{0\}$ by (12.11). Expand $(1 + Q_{j,u})^{-1}$ geometrically; lemma 2.2 controls the whole indexed family, exactly as in proposition 12.9. At each exact exponent only finitely many Taylor and geometric terms contribute, so the dependence on u is polynomial there, and ordinary differentiation in u or in the angular parameters cannot enlarge support. This covers the total pullback including its du terms. $\square$

Theorem 13.5 (Untransformed homotopy invariance). *For every $H \in \mathcal{A}_n$,*

$$\int_{\Phi[\mathbb{T}]}^H \frac{H dz_1 \wedge \cdots \wedge dz_n}{F_1 \cdots F_n} = \int_{Z_0[\mathbb{T}]}^H \frac{H dz_1 \wedge \cdots \wedge dz_n}{F_1 \cdots F_n}. \quad (13.8)$$

Consequently (12.17) holds with $\Phi[\mathbb{T}]$ in place of $Z_0[\mathbb{T}]$: the equation-adapted cycle of theorem 13.2 also represents Res_F .

Proof. A meromorphic top-degree form in n complex variables is closed wherever its denominators are invertible, since differentiating its coefficient produces an extra dz_j which wedges to zero against $dz_1 \wedge \cdots \wedge dz_n$; this is a differential-algebra identity and needs no analytic hypothesis. The support-justified chain rule of lemma 12.2 transports it to the total pullback under Φ_u . By lemma 13.4 that pullback is a strongly summable family of smooth closed n -forms on the compact ordinary cylinder $\mathbb{T} \times [0, 1]$, so theorem 6.1 applies to each coefficient. Since $\partial\mathbb{T} = \emptyset$, the only boundary terms are the two end cycles with opposite orientations, and their difference vanishes. Combining with theorem 12.10 gives the last assertion. $\square$

The ordinary pullback Jacobian $d\Phi_1 \wedge \cdots \wedge d\Phi_n$ is of course present in (13.8); it is the change-of-parameters factor that every pullback of a differential form carries, and it is *not* an auxiliary determinant arising from replacing the equation system F by AF . Neither the integrand nor its list of denominators has been altered, which is the precise difference from theorem 11.4 and theorem 11.5, where $\det A$ and $\det C$ are not optional.

13.4 Degree normalization, a finite-sheet trace, and the two acceptance tests

Write $J_F = \det(\partial F_i / \partial z_j)$ as in (11.9).

Theorem 13.6 (Degree normalization on the adapted cycle). *Under the hypotheses of theorem 12.10,*

$$\text{Res}_F(J_F) = \mu, \quad (13.9)$$

and more generally, for $G \in \mathcal{A}_n$ in the target variables w ,

$$\text{Res}_F(G(F) J_F) = \mu G(0), \quad (13.10)$$

where $G(F)$ is expanded about the ordinary map f in the positive perturbation E and is therefore a well-defined radius-free Hahn germ.

Proof. Work on the adapted cycle, using theorems 13.2 and 13.5. By (13.7) and corollary 13.3,

$$\Phi^* \left(\frac{J_F dz_1 \wedge \cdots \wedge dz_n}{F_1 \cdots F_n} \right) = \bigwedge_{j=1}^n \frac{d(F_j \circ \Phi)}{F_j \circ \Phi} = p^* \left(\frac{dw_1}{w_1} \wedge \cdots \wedge \frac{dw_n}{w_n} \right),$$

whose normalized integral over the degree- μ covering $\mathbb{T}$ is μ . This proves (13.9) *without* importing a Hahn flatness theorem or a Nullstellensatz. For the second identity, $f(0) = 0$ makes each ordinary composition with f a convergent germ and the expansion in E is support-controlled by lemma 12.2; on the adapted cycle $G(F \circ \Phi) = G(t^{\kappa_1 \rho} p_1, \dots, t^{\kappa_n \rho} p_n)$, whose nonconstant target Taylor monomials integrate to zero against $\bigwedge_j dp_j/p_j$ on the torus, while its constant monomial is $G(0)$. Finite covering degree and strong summability give (13.10). $\square$

The ordinary degree-residue identity behind (13.9) is classical. What is added is that it survives for entirely unrestricted radius-free positive-support perturbations and follows from an actual admissible parameterized cycle rather than from a finite-algebra package. *Calling (13.9) a conservation law for local algebra length would additionally require the relevant finite-flatness identification; that identification is not smuggled into this proof.*

Corollary 13.7 (Protected trace representation). *For $H \in \mathcal{A}_n$ the fibrewise sum*

$$\Theta_H(\xi) = \sum_{q \in p^{-1}(\xi)} \frac{H(\Phi(q))}{J_F(\Phi(q))}$$

is a single-valued Hahn-holomorphic function on the polyannulus $\mathbb{A}$, and

$$\text{Res}_F(H) = \frac{1}{(2\pi i)^n} \int_{(S^1)^n}^H \Theta_H(\xi) \frac{d\xi_1}{\xi_1} \wedge \dots \wedge \frac{d\xi_n}{\xi_n}. \quad (13.11)$$

Proof. Invertibility of $D_z F(\Phi)$ and the differential of (13.7) give

$$\Phi^* \left(\frac{H dz_1 \wedge \dots \wedge dz_n}{\prod_j F_j} \right) = \frac{H(\Phi)}{J_F(\Phi)} \bigwedge_j \frac{dp_j}{p_j},$$

and integration over a finite covering is integration of the fibrewise sum, valid for each ordinary coefficient and hence for the Hahn family. The matrix inverse used here is a fixed finite monomial shift times a positive-support expansion, so it adds no summability question. $\square$

This is a trace over *protected regular fibres*. It is not a claim that all individual zeros of F at target value 0 have been separated or parameterized, and it does not identify Θ_H with a multiplication trace on a finite algebra.

Proposition 13.8 (Annihilation of the equation ideal, and descent). *For every $H \in \mathcal{A}_n$ and every j , $\text{Res}_F(F_j H) = 0$. Hence Res_F descends to a nonzero K -linear functional on $\mathcal{A}_n/(F_1, \dots, F_n)$.*

Proof. Classical residue annihilation gives $\text{res}_{(f_1^{k_1+1}, \dots, f_n^{k_n+1})}(f_j h) = 0$ when $k_j = 0$, and equals $\text{res}_{(\dots, f_j^{k_j}, \dots)}(h)$ — the same functional with k_j lowered by one — when $k_j > 0$. Substituting these identities into (2.4) and shifting the j th summation index gives $\text{Res}_F(f_j H) = -\text{Res}_F(E_j H)$, every reindexing being licensed by positive support and lemma 2.2; adding $\text{Res}_F(E_j H)$ gives $\text{Res}_F(F_j H) = 0$. The functional is nonzero by (13.9) in characteristic zero. *This is descent only, and not nondegeneracy of a duality pairing.* $\square$

The two acceptance tests, checked. Section 17.3 fixed two compatibility tests for any proposed cycle: its integral should annihilate (F) , and it should satisfy $\text{Res}_F(HJ_F) = \text{Tr}(M_H)$. The first is now proved for the cycle itself, by proposition 13.8, from classical residue annihilation and nothing else. The second is a statement about the functional Res_F , and it is the reported companion result (11.9); what the new theorem contributes is that the untransformed cycle integral *is* that functional, so the test is met exactly to the extent that (11.9) holds — and its $H = 1$ case, $\text{Res}_F(J_F) = \mu = \text{Tr}(M_1)$, is now proved directly on the cycle by theorem 13.6, without the finite-algebra package. *Neither test is offered as a substitute for the construction, and passing them was never in doubt for a functional defined by (2.4); their role is to certify that the cycle computes the right functional and not some near neighbour of it.*

13.5 Formal coefficient germs

The radius-free theorem needs no common radius but still assumes each coefficient germ converges. That hypothesis can be removed from E and H , by a separate argument. The leading map f remains an ordinary *convergent* map with an isolated zero throughout; its convergence is what produced the bounded finite covering, and no formal substitute for that object is offered here.

Lemma 13.9 (Finite-jet extension of an ordinary local residue). *For fixed f and k the functional $\text{res}_{(f_1^{k_1+1}, \dots, f_n^{k_n+1})}$ depends only on a finite Taylor jet of its numerator, and has a unique extension to $\mathbb{C}[[z]]$ continuous for the formal maximal-ideal filtration.*

Proof. The ideal $I_k = (f_1^{k_1+1}, \dots, f_n^{k_n+1}) \subset \mathbb{C}\{z\}$ has the same radical as (f) , hence is primary to the maximal ideal $\mathfrak{n} = (z_1, \dots, z_n)$ of $\mathbb{C}\{z\}$ — which is not the symbol $\mathfrak{m}$ of section 1.5 — so $\mathfrak{n}^{a_k} \subset I_k$ for some finite a_k . Ordinary residue annihilation makes the functional vanish on I_k and therefore on $\mathfrak{n}^{a_k}$, so evaluation on the Taylor polynomial of degree less than a_k defines the extension, and any higher truncation gives the same value. Uniqueness follows because polynomials are dense for that filtration. This is the classical finite-jet property of local residue functionals [21, 34], not a new duality theorem. $\square$

For E and H with formal coefficient germs, (2.4) still defines Res_F using these extended functionals, and positive Hahn support still makes the series strongly summable.

Theorem 13.10 (Formal-coefficient contour theorem). *In theorems 12.10, 13.2 and 13.5 the hypotheses $E \in \mathcal{A}_n^n$ and $H \in \mathcal{A}_n$ may be replaced by $E \in \mathbb{C}[[z]]((t^\Gamma))^n$ and $H \in \mathbb{C}[[z]]((t^\Gamma))$, with f still convergent with an isolated zero. The same cycle, the same support certificates and the same homotopy apply, the pullback coefficients are still ordinary holomorphic functions on the fixed parameter domain, and (12.17) holds with the finite-jet definition of the residue.*

Proof. The construction of p, X, T, Z uses only f and is unchanged. Every substitution of a formal coefficient germ into Z_0 , into Φ or into the homotopy is licensed by lemma 12.2, which was stated for formal h_λ precisely for this purpose; the lift of theorem 13.2 uses only such substitutions, derivatives of formal germs, and finite coefficient operations, so its proof and its support estimates survive verbatim. What must be redone is the residue equality, because lemma 12.8 cannot be quoted for a divergent formal numerator.

Fix an output exponent ν . On the contour side the accounting (12.16) admits only finitely many numerator exponents, perturbation exponents and radial degrees at ν , and each contributing radial coefficient reads only a finite Taylor jet of the selected formal germs. On the series side lemma 2.2

admits only finitely many coefficient selections and multi-indices k at ν , and by [lemma 13.9](#) their residues also read only finite Taylor jets. Take the union of these two finite lists of jet requirements, and replace each selected formal germ by a Taylor polynomial meeting all of them, setting the other coefficients to zero. This introduces no new Hahn exponents and changes the coefficient at ν on neither side. The already proved analytic theorem applies to these polynomial coefficient data and identifies the two coefficients at ν ; as ν was arbitrary, the protecting-cycle identity follows. Finally the homotopy proof is coefficientwise ordinary Stokes applied to expressions that are polynomial in the real parameter u at each exponent, and never uses convergence of an original coefficient germ, so it identifies the adapted period with the protecting period as before. $\square$

The finite-jet argument is a bridge between two of the rings of [section 2.3](#), not an identification of them. It does not assert that a formal germ is an ordinary function on a complex neighborhood; it becomes a Hahn family of holomorphic functions only after the positive-support substitution. Nor does it construct the initial ramified covering when f itself is formal: that is a different extension problem, recorded in [section 13.9](#).

13.6 A mixed six-sheeted example, and the shortcut that fails

Example 13.11 (A six-sheeted singularity, its adapted cycle, and all its monomial residues). Work over $\Gamma = \mathbb{Q}$ and take

$$f_1 = x^2 + y^3, \quad f_2 = x^2 + 3y^3. \quad (13.12)$$

Then $(f_1, f_2) = (x^2, y^3)$, so $\mu = 6$, and the ordered-denominator transformation matrix is $\begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$ with determinant 2; already at the ordinary origin $\text{res}_{(f_1, f_2)}(xy^2) = 1/2$. *That factor 2 is the tripwire of this example:* any calculation that silently replaces the integrand by an ordered-denominator transform of it, without the compensating determinant of [lemma 11.3](#), will differ from the correct answer by exactly this factor.

For a target point w the inverse equations are $x^2 = (3w_1 - w_2)/2$ and $y^3 = (w_2 - w_1)/2$, so the branch divisor lies in $(3w_1 - w_2)(w_2 - w_1) = 0$. Choose target weights $(1, 2)$ and radial ramification $N = 6$, giving $(\kappa_1, \kappa_2) = (6, 12)$, and on a polyannulus in coordinates (a, b) put $p(a, b) = (a^6, b)$, a covering of degree 6. The ordinary family of [theorem 12.7](#) may be written explicitly as

$$Z_x = \sqrt{\frac{3}{2}} s^3 a^3 \left(1 - \frac{s^6 b}{3a^6}\right)^{1/2}, \quad Z_y = -2^{-1/3} s^2 a^2 \left(1 - \frac{s^6 b}{a^6}\right)^{1/3}, \quad (13.13)$$

both binomial branches being 1 at $s = 0$, single-valued and holomorphic for small ordinary s on a fixed collar of $|a| = |b| = 1$. Substitution gives $f_1(Z) = s^6 a^6$ and $f_2(Z) = s^{12} b$. As a runs over the six preimages of a fixed a^6 , the powers a^3 and a^2 realize every pair of square-root and cube-root choices exactly once, *so this is the full six-sheeted cycle and not one selected branch.*

Now deform by

$$F_1 = x^2 + y^3 - t, \quad F_2 = x^2 + 3y^3 - 5t, \quad (13.14)$$

so $\delta = 1$ is an admissible threshold and $J_F = \det \begin{pmatrix} 2x & 3y^2 \\ 2x & 9y^2 \end{pmatrix} = 12xy^2$. On [\(13.13\)](#) the coordinate x has radial order 3 and y has radial order 2, the inverse Jacobian has pole order at most 4, and distinct sheets separate at radial order at most 3; so one may take $b_{\text{pole}} = 4$, $c_{\text{sep}} = 3$, $L = 13$ and $\rho = 1/40$, the inequality [\(13.2\)](#) reading $17/40 < 1$. With $s = t^{1/40}$ the adapted roots of [theorem 13.2](#) are characterized by

$$\Phi_x^2 = \frac{3s^6 a^6 - s^{12} b}{2} - s^{40}, \quad \Phi_y^3 = \frac{s^{12} b - s^6 a^6}{2} + 2s^{40}, \quad (13.15)$$

with the branches selected by their leading terms, and then $F_1(\Phi) = s^6 a^6$, $F_2(\Phi) = s^{12} b$ exactly. Their leading displacements are

$$\Phi_x - Z_x = -\frac{\sqrt{6}}{6a^3} s^{37} + \dots, \quad \Phi_y - Z_y = \frac{2 \cdot 2^{2/3}}{3a^4} s^{36} + \dots,$$

the omitted terms having strictly larger integer radial support — *this is a statement about supports, not a limit in any surreal topology*. The general estimate $v_H(\Upsilon) \geq \delta - b_{\text{pole}}\rho = 36/40$ is attained by the second coordinate, whereas the uniqueness ball $v_H(\Upsilon) > 13/40$ of [theorem 13.2](#) is much larger than the actual correction: it is a uniqueness region, not a sharp bound.

The displaced zeros of (13.14) satisfy $x^2 = -t$ and $y^3 = 2t$, six distinct points of K^2 , and the monomial residues are

$$\text{Res}_F(x^\alpha y^\beta) = \begin{cases} \frac{1}{2}(-t)^p(2t)^q, & \alpha = 2p + 1, \beta = 3q + 2, p, q \in \mathbb{N}, \\ 0, & \text{otherwise,} \end{cases} \quad (13.16)$$

for $\alpha, \beta \in \mathbb{N}$. An independent verification transforms the ordered equations to $(x^2 + t, y^3 - 2t)$, includes the factor $1/2$, and uses the one-variable residues $(-t)^p$ of $x^\alpha/(x^2 + t)$ and $(2t)^q$ of $y^\beta/(y^3 - 2t)$; equivalently it reduces in the quotient basis $1, y, y^2, x, xy, xy^2$ by $x^2 = -t$ and $y^3 = 2t$ and halves the coefficient of xy^2 . In particular

$$\text{Res}_F(xy^2) = \frac{1}{2}, \quad \text{Res}_F(x^3y^2) = -\frac{t}{2}, \quad \text{Res}_F(xy^5) = t, \quad \text{Res}_F(J_F) = 6,$$

the last checking the total degree of the cycle and detecting the loss of a sheet or a reversed orientation. For the entire numerator $H = e^{x+y}$ the positive substitution is legitimate and gives the exact Hahn series

$$\text{Res}_F(e^{x+y}) = \frac{1}{2} \sum_{p,q \geq 0} \frac{(-1)^p 2^q}{(2p+1)!(3q+2)!} t^{p+q}, \quad (13.17)$$

whose constant coefficient is $1/4$, not $1/2$, because the xy^2 coefficient of the numerator is $1/2$. *This is a strong Hahn sum; ordinary numerical convergence at a complex value of t is irrelevant to the statement*. Neither of the two independent verifications substitutes for the arbitrary-support proof of [theorem 12.10](#).

Example 13.12 (The original coordinate torus is the wrong shortcut). In the setting of (13.12), integrate the original form over a small coordinate product circle in (x, y) instead. On a torus with $|y|^3 \ll |x|^2$ both denominators are dominated by x^2 ; on one with the opposite inequality both are dominated by y^3 . Neither is the equation-adapted Leray cycle. In the first region

$$\frac{xy^2}{(x^2 + y^3)(x^2 + 3y^3)} = x^{-3}y^2 \sum_{r,\sigma \geq 0} (-1)^r (-3)^\sigma \frac{y^{3(r+\sigma)}}{x^{2(r+\sigma)}},$$

which has no $x^{-1}y^{-1}$ term, so its coordinate-torus integral is 0 while $\text{Res}_F(xy^2) = 1/2$. *This is why definition 12.1 must allow a finite ramified cover*: the ramification fixes the geometry without multiplying the integrand by an auxiliary determinant, and a Jacobian from the actual parameterization remains, as it must for any pullback of a differential form. The example also shows that the objection recorded in [section 11.4](#) was correct about the coordinate torus: it is the *cycle*, not the theorem, that had to change.

Remark 13.13 (What the executable checks do and do not establish). The source archive ships an exact symbolic suite, preserved in this repository as `code/05-untransformed-leray-cycles-checks.py` with its recorded run in `data/05-untransformed-leray-cycles-checks.json`: 632 exact checks pass under SymPy 1.14.0, in eight groups — 180 binomial-jet identities, 225 monomial residues, 200 ideal-annihilation instances, 9 coefficients of (13.17), 7 quotient relations, 5 target equations, 4 leading displacements and 2 normalizations. The arithmetic is exact over $\mathbb{Q}(t)$ with no floating point, and the matrix route is deliberately independent of the contour derivation: with M_x, M_y the multiplication matrices on the basis $1, y, y^2, x, xy, xy^2$, the trace $\text{Tr}(M_x^\alpha M_y^\beta (12M_x M_y^2)^{-1})$ computes the sum of the simple-zero residues of $x^\alpha y^\beta$ and is compared with (13.16), which tests the determinant factor and all six sheets at once. *These are finite examples only.* They do not verify well-ordering for every input support, parameterized removable singularities for every germ, the rooted-tree construction of lemma 12.4, or novelty, and the source’s own README and recorded output say so.

13.7 Radius collapse and non-Archimedean supports

Two pathologies exercise the hypotheses, and both are handled by the same cycle.

Example 13.14 (Infinitely shrinking coefficient radii in the perturbation). Keep (13.12) and take

$$E_1 = -\sum_{j \geq 1} \frac{t^j}{1 - jx} \in \mathcal{A}_2, \quad E_2 = 0,$$

that is, the witness (2.2) of section 2.3 used now as a *perturbation* rather than as a numerator. Its support is $\mathbb{N}_{>0}$, so $\delta = 1$, but its j th coefficient has a pole at $x = 1/j$, so there is no common ordinary neighborhood of the origin carrying all of its coefficient germs, and any argument resting on a uniformly convergent family on a fixed source disk fails outright. The protecting cycle (13.13) with $s = t^{1/40}$ nevertheless works: on it

$$\frac{1}{1 - jZ_x} = \sum_{r \geq 0} j^r Z_x^r$$

is a strongly summable Hahn identity, each exact output exponent sees only finitely many pairs (j, r) and finitely many radial coefficient choices, and every coefficient function is a finite polynomial expression in the coefficient functions of Z_x and therefore holomorphic on the same annular parameter domain. *Their sizes may grow without bound as the exponent varies; no coefficient norm was imposed or used.* The exact adapted lift also exists, by theorem 13.2, although no closed formula such as (13.15) is available: its coefficients come from the finite rooted trees of lemma 12.4, all on the same parameter domain, with the certificate (13.6).

Example 13.15 (A value group in which radial powers are not cofinal). Let $\Gamma = \mathbb{Q}^2$ with the lexicographic order, and put $\rho = (0, 1)$ and $\eta = (1, 0)$, so that $r\rho < \eta$ for every positive integer r . No sequence of radial Taylor truncations indexed by $r \in \mathbb{N}$ is then a valuation approximation at all scales — the phenomenon of example 2.3 — yet $\sum_{r \geq 0} t^{r\rho}$ is a perfectly good Hahn sum with well-ordered support and a single radial degree at each exact exponent. Combining this with radius collapse and a support that lies in no finitely generated positive semigroup, take

$$S = \{(0, 1 - 1/j) : j \geq 2\} \cup \{(1, 0)\}$$

as the perturbation support, of order type $\omega + 1$ with least element $\delta = (0, 1/2)$, and assign to its first part the coefficient germs $(1 - jx)^{-1}$. Choosing $\rho = \delta/40$ supplies the same protecting and

lifting inequalities as before, and the support proofs remain valid although infinitely many radial degrees lie below $(1, 0)$. *Every proof above uses finiteness at an exact exponent, never finiteness of an initial segment*, which is exactly the distinction recorded after [example 2.3](#) and in [remark 12.5](#).

Remark 13.16 (Why positivity cannot simply be dropped). The positive-support condition is structural, not cosmetic. If arbitrary Hahn perturbations were allowed one could take $E = -f$, leaving $F = 0$ and no denominator at all. Less crudely, a support unbounded below destroys the finite-jet property of substitution: an expression such as $\sum_{r \geq 0} t^{-r^2} z^r$ does not lie in $\mathbb{C}[[z]]((t^\Gamma))$, its coefficient support failing to be well ordered, and it must not enter the argument merely because each individual coefficient exists. *The results above do not assert that positivity is necessary in every individual example*; they give one uniform sufficient condition under which the cycle, the substitution, the geometric inversion and the homotopy all carry verifiable support certificates.

13.8 A construction guide, and an explicit refusal of effectivity

When the leading germ is polynomial or effectively algebraic the proof is an actionable recipe, and it is worth stating separately from the workflow of [section 17](#) because its steps are specific to this cycle. Compute a finite-map representative or a local elimination description; find a nonzero discriminant equation; choose positive integer weights exposing one of its monomials, as in [lemma 12.6](#); compute the radial monodromy, or simply use an exponent dividing $\text{lcm}(1, \dots, \mu)$ that kills it; retain the finite annular cover rather than forcing globally single-valued branches over the target; and expand the source-coordinate map in the ramified radial variable. For a protecting cycle only [\(12.11\)](#) is required. For an adapted cycle, compute the inverse-Jacobian pole bound b_{pole} and the sheet-separation bound c_{sep} of [lemma 13.1](#), choose L and ρ by [\(13.2\)](#), and solve [\(13.4\)](#) by coefficient recursion; the certificate [\(13.3\)](#) is part of the output.

This is not an effectivity claim, and the difference matters for computer algebra. A requested individual Hahn coefficient uses only finitely many tree terms, but the existence of a finite dependency set supplies no decision procedure for membership in an arbitrary well-ordered subset of a general ordered group — the same caution already recorded for Sturm and sign algorithms in [section 4](#). Practical implementations should use rational lattices, explicitly presented support semigroups, algebraic coefficient germs, and exact branch specifications. Arbitrary convergent germs need not admit finite effective descriptions, no uniform complexity bound is asserted for any class of inputs, and the scale estimates [\(13.2\)](#) are transparent but can be far from sharp, as [example 13.11](#) shows.

Step	Established input	What must be checked here
Finite ordinary map	Isolated convergent zero; finite mapping theorem	Proper bounded representative, degree μ , a nonzero discriminant
Target weighting, lemma 12.6	Convergent nonzero discriminant	Unique least-weight monomial and uniform annular avoidance
Radial ramification, theorem 12.7	Finite covering monodromy	Kill the <i>radial</i> permutation only, not all angular monodromy
Extension across $s = 0$	Bounded source representative	Holomorphic parameter extension and $Z(q, 0) = 0$
Radius-free substitution, lemma 12.2	Positive well-ordered support	Finitely many contributions at each <i>exact</i> exponent
Protecting cycle, (12.11)	$\kappa_* \rho < \delta$	Positive denominator errors and the certificate (12.13)

Step	Established input	What must be checked here
Residue equality, theorem 12.10	Fixed-germ Laurent period identity	Strong regrouping, with no common radius for all germs
Adapted cycle, theorem 13.2	Inverse-Jacobian bound b_{pole} ; $L > b_{\text{pole}}$	Positive implicit coefficients; the nullary-vertex support bound
Homotopy, theorem 13.5	$L > \kappa_*$ and positive support	No denominator zero; coefficientwise closedness and Stokes
Formal extension, theorem 13.10	Finite jets of ordinary residues	A separate finite-jet comparison, not an analytic assertion about a formal germ

13.9 What is classical, what is new, and what remains open

Ordinary Leray cycles and the Grothendieck residue are classical [21, 34, 36], as are finite holomorphic maps, analytic discriminants and Weierstrass preparation [20]. The passage from a finite unramified cover to ramified single-valued analytic data is an established ingredient of Abhyankar–Jung theory; Parusiński and Rond use a finite covering, a power substitution and removable singularities in their analytic proof [30, Prop. 2.1]. The positive support lemma is classical [28, 22]. *None of these ingredients is claimed here as an independent discovery*, and the particular one-radial-parameter version of the ramification step is proved above rather than quoted.

What is proposed as new is their combination: an original-form period on a single fixed finite-cover cycle, uniformly for every perturbation above one specified positive valuation and for every radius-free or formal numerator, with an explicit well-ordered support certificate, an exact adapted lift, and an admissible homotopy inside the same chain category. *The targeted literature search of September 21, 2026 did not locate a published theorem with these arbitrary-support, radius-free, uniform-protecting-cycle hypotheses; that is not a priority certification*, and it is subject to the same consolidated status statement as the rest of this article, [section C](#).

Three comparisons must not be collapsed. Müller and Strohmaier’s Hahn-holomorphic and Hahn-meromorphic functions are another genuine use of Hahn series in complex analysis [27], but their analytic summation category is *not* identified with the coefficientwise ring $\mathcal{A}_n$ of arbitrary radius-free germs used here; the two are different objects and no comparison theorem between them is asserted. Second, the determinant-transformed formula (11.8) remains valid and useful; the new theorem answers a different question and does not supersede it. Third, the lift of [theorem 13.2](#) refines, but does not reprove, the support-preserving lift of the ordinary root cover with unchanged monodromy of the companion report *Finite Hahn Deformations: Division, Conservation of Multiplicity, and Residue Duality* [3] at `docs/surcomplex/finite-deformations`: there a coherent root cover is lifted with its monodromy intact, whereas here the finite *radial* monodromy is deliberately killed, before any Hahn substitution, in order to obtain one fixed cover. That report is cited for the lift rather than having it rebuilt.

Exact boundaries. All supports, covers and parameter manifolds are set-sized, and every coefficientwise integral is an ordinary integral over a compact real manifold. The Hahn parameterization is *not* asserted continuous for the fine surreal topology, and [theorem 13.5](#) is not a fine-topological homotopy; no surreal-valued measure and no unrestricted Riemann-sum construction is invoked, consistently with [propositions 3.1](#) and [3.5](#). The input must have an isolated *convergent* ordinary

leading map and a positive-support perturbation: an arbitrary Hahn tuple with no such leading map, and a non-isolated leading zero, are both out of scope. The theorems produce no canonical choice-independent contour; they prove that the *period* is independent of the choices, by [corollary 12.11](#). The parameter space may be disconnected, and no claim is made that every admissible cycle is homologous to the chosen one. [Corollary 13.3](#) gives μ distinct regular points over each protected target point but does not assert that every zero of F is simple or that the deformed zero algebra is reduced, and [proposition 13.8](#) establishes nonzero descent without a perfect-pairing or flatness theorem.

What is now more precise, and still open. First, whether the convergent leading map can be replaced by an arbitrary *formal* isolated map while keeping a fixed ordinary compact parameter manifold. [Theorem 13.10](#) removes convergence from E and H only; the construction of the cycle depends on a bounded analytic finite covering obtained before any Hahn substitution, and formal algebra alone does not supply that object. Second, effective scale selection: an intrinsic invariant of the finite map controlling the best protecting and lifting scales would improve both the mathematics and exact symbolic implementations, and the attained bound 36/40 in [example 13.11](#) shows that the inverse-Jacobian pole order carries real information even where the uniqueness radius is conservative. Third, a homology comparison among different protecting weights, covers and microscopic scales: equality of the periods is proved by [corollary 12.11](#), but a chain-level comparison in a precisely specified larger Hahn category is strictly stronger and does not follow from period equality. This last item is the surviving part of the *admissible scale atlas* question of [section 17.3](#).

14 An algebraic contour pairing for rational differentials

The Hahn constructions above rely on an ordinary base or a useful infinitesimal chart. Rational functions admit a different, fully algebraic closed-period theory over any fixed real closed workspace, requiring neither an infinite sum nor a limiting partition. In this section the coefficient ring is the rational function field $K(z)$ over $K = R[i]$, R real closed containing $\mathbb{R}$; it is none of the four rings of [section 2.3](#), and no statement below is transferred to or from them except through the explicitly proved comparison [proposition 14.4](#).

14.1 Definition from definable winding

Let $\gamma : [0, 1]_R \rightarrow K$ be a semialgebraic continuous loop, piecewise smooth when differential notation is used, or more generally a finite oriented semialgebraic cycle. For $a \notin \gamma$ write $\text{wind}_{\text{sa}}(\gamma, a) \in \mathbb{Z}$ for its semialgebraic winding number as in [section 4](#), normalized to 1 for a positively oriented circle about an inside point and 0 outside.

Definition 14.1 (Algebraic rational contour functional). For a rational differential $\omega = Q(z) dz$, $Q \in K(z)$, with no pole on γ , define

$$I_{\text{alg}}(\gamma, \omega) := 2\pi i \sum_{a \in K} \text{wind}_{\text{sa}}(\gamma, a) \text{res}_a \omega. \quad (14.1)$$

Only the finitely many finite poles contribute, so the sum is finite. As fixed in [section 2.4](#), the factor $2\pi i$ is a normalization selected to match ordinary complex integration using the distinguished copy of $\mathbb{C}$ inside K ; *it is not forced by ordered-field arithmetic*, and over an abstract real closed

field with no distinguished copy of $\mathbb{R}$ one would normalize formally or use $I/(2\pi i)$. A residue at infinity could be included after compactifying and changing charts, but is unnecessary for this affine formula and supplies no new generator, its residue being the negative sum of the finite ones.

Equation (14.1) is the definition of a pairing. It is not a residue theorem deduced from a previously constructed Riemann integral, and it is not a claim that a finite-partition Riemann net converges to this value. Its ingredients are algebraic local residues and the independently defined definable degree. It is useful because it requires neither ordinary leading separation nor a global logarithm: poles and contour coefficients may lie at arbitrary Hahn scales.

Theorem 14.2 (Characterization and algebraic Cauchy theory). *For a fixed semialgebraic loop γ , (14.1) is the unique K -linear pairing on rational differentials without poles on γ which vanishes on exact rational differentials and satisfies*

$$I_{\text{alg}}\left(\gamma, \frac{dz}{z-a}\right) = 2\pi i \text{wind}_{\text{sa}}(\gamma, a).$$

It is additive under concatenation of loops and additive in cycles, changes sign under orientation reversal, and is invariant under definable homotopies avoiding the poles. If all indices at the poles vanish, the integral is zero. If γ is a positively oriented semialgebraic Jordan curve and Q has no poles on its bounded side or on its boundary, then $I_{\text{alg}}(\gamma, Q dz) = 0$; and if a lies on that bounded side, then

$$I_{\text{alg}}\left(\gamma, \frac{Q(z)}{z-a} dz\right) = 2\pi i Q(a).$$

Finally, for a nonzero $Q \in K(z)$ with neither zeros nor poles on γ ,

$$\frac{1}{2\pi i} I_{\text{alg}}\left(\gamma, \frac{Q'}{Q} dz\right) = \sum_a \text{wind}_{\text{sa}}(\gamma, a) \text{ord}_a(Q), \quad (14.2)$$

negative orders counting poles.

Proof. Since K is algebraically closed of characteristic zero, partial fractions give $Q(z) = P(z) + \sum_a \sum_{j=1}^{m_a} c_{a,j}(z-a)^{-j}$. The polynomial part has a polynomial primitive, and for $j \geq 2$

$$\frac{dz}{(z-a)^j} = d\left(\frac{(z-a)^{1-j}}{1-j}\right).$$

Thus a pairing with the stated properties must retain exactly the $j = 1$ terms with the displayed winding coefficients; this proves uniqueness and simultaneously verifies that (14.1) has those properties, since the residue of a rational derivative is zero (differentiating a Laurent term cannot produce exponent -1). Additivity, orientation behaviour and homotopy invariance follow from the corresponding properties of wind_{sa} . For a positively oriented Jordan curve, exterior poles have winding zero; in the Cauchy-kernel expression the only contributing interior pole is a , with residue $Q(a)$. If $Q = (z-a)^m V$ with $V(a) \neq 0$, then $Q'/Q = m/(z-a) + V'/V$ has residue m at a , giving (14.2). $\square$

Equation (14.2) is a rational argument principle over the full real closed workspace; for a simple positively oriented Jordan cycle its right side is the number of inside zeros minus inside poles, with multiplicities, an exact arbitrary-scale counterpart of the classical statement. Eisermann's polynomial Cauchy and root-counting theorems provide the effective winding component for polynomially described contours [17, 32], subject to the effectivity caveat of section 4.

14.2 Residues are the complete rational obstruction

Proposition 14.3 (Rational de Rham quotient). *Over the algebraically closed characteristic-zero field K ,*

$$\frac{K(z) dz}{dK(z)} \simeq \bigoplus_{a \in K} K \left[\frac{dz}{z-a} \right], \quad (14.3)$$

and the coordinates of a class are its residues at the finite points.

Proof. Partial fractions express every rational differential as a polynomial differential plus finitely many pole terms. Polynomial terms integrate by division by positive integers; for $j \geq 2$, $(z-a)^{-j} dz$ is the derivative of $-(z-a)^{-j+1}/(j-1)$. Only simple poles remain. The residue of a derivative is zero, so a nonzero linear combination of distinct simple-pole terms cannot be exact. This proves both spanning and independence. $\square$

Consequently (14.1) is the winding pairing with the *entire* rational de Rham quotient, not an arbitrary selection of part of the rational information. The corresponding local statement is purely formal: for $\omega = \sum_{j \geq j_0} c_j u^j du \in K((u)) du$ there is a formal Laurent primitive exactly when $c_{-1} = 0$, since every other term integrates by division by $j+1$ while differentiating a Laurent series can never produce a u^{-1} coefficient. Hence

$$K((u)) du / dK((u)) \simeq K, \quad [\omega] \mapsto c_{-1}, \quad (14.4)$$

and the residue is invariant under an invertible formal coordinate change $u = v h(v)$, $h(0) \neq 0$: exact differentials remain exact, and $du/u = dv/v + h'(v) dv/h(v)$, whose second term has zero residue. Residues therefore give a coordinate-invariant obstruction to exactness independently of any connected topology. This also explains the difference from the fine-local category: a global fine-analytic primitive of $1/z$ may exist after a phase convention, but no rational primitive exists, and the cohomology computed in (14.3) is that of a differential algebra, not unrestricted fine-local cohomology.

Finite rational and semialgebraic data over **SC** can always be placed in a set-sized real closed workspace: the relevant roots lie in its algebraic closure K , and enlarging the workspace changes neither finite sign computations nor the residue formula. This gives an especially robust full-surcomplex interpretation for finite algebraic input.

14.3 Agreement with Hahn integration on a verified overlap

Proposition 14.4 (Comparison on separated charts). *Suppose a semialgebraic loop γ has an affine normalized description $C = a + r\gamma_R$, where γ is an ordinary piecewise-polynomial closed loop and γ_R its semialgebraic extension to field-valued parameters, whose restriction to ordinary parameters is an admissible Hahn-deformed contour $\gamma_0 + \eta$. Assume that the same description on the full field interval is uniformly infinitesimally close to its ordinary real-coefficient leading loop. Suppose each pole b of $\omega = Q dz$, $Q \in K(z)$, has either finite normalized position $(b-a)/r$ with standard part off the ordinary trace of γ_0 , or infinite normalized position. Then, whenever the rational pulled-back differential is a coherent family on a common ordinary collar,*

$$I_{\text{alg}}(\gamma, \omega) = \int_{\gamma}^H \omega.$$

The same holds after a polynomial infinitesimal deformation for which a definable pole-avoiding homotopy to this loop is verified.

Proof. Use finite partial fractions after the affine substitution. Polynomial terms have coherent primitives and zero period; a term of order greater than one has a rational primitive, coherent on the coefficient collar, and also zero period. For a finite pole b with normalized position $c + \varepsilon$, ordinary compactness gives a positive ordinary separation between γ_0 and c ; the real-coefficient semialgebraic description has the same lower bound on the field interval, and uniform infinitesimal closeness makes the straight definable homotopy from (γ, b) to (γ_0, c) avoid zero differences throughout, so $\text{wind}_{\text{sa}}(\gamma, b) = \text{wind}(\gamma_0, c)$. [Theorem 8.4](#) then gives equality of the two pairings on $dz/(z - b)$. If the normalized pole is infinite, its inverse is infinitesimal, the reciprocal has a coherent expansion in nonnegative powers of the finite chart variable with zero period, and the field-valued polynomial loop lies in the bounded part of that chart so the pole has index zero. Exact rational differentials have zero period by the coefficientwise fundamental theorem on the admitted pullbacks. Partial fractions now prove the assertion term by term, the affine scale factors cancelling as in [\(7.6\)](#). For the last sentence, use the specified definable homotopy for algebraic indices and [theorems 7.6 and 8.2](#) for the coherent integrals. $\square$

The explicit overlap hypotheses prevent an unwarranted claim that every smooth Hahn-deformed path extends to a semialgebraic map on $[0, 1]_{R^+}$; most do not. The proposition says the two theories agree when their objects and their homotopies really can be compared. Agreement of the two constructions is *proved where both are defined*, not postulated across different chain categories.

[Propositions 9.1 and 14.4](#) are two separate compatibility results and must not be conflated. The first compares two *chart descriptions of one chain* within the Hahn category; the second compares *two different integration theories*. Here is the explicit case where conflating them fails. For the full semialgebraic unit circle, [\(14.1\)](#) gives immediately

$$I_{\text{alg}}\left(S^1, \frac{dz}{z - (1 - t)}\right) = 2\pi i, \quad I_{\text{alg}}\left(S^1, \frac{dz}{z - (1 + t)}\right) = 0. \quad (14.5)$$

These are legitimate values *in exactly the case where the unmodified macroscopic Hahn pullback fails*, namely the boundary-layer situation of [section 3.6](#) and [example 4.2](#). They do not contradict [proposition 14.4](#), whose leading-separation hypothesis is absent here; and they are not obtainable from [proposition 9.1](#), which presupposes admissible pullbacks that do not exist in this case.

14.4 What this pairing does not determine

For an *open* path, the differential $dz/(z - a)$ requires a logarithmic increment, not just a residue and a winding integer; a choice of path continuation or a more elaborate period object is needed, and a global fine logarithm is not automatically that choice by [section 3](#). For nonrational analytic functions, partial fractions need not leave a rational exact remainder, so [\(14.1\)](#) is *not* a definition on all of $\mathcal{H}(U)$ or $\mathcal{A}_n$; those larger classes are handled by the coherent and microscopic constructions, not by pretending that the rational formula already covers them. The functional also does not by itself extend to a general Hahn-holomorphic function with an infinite collection of coefficient singularities.

15 Root-of-unity sampling: a positive theorem and its indispensable hypothesis

Two of the merged manuscripts leave opposite impressions about root-of-unity quadrature: one presents it as a negative result, the other devotes a section to its positive side. They are compatible,

and the hypothesis is doing all the work. This section therefore states the positive theorem *with* that hypothesis and keeps the negative example immediately alongside as the demonstration that the hypothesis cannot be dropped.

15.1 What the classical construction assumes

For a complete algebraically closed field with a real-valued non-Archimedean absolute value, a useful circle integral is a limit of the weighted samples

$$S_N(f; a, r) = \frac{r}{N} \sum_{\xi^N=1} f(a + r\xi) \xi, \quad (15.1)$$

the ξ being the ordinary complex roots of unity. Cherry's presentation of the Schnirelman integral uses this normalization, takes $N \rightarrow \infty$ with $|N| = 1$, and defines the integral only when the field-valued limit exists [13, §1.3]; its residue normalization is 1, not $2\pi i$, and the chosen r is part of the datum for arbitrary functions, not just its absolute value. In a Hahn field with coefficient field $\mathbb{C}$ the natural valuation has residue characteristic zero and $v(N) = 0$ for every positive integer. Related averages appear in p -adic line integral theories [16, 18].

The basic algebraic fact is exact and requires no limit. If $f(a + w) = \sum_{j=j_0}^{j_1} c_j w^j$ is a Laurent polynomial, then

$$S_N(f; a, r) = \sum_{\substack{j_0 \leq j \leq j_1 \\ N|j+1}} c_j r^{j+1}, \quad (15.2)$$

from $N^{-1} \sum_{\xi^N=1} \xi^q = 1$ when $N \mid q$ and 0 otherwise. When N exceeds every nonzero $|j + 1|$ in the finite support, the average equals c_{-1} , which is exactly the corresponding normalized contour integral. *This is exact algebraic quadrature for finite Laurent data; it is not a convergence theorem.* The words “complete Hahn field” alone do not justify a limit: a common ordinary analytic coefficient expansion and a convergent non-Archimedean Laurent expansion are different hypotheses.

15.2 A rank-independent sufficient condition for a valuation limit

Proposition 15.1 (A precise sampling criterion). *Suppose $A_k \in K$, $k \in \mathbb{Z}$, satisfy the decay condition*

$$\text{for every } \gamma \in \Gamma, \quad \{k \in \mathbb{Z} : v(A_k) \leq \gamma\} \text{ is finite.} \quad (15.3)$$

Then $f(\xi) = \sum_{k \in \mathbb{Z}} A_k \xi^k$ is strongly defined at every root of unity, and

$$S_N(f; 0, 1) = \sum_{\ell \in \mathbb{Z}} A_{\ell N - 1} \longrightarrow A_{-1} \quad (15.4)$$

in the valuation topology as the ordinary integer N tends to infinity. This holds for Γ of arbitrary rank.

Proof. For a fixed exponent bound γ , only finitely many A_k can contribute below that bound; their supports form a finite union of well-ordered sets. This proves that the full union is well ordered — a descending sequence would be bounded above by its first term and hence lie in that finite union — and also proves local finiteness of contributions. Multiplication by a root of unity does not change support, so the series defining $f(\xi)$ is strong. The finite root-of-unity sum gives $N^{-1} \sum_{\xi^N=1} \xi^{k+1} = 1$ when $N \mid k + 1$ and 0 otherwise, and strong summability permits applying

this finite linear operation, proving the alias formula. Given γ , choose N_0 so large that the finitely many $k \neq -1$ with $v(A_k) \leq \gamma$ all satisfy $|k + 1| < N_0$; for $N \geq N_0$ none of them has $N \mid k + 1$, so every remaining nonprincipal term in (15.4) has valuation greater than γ , and so does the error. This is exactly valuation convergence. $\square$

The hypothesis is doing all the work, and its strength should be stated plainly. Condition (15.3) demands eventual domination of *every* value-group threshold; it is strictly stronger than the existence of a strong sum. Moreover, when no countable subset is cofinal in the positive value group, a nontrivial infinite sequence of finite valuations *cannot* satisfy this tail condition. This is one reason sequential constructions have a limited role at very large ranks.

15.3 Coefficientwise recovery always has a different meaning

Let $F = \sum_{\lambda \in S} f_\lambda t^\lambda$ have one well-ordered support and ordinary continuous coefficient functions on the unit circle. In the fixed support space $\mathbb{C}^S$ use the product topology of the ordinary complex coefficient topologies; this is a topology on *coefficient presentations* and is not the natural valuation topology of K .

Theorem 15.2 (Coefficientwise recovery of a Hahn contour). *The samples $S_N(F; 0, 1)$ have support contained in S and*

$$\lim_{N \rightarrow \infty}^{\text{coefficientwise}} S_N(F; 0, 1) = \frac{1}{2\pi i} \oint_{|z|=1}^H F(z) dz, \quad (15.5)$$

the right side being a Hahn series with support contained in S . If the f_λ are holomorphic on a common ordinary annulus — that is, if $F \in \mathcal{H}$ of that annulus — the coefficient at λ is their ordinary Laurent residue.

Proof. At exponent λ the sample is $N^{-1} \sum_{j=0}^{N-1} f_\lambda(e^{2\pi i j/N}) e^{2\pi i j/N}$, an ordinary Riemann sum for $(2\pi)^{-1} \int_0^{2\pi} f_\lambda(e^{i\theta}) e^{i\theta} d\theta$, which is the normalized ordinary contour integral; its convergence requires no uniformity in λ . Taking these limits creates no exponents outside the fixed well-ordered S , so the result is a legitimate element of K . The annulus assertion is the ordinary Laurent coefficient formula. $\square$

This realizes the already defined coefficientwise integral by finite samples in a *specified coefficient topology*. It does not make that integral a fine limit, and in general it does not make it a valuation limit either. Taking ordinary coefficient limits is a legitimate alternative description of the Hahn contour functional; it is not a limit in the internal valuation topology.

15.4 Why the hypothesis cannot be dropped

Example 15.3 (A failed valuation-limit contour definition). For $a = 0$, $r = 1$ and $f(z) = 1/(z - b)$ with ordinary real $b = 1/2$, logarithmic differentiation of $X^N - 1$ gives

$$S_N(f; 0, 1) = \frac{1}{N} \sum_{\xi^N=1} \frac{\xi}{\xi - b} = \frac{1}{1 - b^N}.$$

The ordinary complex limit is 1, but $v(S_N(f) - 1) = 0$ for *every* N , so there is no convergence to 1 in the Hahn valuation topology. In the full fine topology it cannot converge at all, since it is not eventually constant. If instead $b = t^\rho$ with $\rho > 0$, then $v(S_N(f) - 1) = N\rho$: a rank-one valuation can see convergence, but a higher-rank group containing an element larger than every $N\rho$ still cannot.

Example 15.4 (Ordinary convergence invisible to the Hahn valuation). Take the ordinary $f(z) = 1/(1 - z/2)$ on a neighborhood of the unit disk, embedded as the sole coefficient at exponent zero. Then

$$S_N(f; 0, 1) = \frac{2}{2^N - 1}. \quad (15.6)$$

It tends to 0 ordinarily and coefficientwise, agreeing with its zero contour integral, but every displayed nonzero number has Hahn valuation 0 and distinct samples differ by nonzero real amounts of valuation 0, so the sequence is not valuation Cauchy. Its ordinary Taylor coefficients 2^{-k} do not tend to zero in the natural non-Archimedean valuation. *No classical Schnirelman theorem on valuation-convergent analytic series is being contradicted.*

Example 15.5 (An obstruction occurring only in higher rank). In $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ let $f(z) = 1/(1 - tz)$. Its coherent unit-circle integral is zero, but

$$S_N(f; 0, 1) = \frac{t^{N-1}}{1 - t^N}, \quad v(S_N) = N - 1 < \omega. \quad (15.7)$$

The samples fail to converge to zero in the valuation topology, although each fixed Hahn coefficient is eventually zero. In a rank-one workspace with value group $\mathbb{Q}$ the same formula does converge to zero. In the full fine topology it cannot converge, since it is never eventually zero. *One exact formula therefore separates three notions of convergence.*

Both sample identities follow without any infinite-series argument: from $\prod_{\xi^N=1} (1 - q\xi) = 1 - q^N$, logarithmic differentiation gives $N^{-1} \sum_{\xi^N=1} \xi/(1 - q\xi) = q^{N-1}/(1 - q^N)$; substitute $q = 1/2$ and $q = t$. This calculation is also an effective diagnostic: before interpreting root-of-unity samples as an integral in a chosen topology, check whether their error valuations dominate the thresholds that topology actually uses. Finite Laurent data give exact algebraic quadrature; general coherent coefficients give ordinary coefficient limits; valuation convergence needs the additional cofinality and regularity hypothesis (15.3).

16 Other established approaches and their exact scope

The following theories are genuine precedents or alternatives. They do not all integrate the same forms, take values in the same field, or use the same notion of path. The comparisons are intentionally narrower than a claim of equivalence, and importing a theorem name without its hypotheses would be misleading.

16.1 Semialgebraic chains, Stokes, and the target of integration

Ohmoto and Shiota prove a C^1 triangulation theorem for semialgebraic sets and formulate integration and Stokes using finite differentiable simplices [29, Thm. 4.1, Prop. 4.2]. This is an important existing counterpart, not merely a proposal. Their non-Archimedean discussion explicitly warns that integral values need not remain in the given field and that a larger saturated target may enter; it should not be read as a construction of the particular K -valued contour operator defined here.

There is a further codomain distinction in the cited literature. Maříková and Shiota construct a measure with values in an ordered semiring, a Dedekind completion of a quotient, with a volume-preserving change of variables [26]; Berarducci and Otero's earlier additive measure is real-valued on definable subsets of the finite part [9]; Kaiser develops semialgebraic measure and integration over

non-Archimedean real closed fields containing $\mathbb{R}$ with *Archimedean* value group, using valuation-section data, and his construction can enlarge the value ring — for Puiseux series a polynomial ring in an additional variable occurs [23]. These are different targets retaining different information, and *a semiring-valued measure is not automatically a signed K -valued complex line integral*. A reference to any of them is not a proof that every signed or complex differential form has a canonical integral in the original Hahn field. Kaiser’s setting also warns that integrals of rational functions may require logarithmic values outside the initial workspace — which the polynomial theorem [theorem 5.1](#) avoids by using finite factorial identities, and which [theorem 14.2](#) avoids by using residues rather than open-path logarithms.

For a surcomplex application one must therefore specify the scalar integration operator on each simplex, its target, and an embedding or comparison map if values in **SC** are wanted. *Triangulation supplies a finite chain decomposition; it does not by itself solve the scalar integration problem*. In particular, saturation alone does not make an ordinary sequence $1/N$ converge to zero in the full ordered topology of a field containing infinitesimals, so the finite-mesh obstruction [proposition 3.5](#) remains unless the limiting notion has been changed. A promising combination is to use definable triangulation to organize domains and then impose Hahn-coherent pullback conditions on each simplex; then [theorems 6.1](#) and [7.4](#) give Stokes whenever those checks succeed. *A theorem that all desired definable data admit such a finite coherent atlas would be an additional result, not a consequence of triangulation alone*.

16.2 Berkovich geometry changes the space, not just the topology

For a rank-one subgroup $\Gamma \subset \mathbb{R}$ put $|x|_v = \exp(-v(x))$. The Hahn field is then complete and algebraically closed, hence a legitimate base for Berkovich geometry. The Berkovich line adds multiplicative seminorms on $K[T]$, not merely more field points; it is tree-like and uniquely arc-connected, and deleting a branch point can create infinitely many components [35, §2.3.3], [10]. Thus the enriched line acquires meaningful paths but *not* a planar Jordan circle: two distinct arcs around a circle would contradict uniqueness, and an annulus has an interval skeleton rather than a circular one. The appropriate boundary intuition is a disk point with outgoing directions, or the ends of an annular skeleton, not a fine-connected curve in K .

The rank restriction is substantive. An arbitrary higher-rank Γ cannot be embedded order-preservingly into $\mathbb{R}$; passing to a real-valued coarsening may lose infinitesimal scales, while retaining them requires a different higher-rank framework, and [example 9.3](#) exhibits distinctions that no single real-valued valuation can retain. Also $|c|_v = 1$ for every nonzero ordinary complex constant c , however small its usual complex modulus, so ordinary coefficient convergence is not non-Archimedean convergence. *No canonical reduction of all surreal scales to one rank is asserted here*.

16.3 Tropical differential forms and Stokes on Berkovich spaces

Chambert-Loir and Ducros construct real (p, q) -forms and currents on Berkovich spaces using tropicalizations and calibrated skeleta, with integration of top-bidegree forms and a Stokes formula; the expanded 2025 revision develops this in a broad tropical setting and includes a tropical Stokes formula [14, Prop. 5.11.1]. This is a powerful established analogue of part of complex differential geometry, and a reason not to regard $d^2 = 0$ as an integration theorem by itself.

Its integrals are real-valued and its geometric objects are enriched analytic spaces. The K -valued periods of $F(z)dz$ constructed here, with the chosen normalization $2\pi i$, are different objects. A comparison could associate tropical quantities to valuations or norms of analytic data, but one should

not expect that operation alone to recover complex phase or the cancellation of opposite residues such as the $\pm 1/(2t)$ of [example 8.8](#). A theorem connecting the pairings would need additional hypotheses and an explicit map on both forms and chains. *A Berkovich Stokes theorem cannot be quoted as the proof of the specific contour identities in the supplied manuscripts.*

16.4 Perfectoid analytic cycles: an actual Stokes theorem, but a different base

Mihara constructs analytic singular homology from a cosimplicial perfectoid object and integration over a restricted class of integrable analytic simplices, with [\[25, Thm. 5.15\]](#) proving a Stokes formula for those simplices; in the integration part of the current text the base field is a finite extension of $\mathbb{Q}_p$ and the target is the de Rham period ring B_{dR} . This is directly relevant as evidence that replacing ordinary simplices and carefully restricting integrability can produce a non-Archimedean cycle pairing.

It is *not* a theorem for the present natural Hahn valuation with residue field $\mathbb{C}$: the residue characteristic and the integration target both differ, and the positive-residue-characteristic hypothesis is decisive. Citing an older title or an abstract about “non-Archimedean integration” would not justify silently applying its period-ring construction to all surcomplex numbers. The common lesson is methodological: a chain category and its face compatibility must be constructed before a Stokes assertion is meaningful.

16.5 Surreal antiderivatives, resurgent extensions, and nonstandard bridges

Costin and Ehrlich construct extensions and integration for substantial classes of resurgent real functions, with treatments of singularities and ordered exponential subfields [\[12\]](#). Their framework is a natural candidate when a contour is parameterized by functions whose real and imaginary pullbacks belong to those classes, and it can retain asymptotic information that a mere standard-part integral would lose. It is not merely the coefficientwise integral on a fixed ordinary domain.

To turn that into a contour theorem, however, one must prove closure under the required compositions $F(\gamma(s))\gamma'(s)$, specify logarithmic branches and endpoint conventions, and establish substitution and homotopy theorems; agreement with Hahn integration should first be tested on common analytic germs, rational forms, and the microscopic residues above. *Extending a real one-variable integral to surreal endpoints is not automatically a construction of complex contour integration*: complexification, phase and branch data, pullback along a path, and compatibility with a two-dimensional boundary operator remain to be established. Outside such an overlap, equality of integration operators should be a theorem, not an assumption. There is no conflict between the breadth of these function-extension theories and the arbitrary smooth coefficient forms used on ordinary parameter manifolds in [section 6](#): the latter construction integrates ordinary functions first and stores their values as Hahn coefficients, and does not assert a canonical fine-smooth extension of every ordinary smooth function to every surcomplex point.

Ducros, Hrushovski, and Loeser construct a comparison between complex and non-Archimedean integrals using nonstandard models and compatible differential-form structures [\[15\]](#). Their work shows that rigorous bridges exist, and also how much additional structure a bridge requires. Real closedness alone is not a transfer principle for arbitrary analytic functions, manifolds, or hyperfinite sums: a nonstandard model with a specified star map can transfer ordinary smooth integration and use internal hyperfinite sums, but the transferred domain, functions, and partitions must be internal, and standard-part or summation operations then determine what information the output retains. *It is invalid to call a proper-class surreal interval an internal hyperfinite interval merely*

because both contain infinitesimals. This identifies the extra data such a comparison would have to carry; it does not rule one out.

16.6 Comparison of the integration strategies

Strategy	What it supplies	What must not be assumed
Polynomial simplex integration	Finite algebraic Stokes identities at arbitrary field scales	That it defines an integral for every semialgebraic function, or for forms with poles on the chain
Hahn coefficient integration	Exact K -valued periods and Stokes on admitted parameter chains	That fine continuity, or mere pole avoidance, is sufficient
Deformed chains and homotopy	Exact endpoint transport, integer winding, deformed Cauchy formulae	That all contour integrals are infinitesimally deformation invariant; only closed forms are
Microscopic rescaling	Contour realizations for radius-free analytic-germ residues	That the original coefficient germs share an ordinary radius
Determinant transformation	Coordinate-torus representation of a general isolated complete-intersection residue	That the untransformed integrand has the same torus integral, or that $\det A$ is optional
Algebraic winding and residues	Rational closed-contour pairing at arbitrary scales	A prior Riemann integral, or canonical open-path logarithms
Semialgebraic integration	Finite triangulated chains and Stokes formulations	That every version has the same scalar target
Berkovich–tropical integration	Real differential forms on enlarged analytic spaces	That the original point plane, or all higher valuation ranks, are preserved
Perfectoid cycle integration	An actual chain category with a Stokes formula	That residue characteristic and period-ring target can be changed for free
Surreal antiderivatives	Integration on distinguished extendable function classes	Automatic closure under arbitrary contour substitutions
Roots-of-unity averaging	Exact Laurent extraction; coefficientwise quadrature	That ordinary coefficient limits are valuation limits

17 A usable procedure, and research directions with exact missing steps

17.1 Choose the object before choosing the formula

The most effective first decision is whether the input is a polynomial differential on a nonstandard simplex, a rational function on a semialgebraic loop, a coherent analytic family on an ordinary base, or a radius-free germ. Those choices lead respectively to (5.1), (14.1), (8.2), and a small-scale realization such as [theorems 10.4, 11.1 and 11.4](#). *Merely declaring a function fine-holomorphic selects none of these integrals.* A reliable workflow is the following finite sequence of checks.

1. **Specify the workspace and the category of chain.** Record Γ , the chosen monomials, and whether the parameters are ordinary real parameters, field-valued semialgebraic parameters, or

- polynomial simplex data. State which notion of interior or component is used, and which of the four rings of [section 2.3](#) the coefficients live in.
2. **Normalize the scale and the integrand.** In a chart $z = a + rw$, compute the pulled-back differential *including* its factor r , as in [\(7.6\)](#). For a meromorphic function, normalize its denominator by a leading monomial and inspect the resulting ordinary leading coefficient on the base contour.
 3. **Prove the support certificate.** Identify the source support S and the positive perturbation set E , and check $S + E^*$. Positivity without well-ordering is insufficient; a lower valuation bound without finite contribution at each exponent is also insufficient.
 4. **Choose the simplification justified by the hypotheses.** Use Stokes or a coherent primitive for a bounding contour; [theorem 8.5](#) for an evaluation; preparation and residues for finite moving poles; algebraic indices when rational semialgebraic geometry is more informative than a coarse standard part.
 5. **Resolve a boundary layer rather than assigning it a side.** When a normalized pole reduces to the base contour, the coefficient inverse may not exist on a common collar ([section 3.6](#)). Move the contour by an explicitly admissible detour, zoom into the offending scale, or use [\(14.1\)](#). State the resulting convention.
 6. **Keep equality distinct from convergence.** A strong sum, a coefficientwise limit, a rank-one valuation limit, and a full fine limit are different assertions. Return the exact Hahn result with its support justification, and add an asymptotic or numerical approximation only in an explicitly chosen topology.

At each step record the coefficient algebra, exponent group, parameter domain, differential, and integral target. Those data remove most apparent paradoxes. A proof that uses a derivative-zero fine function, an ordinary real contour, a definable Jordan interior, and a valuation limit without comparison theorems is generally mixing four different constructions.

17.2 A compact decision table

Input	Natural method	Check before applying it
Polynomial differential on a nonstandard simplex or polytope	Factorial simplex integral and theorem 5.1	Finite polynomial presentation; matching oriented faces.
Coherent holomorphic $F \in \mathcal{H}(U)$ on an ordinary base	Hahn contour integral; theorems 8.1 and 8.5	One common coefficient domain and one well-ordered support.
Uniform positive-support contour deformation	Taylor pullback and theorem 8.2	Common support throughout the deformation; nonsingular coefficient collar.
Rational $Q \in K(z)$ on a semialgebraic loop	Algebraic winding weighted by local residues	No pole on the loop; normalization and orientation fixed.
Radius-free one-variable germs in $\mathcal{A}_1$	Rescale to a protecting circle, theorem 10.4	The strict inequality $m\rho < v_H(E)$.
Radius-free coordinate-power system in $\mathcal{A}_n$	Scaled product torus, theorem 11.1	All $d_j\rho_j < v_H(E_j)$.
General isolated complete intersection in $\mathcal{A}_n$	Determinant transformation, theorem 11.4	The integrand must be the <i>transformed</i> one; see section 11.4 .

Input	Natural method	Check before applying it
The same, with the <i>untransformed</i> integrand	Ramified finite-cover cycle, theorem 12.10 ; adapted lift theorem 13.2	A convergent isolated f and one threshold δ with $\kappa_*\rho < \delta$; the cycle is not a coordinate torus (example 13.12).
Root-of-unity sample sequence	Coefficientwise recovery, or a verified valuation-tail criterion	Never infer the topology of the limit from the sampling formula alone.

17.3 Research problems sharpened by these results

A single-cycle realization for arbitrary finite Hahn zero schemes — posed here, and now solved. The problem, as this article posed it, was: given arbitrary isolated f and radius-free positive-support E , construct one admissible n -cycle on which the *untransformed* integrand integrates to (2.4). A solution may require a non-diagonal analytic change of variables, a finite resolution-type decomposition, or chains on an enlarged analytic space, and it must prove that no omitted summation or radius assumption is hidden in the parameterization. The finite-algebra and trace identities (11.9) offer a robust target: any proposed cycle integral should annihilate (F) and satisfy $\text{Res}_F(HJ_F) = \text{Tr}(M_H)$. These are strong compatibility tests; they do not by themselves construct the cycle. See [section 11.4](#) for exactly how this problem relates to [theorem 11.4](#), which solves the transformed version.

The statement of the problem is retained above because it is the record of what was asked. It is answered by [theorems 12.10](#) and [13.2](#), under the three hypotheses set out in the closing paragraph of [section 11.4](#): a convergent isolated leading map, one specified positive valuation lower bound δ on the perturbation support, and a radius-free — or, by [theorem 13.10](#), formal — numerator. The solution takes the first of the three routes the problem allowed, in a form the problem did not name: not a change of variables in the source but a monomial direction in the *target* combined with a ramified finite covering of the coordinate torus, [lemma 12.6](#) and [theorem 12.7](#). Both acceptance tests are met, and are discussed where they are proved, [section 13.4](#): annihilation of (F) is [proposition 13.8](#), and the trace test is met because the cycle computes Res_F itself, with its $H = 1$ case $\text{Res}_F(J_F) = \mu$ now proved directly on the cycle by [theorem 13.6](#) without the finite-algebra package. What was *not* achieved, and remains open, is a canonical choice-independent cycle and a chain-level homology comparison across protecting weights, covers and scales; only equality of the periods is proved, by [corollary 12.11](#). The formal-leading-map case, an intrinsic invariant governing the best protecting and lifting scales, and the effectivity question are the three further problems listed in [section 13.9](#).

A boundary-layer atlas theorem. Under explicit tame hypotheses, can rational or selected analytic data with poles infinitesimally close to a leading contour be covered by finitely many scale charts whose pullback integrals glue? The values for rational closed contours are already fixed by (14.1); the issue is realizing them by a uniform chain construction that also accommodates open paths and nonrational forms. More generally, an *admissible scale atlas* would need admissible transition pullbacks, a common-refinement property for contour descriptions, and equality of pulled-back integrands on each overlap; [lemma 7.2](#), [theorem 7.4](#), and [proposition 9.1](#) supply local tools, not that global theorem. If local primitives are used, their overlap differences must lie in a controlled constant or period system, or the fine-local logarithm example [example 8.3](#) destroys path invariance.

A comparison theorem with definable integration. Identify a nontrivial common class of functions and chains on which a chosen semialgebraic integration operator, with its actual codomain specified,

maps naturally to the Hahn functional. Polynomial simplex forms are a necessary initial test; logarithmic forms and infinitesimal boundary layers are more discriminating, because choices or loss of scale become visible there. Agreement of rational periods is a beginning, not a general theorem.

An intrinsic coherent chain category. The present construction uses explicit ordinary parameterizations and support-controlled pullbacks. A global theory would define admissible atlases and equivalence of chains, prove a gluing theorem, and identify their homology independently of a chosen presentation, addressing possibly unbounded collections of supports rather than assuming that local coherence gives global common-support data. A related concrete question is to compare (6.1) with the larger complex of forms whose supports are uniform only near ordinary base points, especially on noncompact bases and for locally finite chains.

Sectorial transcendental contour theory and higher-rank analytic chains. Essential singularities may require a selected summation or extension operator with compatible monodromy: the formal Laurent expansion of $e^{1/z}$ at a nonzero infinitesimal z has descending valuations and is not strongly summable [1, §13], so fine analyticity alone does not identify an extension or its periods. Separately, the microscopic torus formulas retain every exponent in Γ , and any relationship with analytic skeleta or higher-rank valuations should either preserve those exact periods or state precisely which quotient of the scale information it keeps. Rank-one real-valued geometries and positive-residue-characteristic perfectoid chains cannot be substituted unchanged; the finite separating-torus construction provides test cases and exact invariants for any proposed broader chain theory.

These are research directions formulated by the present analysis, *not* statements that a particular named published conjecture remains open or has been solved.

18 Conclusion

There is no single unqualified Jordan–Cauchy–Stokes theorem on the raw surcomplex point class, and no justified reason to put the three classical statements under one unspecified topology on $\mathbf{No}[z]$. There are, however, precise and useful counterparts. Standard-part geometry restores an exact two-component theorem for a thick boundary at any chosen affine scale. Definable topology restores an exact two-component theorem with definable paths and definable connectedness, together with integer winding. Polynomial simplex integration gives finite algebraic Stokes identities at arbitrary field scales. Hahn differential forms give a surcomplex-valued Stokes pairing with computable cohomology, and support-controlled Taylor pullback extends that pairing to infinitesimally deformed chains, yielding exact endpoint transport, Cauchy’s theorem and formula, winding stability, and deformed residue contours — all without relying on fine-continuous paths. Positive microscopic rescaling converts radius-free analytic germs into polynomial-coefficient families and gives genuine small-circle and torus residue realizations; residue transformation with its essential determinant represents arbitrary isolated complete-intersection residues by controlled microscopic tori. Rational winding pairings fill an important part of the gap where macroscopic coefficientwise pullbacks fail. And the question that determinant transformation left open — a single cycle for the *untransformed* integrand — is answered by enlarging the cycle rather than the theorem: a target monomial direction avoids the ordinary discriminant, finite radial ramification supplies one fixed compact parameter manifold, a sufficiently large infinitesimal radius protects the whole perturbation, and the positive support lemma turns every radius-free substitution into finite coefficient algebra. One such cycle, chosen from the leading map and one valuation threshold alone, carries the original form to the independently defined perturbation residue for an entire valuation ball of perturbations whose coefficient germs need share no radius at all.

The unifying principle is constructive rather than prohibitive, and it is not to discard infinitesimals or force every scale into one real-valued metric. It is to specify which information each topology retains, to preserve the support algebra, and to prove that boundary, pullback, and integration commute. Within those conditions, nonstandard scales and infinite residues are manageable. Outside them, changing topology or summation method is a new mathematical step that must be stated and justified.

A Support certificates and logical dependencies

The following table records the containing supports used in the proofs. S is a source support, T a second source support, E a positive well-ordered error support, E_a the union of the coordinate supports of a translation point, and $P = \{\rho_1, \dots, \rho_n\}$ a finite positive rescaling alphabet.

Operation	Containing support or finite mechanism
Coefficientwise d , ordinary integration, differentiation, residue extraction, a chosen linear splitting, or the radial homotopy operator (6.4)	S ; no enlargement whatever.
Product of two Hahn families	$S + T$, with finite convolution at each exponent.
Infinitesimal Taylor evaluation (8.1) or deformed pullback (7.3)	$S + E^*$.
Deformed homotopy operator (7.7)	The same $S + E^*$; ordinary integration in the homotopy parameter adds no exponents.
Endpoint correction (8.4)	$S + (\text{supp } \varepsilon)^*$, with at least one displacement factor in every term.
Positive monomial rescaling (10.1)	$S + P^*$; a fixed output exponent has a <i>polynomial</i> coefficient.
Rescaling after infinitesimal translation (10.2)	$S + (E_a \cup P)^*$.
Normalized one-variable denominator, lemma 10.3	Positive error because $v_H(E) - m\rho > 0$; inverse valid on an ordinary base annulus avoiding $w = 0$.
Normalized coordinate-power denominator tuple, (11.2)	Component j has error support $(\text{supp } E_j - d_j \rho_j) + P^*$, positive by the protecting inequality; inverse valid on a common polyannulus.
Perturbation residue after integration, (11.4)–(11.8)	$S + E^*$; before integration the numerator and volume factors may introduce a fixed shift, but a shift of a well-ordered support is still well ordered.
Substitution of a positive-support Hahn map into a radius-free or formal germ, lemma 12.2	$S + P_W^*$, where P_W is the union of the coordinate supports of the substituted map; each output coefficient is a finite sum of products of ordinary holomorphic functions on the one fixed X .
Radial evaluation $s = t^\rho$ of an ordinary family vanishing at $s = 0$, corollary 12.3	$\mathbb{N}_{>0}\rho$; a fixed output exponent sees finitely many radial degrees.
Untransformed pullback on the protecting cycle, proposition 12.9	$S - \kappa_\Sigma \rho + \mathcal{M}_0$ with $P_0 = \{\rho\} \cup \bigcup_j (S_E - \kappa_j \rho)$ positive by $\kappa_* \rho < \delta$; asserted for the <i>fully expanded</i> family, before any regrouping, via the accounting (12.16).
Rooted-tree solution of a positive implicit equation, lemma 12.4	$W^* \setminus \{0\}$; at a fixed weight the vertex count is bounded, so finitely many trees contribute. <i>Not</i> a limit of iterates.

Operation	Containing support or finite mechanism
Equation-adapted correction, theorem 13.2	$L\rho + (\mathcal{M} \setminus \{0\})$ and also $S_E - b_{\text{pole}}\rho + \mathcal{M}$, with $P = \{\rho\} \cup (S_E - (L + b_{\text{pole}})\rho)$; the second bound comes from the nullary vertex every finite tree has.
Straight homotopy of the two untransformed cycles, lemma 13.4	$S - \kappa_\Sigma \rho + \mathcal{M}$, uniformly on the ordinary compact cylinder $\mathbb{T} \times [0, 1]$; at each exact exponent the dependence on the ordinary parameter is polynomial.
Untransformed period after integration, (12.17)	$S + E^*$; every auxiliary scale in (12.14) cancels after angular integration, by corollary 12.11 .
Algebraic rational contour (14.1)	Finite pole set and integer algebraic winding; no infinite sum or limit.
Polynomial simplex integral (5.1)	Finite rational factorial identities in finitely many coefficients.
Coefficientwise sampling limit (15.5)	A fixed containing support S and ordinary convergence in every coefficient; no valuation-convergence inference.

For clarity, “only finitely many dependencies at an exponent” does not mean “only finitely many exponents below it”; the latter fails below ω in $\mathbb{Q} + \mathbb{Q}\omega$. The proofs use the former, which follows from finite word decompositions. This distinction is essential in [lemma 11.3](#) and in the passage from infinitely many germs to one finite ordinary-parameter calculation.

Several hypotheses have deliberately *not* been weakened. The curve in the topological Jordan theorem may be arbitrary, but integration uses piecewise smooth or polynomial parameterizations. The bounded Jordan domain in the Cauchy formula has closure inside the common coefficient domain. Positive perturbation supports must be well ordered, not just positive. Homotopies share one support on their whole ordinary compact parameter space. [Theorem 11.1](#) assumes coordinate-power leading equations. Rational closed-period integration is a definition, compared to analytic integration only on an explicitly established overlap. [Theorem 12.10](#) assumes a convergent isolated leading map and *one specified* positive lower bound δ on the perturbation support; [theorem 13.10](#) removes convergence from the perturbation and the numerator, but not from the leading map, and [remark 13.16](#) records why positivity of the support cannot simply be dropped.

The dependency structure is explicit. [Lemma 2.2](#) defines the Hahn operations. Ordinary Stokes proves [theorem 6.1](#). Taylor pullback and finite-jet identities prove [theorem 7.4](#). Cauchy follows from closedness and ordinary coefficient Cauchy. Independently, positive rescaling and Laurent coefficient extraction realize the perturbation residue, and classical finite-family residue transformation extends the realization to general isolated complete intersections. Independently again, the untransformed realization runs ordinary finite-map theory, a target monomial direction, radial ramification and a removable-singularity extension entirely in an *ordinary* complex radial variable ([lemmas 12.6, 12.8 and 13.1](#) and [theorem 12.7](#)), and only afterwards substitutes $s = t^\rho$; the Hahn half of that argument uses [lemmas 2.2, 12.2 and 12.4](#) and ordinary Stokes on a compact cylinder, and nothing else. *Reversing that order of operations — substituting the Hahn monomial before the covering and the removable-singularity step — would be exactly the interchange of analytic convergence with Hahn summability that this article refuses throughout.* The rational pairing is defined independently from semialgebraic winding and compared with Hahn integration only afterward. *No Jordan statement*

in the fine topology, and no unproved general fine integration rule, enters this chain of arguments. The new proofs do not depend on the Noetherian or Nullstellensatz claims of [4] except where that manuscript's finite-zero and residue results are expressly invoked — in (11.9) and, conditionally and with that dependence stated, in theorem 11.5. In particular theorems 5.1, 6.1, 6.3, 7.6, 11.4, 12.10, 13.2 and 13.6 and the sampling comparisons stand independently of those deeper analytic-algebra claims; theorem 13.6 in particular proves $\text{Res}_F(J_F) = \mu$ without any Hahn flatness or Nullstellensatz input, and the only place where the new sections rely on [4] at all is the second acceptance test discussed in section 13.4, which is a statement about the functional and not about the cycle.

B Why finite-parameter testing proves residue transformation

This appendix supplies the technical point used in lemma 11.3 and in theorem 11.5, without assuming an infinite-parameter contour for the original germ family. It is stated for a transformation matrix C arising from a linear splitting, which is the harder case; the ordinary analytic matrix A of (11.6) is a special case in which the convergence discussion of section B.2 is unnecessary.

B.1 Finite dependency

In the supplied normal-form construction an ordinary linear splitting for $\mathbb{C}\{z\}/(f)$ is extended coefficientwise, and the perturbed inverse is a Neumann series in operators carrying positive perturbation support. Therefore each coefficient of M_i , of Q_i , and of the chosen C_{ij} is built from a finite sum of positive-support words. The residue expansion (2.4) has the same property: for a numerator with support S , its coefficient at λ uses only decompositions

$$\lambda = \sigma + e_1 + \cdots + e_q, \quad \sigma \in S, \quad e_j \in E,$$

of which there are finitely many, including finitely many possible lengths, by lemma 2.2. *This does not say that only finitely many exponents lie below λ .*

Fix a coefficient to be checked in the transformation identity. Retain all ordinary source germs and perturbation germs occurring in its finite dependency sets on both sides, treating numerator source coefficients separately by linearity. Replace the finitely many retained positive perturbation monomials by independent parameters $u_1, \dots, u_b$. The normal-form formulae now define formal power series in u with ordinary analytic germ coefficients, and their formal identities include $Q(u) = C(u)F(u)$ with $Q_i(0, z) = z_i^\mu$. All omitted input terms are irrelevant to the selected coefficient by construction.

B.2 Why a formal matrix causes no convergence gap

The auxiliary ordinary linear splitting need not be continuous in an analytic norm, so $C(u)$ cannot be asserted convergent as an analytic parameter series. Let N exceed every total parameter degree needed for the selected coefficient and let $C_{\leq N}(u)$ be its finite truncation; its finitely many coefficient germs *do* have a common ordinary representative neighborhood. Define the actual finite analytic family

$$\tilde{Q}(u, z) = C_{\leq N}(u, z) F(u, z),$$

which has ordinary leading tuple $(z_1^\mu, \dots, z_n^\mu)$ and agrees with the formal $Q(u, z)$ modulo $(u)^{N+1}$. The tuple $F(u, z)$ has leading isolated complete intersection f , so after choosing ordinary small representatives both tuples have finite clusters near zero.

The ordinary residue transformation rule, applied to these finite families and their clusters, gives

$$\text{res}_{F(u)}^{\text{cluster}}(H) = \text{res}_{\tilde{Q}(u)}^{\text{cluster}}(H \det C_{\leq N}(u)),$$

the family form of classical residue transformation, which follows from the local transformation law and additivity over the finitely many points [11, 36]. Expanding the reciprocal denominators on ordinary protecting residue cycles identifies their Taylor series with the geometric residue expansions, and a Taylor coefficient through total parameter degree N uses only denominator and numerator coefficients through that degree. Consequently the last display proves the formal transformation law through degree N for the original Q and C .

Finally substitute $u_j = t^{e_j}$ and extract the chosen Hahn coefficient. Several parameter monomials may contribute to the same exponent, but only finitely many do, and their finite sum gives the desired coefficient equality. Repeating the argument at each exponent establishes the transformation identity. The auxiliary ordinary cycles may change from exponent to exponent; *the final microscopic separating torus is constructed independently*, by [theorem 10.1](#) and the radius inequalities (11.2) or (11.13).

C What is claimed, what is imported, and what is not proved

C.1 Consolidated status statement

Established results, statements reported in the two supplied manuscripts, and deductions proved here are distinguished throughout. The two supplied manuscripts [1, 4] are treated as research manuscripts, not as independently refereed references; they are not modified or redistributed, and their source filenames are recorded in the bibliography so that the direction of dependence can be checked. The literature search was conducted on September 21, 2026 and was targeted rather than exhaustive. *This is not a claim of a unique accepted foundation for surcomplex analysis, an exhaustive priority or novelty determination, or a formally verified proof.* No named published conjecture is claimed to be solved. Symbolic checks validate the displayed finite examples — exact expressions, Taylor coefficients, matrix identities, residue traces, finite roots-of-unity averages, polynomial Stokes on monomial forms, and endpoint transport on polynomial deformations — and are *not* machine verification of the general theorems: they do not implement arbitrary Hahn fields, enumerate arbitrary ordered supports, decide arbitrary surreal comparisons, or prove semialgebraic Jordan separation. All sums and computational workspaces are set-sized; no proper-class summation is used. The priority position is that the value of this package is its complete set of hypotheses; *no priority claim is based merely on not finding an identical formulation in the literature*, and the constructions proved here are proposed extensions of the supplied presentation, not a claim that their underlying coefficientwise or residue-theoretic ideas are absent from the literature. This article is not an independent audit of every theorem in either supplied manuscript.

C.2 Per-theorem non-claims, preserved

The following specific non-claims are part of the results they accompany and are repeated here for reference.

- *On the sources.* Neither supplied manuscript establishes a Jordan theorem for arbitrary fine-topological curves, nor generalized Stokes for arbitrary fine-smooth functions and arbitrary surcomplex domains. We do not silently attribute either assertion to them.

- On [proposition 3.1](#). This is not a theorem that a Hahn field with its own valuation topology is discrete; the radius supplied by the cut may lie outside the workspace.
- On [remark 3.3](#). A claim about convergence or connectedness in one of the three topologies is not automatically a claim in either of the others.
- On [theorem 4.1](#). The separating object is the thickened curve, not the ordinary point set; neither component is called fine-connected; two infinitesimally separated points in one boundary monad cannot be assigned different sides, and the theorem cannot distinguish two zeros lying in one standard-part fibre; an unbounded exterior in the internal ordered-field sense has not been obtained.
- On [theorem 4.3](#). This is an application of semialgebraic topology, not a new surreal theorem; the technical finite-topology input is imported, not replaced by an informal appeal to transfer of arbitrary continuous maps; the components are not connected in the unrestricted intrinsic topology; the o-minimal generalization refers to that framework's differentiability and definability hypotheses, not to arbitrary class functions or arbitrary fine-topological subsets; definable results do not supply ordinary singular homology of a proper class.
- On [winding](#). The definable degree is an ordinary integer, not a surreal number of turns; it does not by itself define the integral of a general differential form; the primitive of a semialgebraic function need not be semialgebraic; the availability of a finite algebraic prescription does not make arbitrary surreal inputs computable.
- On [proposition 3.5](#). A polynomial antiderivative still gives $\int_0^1 x dx = 1/2$; the obstruction is to a particular unrestricted finite-mesh definition, not to all integration. Hyperfinite partitions require an additional nonstandard universe and transfer structure, supplied neither by the ordered-field notation nor by the surreal normal form.
- On [theorem 5.1](#). The proof uses classical Stokes to certify a finite algebraic identity; it does not transfer a statement quantifying over arbitrary integrable functions. It does not define an integral for every semialgebraic function, nor for a rational form whose denominator vanishes on the chain.
- On [theorem 6.3](#). For infinite-dimensional ordinary cohomology, $V((t^\Gamma))$ must not be silently replaced by $K \otimes_{\mathbb{C}} V$; and the global uniform-support assignment is not claimed to be a sheaf for arbitrary infinite covers.
- On [definition 7.1](#) and [lemma 7.2](#). Φ describes an admissible parameterized chain, not a fine-continuous map from the ordinary parameter manifold; the jet prolongation is the *specified* extension, with no claim of ordinary Taylor convergence or of a canonical prolongation of every smooth function on every fine domain.
- On [theorem 7.6](#). It is incorrect to claim that all contour integrals are infinitesimally deformation invariant; closedness is the controlling condition.
- On [proposition 7.7](#). Ordered estimates do not imply convergence of set-indexed nets in the full fine topology.
- On [theorem 10.1](#). The conclusion is a new coherent representation in an infinitesimal chart, not ordinary convergence of the original germs on a uniform disk; it neither establishes nor requires all-scale coherence.
- On [theorem 10.5](#). An arbitrary Hahn family of meromorphic germs is larger than a quotient of two elements of $\mathcal{A}_1$, and is not claimed to admit the microscopic contour construction.
- On [theorem 11.4](#). The untransformed form is not claimed to have the same torus integral; $\det A$ is not an optional Jacobian factor, since AF can introduce extra zeros; and F and G are not claimed to have identical zero schemes. [Section 11.4](#) records that one of the merged sources judged the general realization unproved, and that the alternative route [theorem 11.5](#) is conditional on [4].

- *On remark 11.6.* Formula (11.14) does not assert regularity of the original quotient on the chosen torus; omitting the determinant is generally wrong; a real n -torus is not identified with the full boundary of a complex polydisk.
- *On definition 14.1.* Equation (14.1) is a definition, not a claim that a finite-partition Riemann net converges to this value; it does not determine open-path logarithms and is not a definition on all of $\mathcal{H}(U)$ or $\mathcal{A}_n$.
- *On theorem 15.2.* Coefficientwise recovery does not make the Hahn integral a fine limit, and in general does not make it a valuation limit either.
- *On theorem 12.10 — hypotheses.* An isolated *convergent* ordinary leading map and a positive-support perturbation are required. An arbitrary Hahn tuple with no such leading map, a non-isolated leading zero, and a support not bounded below by a positive element are all out of scope. Extending to a *formal* isolated leading map is a different open problem, because the proof needs a bounded analytic finite covering constructed before any Hahn substitution; theorem 13.10 removes convergence only from E and H .
- *On theorem 12.10 — what independence means.* Period independence is proved (corollary 12.11); no canonical choice-independent contour is produced. The parameter space may be disconnected, and it is not claimed that every admissible cycle is homologous to the chosen one. A chain-level homology comparison across weights, covers and scales is listed as open.
- *On theorem 13.2 and corollary 13.3.* The adapted parameterization gives μ distinct *regular* points over each protected target point. It does not assert that every zero of F is simple, nor that the deformed zero algebra is reduced, and *its image is not identified with the entire class of surcomplex solutions of $|F_j| = t^{\kappa_{j\rho}}$* — no such identification is needed for any integral proved here.
- *On proposition 13.8.* Nonzero descent of Res_F to $\mathcal{A}_n/(F)$ is established; a perfect-pairing or flatness theorem is not inferred from it. Theorem 13.6 is not called a conservation law for local algebra length, which would additionally require the finite-flatness identification.
- *On theorem 13.5.* The Hahn parameterizations are not asserted continuous for the fine surreal topology, and the homotopy is not a fine-topological homotopy; the parameter u is an ordinary real parameter. All supports, covers and parameter manifolds are set-sized.
- *On section 13.8.* A finite coefficient dependency set supplies no decision procedure for membership in an arbitrary well-ordered subset of a general ordered group; no uniform complexity bound is claimed; arbitrary convergent germs need not have finite effective descriptions; and the scale estimates (13.2) are transparent but can be far from sharp.
- *On remark 13.13.* The 632 exact symbolic checks are finite examples only. They do not verify arbitrary-support Hahn summability, parameterized removable singularities, the rooted-tree construction of lemma 12.4, or novelty; the source’s README and its recorded output both say so.
- *On section 13.9.* Müller and Strohmaier’s Hahn-meromorphic category [27] is explicitly *not* identified with the coefficientwise ring $\mathcal{A}_n$ of radius-free germs, and no comparison theorem between them is asserted. The targeted literature search of September 21, 2026 found no published theorem with the arbitrary-support, radius-free, uniform-protecting-cycle hypotheses proved here, and *that is not a priority certification*. The publisher endpoint for the Neumann reference [28] could not be fetched during that search and its details were corroborated indirectly.

D Notation

Symbol	Meaning
$\mathbf{SC} = \mathbf{No}[z]$	The full surcomplex class field.
Γ, R_Γ, K	A set-sized divisible ordered subgroup of $\mathbf{No}$, and the real and complex Hahn workspaces $\mathbb{R}((t^\Gamma))$ and $\mathbb{C}((t^\Gamma))$, with $t^\gamma = \omega^{-\gamma}$.
v, v_H	Least exponent of a field element; least exponent of a Hahn coefficient-family datum.
$K^\circ, \mathbf{m}_K, \mathbf{m}, \text{st}$	Finite elements; infinitesimals of the workspace; the class of all infinitesimals of $\mathbf{SC}$; the standard-part (reduction) map $K^\circ \rightarrow \mathbb{C}$.
$U^\sharp$	$\text{st}^{-1}(U) = \{c + \varepsilon : c \in U, \varepsilon \in \mathbf{m}_K\}$.
Standard-part topology	The initial topology of st ; called the residue, reduction, or shadow topology elsewhere.
E^*	The additive monoid of finite words in a well-ordered positive support E , including 0.
$\mathcal{H}(U)$	$\mathcal{O}(U)((t^\Gamma))$: one common ordinary coefficient domain.
$\mathcal{A}_n$	$\mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$: radius-free coefficient germs.
$\mathbb{C}[w]((t^\Gamma))$	Polynomial-coefficient ring; the target of theorem 10.1 .
$\Omega_H^q(M)$	$\Omega^q(M; \mathbb{C})((t^\Gamma))$: globally supported Hahn series of ordinary smooth complex q -forms.
$\Phi = \phi_0 + \eta$	Ordinary base parameterization plus a uniformly supported infinitesimal deformation, definition 7.1 .
$\int^H, \oint^H$	Hahn coefficientwise integration and its closed-contour form.
$\int^{\text{pol}}, I_{\text{alg}}$	Polynomial-simplex integration (5.1); the algebraic rational contour pairing (14.1).
$T_F(c, \varepsilon)$	Microscopic endpoint Taylor correction (8.4).
wind, wind _{sa}	Ordinary complex winding; definable/algebraic integer winding.
res _a , Res _F	Ordinary local residue (Laurent or Grothendieck); the strong Hahn perturbation residue (2.4).
m, μ, e_a	Order of a one-variable germ; global colength; local multiplicity at an actual displaced zero, $\sum_a e_a = \mu$.
$\rho_j, r_j = t^{\rho_j}$	Rescaling exponents and the corresponding microscopic radii.
d_j	Coordinate powers in a coordinate-power leading system (11.1).
J_F, M_H, B_F	Jacobian determinant; multiplication operator; the finite quotient algebra $\mathcal{A}_n/(F)$.
$\mathcal{H}(X)$	$\mathcal{O}(X)((t^\Gamma))$ for one fixed ordinary complex manifold X , possibly disconnected; $\mathcal{H}(X)_+$ for strictly positive support, (12.3).
A, p, X, T	A polyannulus around the ordinary torus; a finite unramified covering $p : X \rightarrow A$ of degree μ ; its total space; and $T = p^{-1}((S^1)^n)$, the compact oriented ordinary parameter manifold of theorem 12.7 .
$\kappa_j, \kappa_*, \kappa_\Sigma$	Target radial weights with $f_j(Z(q, s)) = s^{\kappa_j} p_j(q)$, their maximum and their sum. <i>Written m_j and M in the source manuscript</i> ; renamed because m is the order of a one-variable germ.
s, ρ, δ	The <i>ordinary</i> complex radial variable of section 12.3 ; the protecting exponent with $\kappa_* \rho < \delta$; the specified positive lower bound on $\text{supp } E$.
Z, Z_0, Φ, Υ	The ordinary ramified family; the protecting Hahn cycle $\sum_{\ell \geq 1} t^{\ell \rho} Z_\ell$; the equation-adapted lift $Z_0 + \Upsilon$; and its correction.
$b_{\text{pole}}, c_{\text{sep}}, L$	Inverse-Jacobian pole order and sheet-separation degree of lemma 13.1 ; the lift exponent $L = \max\{b_{\text{pole}}, c_{\text{sep}}, \kappa_*\} + 1$.
$S_E, P_0, \mathcal{M}_0, P, \mathcal{M}$	$\text{supp } E = \bigcup_j \text{supp } E_j$; the protecting and lifting generating sets (12.13) and (13.3), and the monoids they generate.

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